Lecture Notes in Mathematics

1595

Robert D. M. Accola

Topics in the Theory of Riemann Surfaces

Springer

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Editors:
A. Dold, Heidelberg
B. Eckmann, Zürich
F. Takens, Groningen



Robert D.M. Accola

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Springer-Verlag
Berlin Heidelberg New York
London Paris Tokyo
Hong Kong Barcelona
Budapest

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Mathematics Subject Classification (1991): 30F10; 14H15

ISBN 3-540-58721-7 Springer-Verlag Berlin Heidelberg New York

CIP-Data applied for

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Printed in Germany

Typesetting: Camera-ready out put by the author
SPIN: 10130221 46/3140-543210 - Printed on acid-free paper

Preface

These are lecture notes for a course given during the Fall of 1988 at Brown University. The students were assumed to have had a previous course on the theory of Riemann surfaces or algebraic curves. Chapter One of these notes gives a review of most of the basic material needed later, but few proofs are given, and the reader is assumed to have some previous acquaintance with the material.

Besides giving the author's point of view and notational conventions, the introduction gives a few proofs (or rather demonstrations) intended to bridge the gap between the more analytic approach to the subject as exemplified by the books of Ahlfors-Sario [4], Farkas-Kra [11] or Forster [12] and the more algebraic approach as exemplified by Walker [26]. In particular the proof of the genus formulas for a plane curve based on the Riemann-Hurwitz formula and other materials in Section 1.3 were inspired by a seminar given in the early '60's at Brown by S. Lefschetz.

Despite the recent appearance of several excellent books on the theory of compact Riemann surfaces (which is, of course, basically the same as the theory of algebraic curves over the complex numbers), most of the material in these notes does not appear in these books. The two main subjects treated here are exceptional points on Riemann surfaces (Weierstrass points, higher-order Weierstrass points) and automorphisms of Riemann surfaces. A foundational treatment of the theory of automorphisms from the viewpoint of Galois coverings of Riemann surfaces is given in Chapters Four and Five, following and expanding to some extent the treatment of Ahlfors-Sario [4] and Seifert-Threlfall [25]. The treatment is technically different from that of A. M. Macbeath [19] and his students, although fundamentally it is the same. In the treatment here, no mention of Fuchsian groups occurs. Also a treatment of the extremely useful inequality of Castelnuovo-Severi is included, a treatment for which the author knows no reference. (It is difficult, however, to believe that anything on this venerable subject can be really new.)

The treatment of all subjects is basically elementary.

The author heartily thanks Natalie Ruth Johnson for an excellent job in preparing this manuscript.

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Chapter 1. Review of some basic concepts in the theory of Riemann surfaces.

1.1 Coverings. A surface X is a connected Hausdorff space which satisfies the second axion of countability and has a basis for the open sets of sets homeomorphic to open sets in $\mathbb{C}$, the complex numbers. A Riemann surface is a surface with an open cover $\{U_\alpha \mid \alpha \in A\}$, for some index set A, and homeomorphisms $\{\varphi_\alpha\}$, $\varphi_\alpha : U_\alpha \to \mathbb{C}$, where $\varphi_\beta \circ \varphi_\alpha^{-1}$ is biholomorphic wherever it is defined. A pair $(U_\alpha, \varphi_\alpha)$ is called a chart and the set $\{(U_\alpha, \varphi_\alpha) \mid \alpha \in A\}$ is called an atlas. φ_α will be called a local parameter. All Riemann surfaces are orientable.

A continuous function $f : X \to Y$ between Rieman surfaces will be said to be holomorphic if it is holomorphic (analytic) when expressed in local parameters. If f is holomorphic, injective, and surjective then f is said to be biholomorphic and X and Y are said to be conformally equivalent. A biholomorphic map of a Riemann surface onto itself will be called an automorphism, and the group (under composition) of automorphisms of a Riemann surface X will be denoted $\mathrm{Aut}(X)$. An automorphism of period 2 will be called an involution.

A holomorphic (meromorphic) function on a Riemann surface X is a holomorphic mapping of X into $\mathbb{C}$ ($\mathbb{P}^1$, the projective line, or Riemann sphere).

If $f : X \to Y$ is a non-constant holomorphic mapping of Riemann surfaces and $x \in X$, then in suitable local coordinates at x and $f(x)$, f looks like $z \to z^n$, n a positive integer. We say f is equivalent to $z \to z^n$ at x. (In particular, f is an open mapping.) n is said to be the multiplicity of f at x, and $n - 1$ is said to be the branching or ramification of f at x, denoted $\mathrm{ram}_x(f)$. The ramified points x, where $\mathrm{ram}_x(f) > 0$, are a discrete set in X.

Theorem. Let $f : X \to Y$ be a non-constant holomorphic map between compact Riemann surfaces. Then there exists a positive integer n so that for any $y \in Y$

$$n = \mathrm{card}\left\{f^{-1}(y)\right\} + \sum_{f(x)=y} \mathrm{ram}_x(f) .$$

n will be called the number of sheets in the covering $X \to Y$. If $Y = \mathbb{P}^1$ then f is a non-constant meromorphic function on X and

n is called the order of f, denoted o(f).

A non-constant holomorphic map of compact Riemann surfaces will often be called a covering.

The Riemann-Hurwitz Formula for coverings.

Let $f : X \to Y$ be a non-constant n-sheeted holomorphic map between compact Riemann surfaces. Let p and q be the genus of X and Y respectively. Let $\text{ram}(f) = \sum_{x \in X} \text{ram}_x(f)$. Then

$$2p - 2 = n(2q - 2) + \text{ram}(f).$$

Definition. If X_p is a compact Riemann surface, the subscript p will always denote the genus of X_p.

Corollary. Let $f : X_p \to X_q$ be a non-constant n-sheeted holomorphic mapping of compact Riemann surfaces where $n \geq 2$. Then $p \geq q$ with equality possible only if $p = 0$ or 1.

1.2 Function Fields. If X is a Riemann surface let M(X) denote the field of meromorphic functions on X. If $f : X \to Y$ is an n-sheeted holomorphic map then $f^* : M(Y) \to M(X)$ maps M(Y) onto a field which is of index n in M(X). Conversely, if X is a compact Riemann surface and K is a subfield of M(X) of index n, then there exists an n-sheeted covering $f : X \to Y$ of compact Riemann surfaces and $f^*(M(Y)) = K$.

A rational function field is a field isomorphic to $M(\mathbb{P}^1)$, that is, the rational functions. A meromorphic function field on a Riemann surface of genus 1 wil be called an elliptic function field. A Riemann surface X_p, $p \geq 2$, will be called hyperelliptic if $M(X_p)$ admits a rational subfield of index 2. This is equivalent to X_p admitting a 2-sheeted covering $X_p \to \mathbb{P}^1$. The interchange of the two sheets of this covering is an automorphism of X_p of order 2 called the hyperelliptic involution. A hyperelliptic Riemann surface, X_p, is the Riemann surface for a polynomial $P(z,w) \in \mathbb{C}[z,w]$ where

$$P(z,w) = w^2 - \prod_{j=1}^{2p+2} (z - a_j) \text{ where } a_i \neq a_j.$$

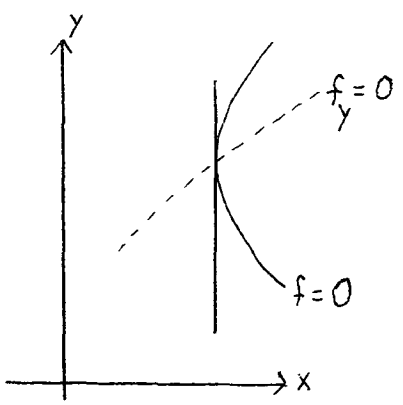
A Riemann surface of genus greater than one will be called elliptic-hyperelliptic if it admits a 2-sheeted covering of a Riemann

surface of genus one ; that is, its function field admits an elliptic function field of index 2 .

In general, if X is a compact Riemann surface then M(X) is the Riemann surface for some (many) irreducible polynomial $P(z,w) \in \mathbb{C}[z,w]$.

1.3 Plane Curves [26]. Let $f(x,y,z)$ be an irreducible homogeneous polynomial of degree n . Let $C_n = \{(x,y,z) \in \mathbb{P}^2(\mathbb{C}) \mid f(x,y,x) = 0\}$, an irreducible plane curve. Suppose C_n is non-singular, that is, $f_x(x,y,z) = 0 = f_y(x,y,z) = f_z(x,y,z)$ implies $(x,y,z) = (0,0,0)$.

We show that as a Riemann surface, C_n has genus $\frac{(n-1)(n-2)}{2}$. C_n has only a finite number of inflection points (where the tangent line has more than two intersections with C_n). Choose coordinate axes in $\mathbb{P}^2$ so that none of the inflectional tangent lines pass through $(0,1,0)$ (i.e. no inflectional tangent line is parallel to the y-axis). Then each tangent line to C_n passing through $(0,1,0)$ has only two intersections where it is tangent. Such points occur where $f = 0$ and $f_y = 0$. At such points the projection of C_n onto the x-axis is locally two-to-one; that is, this projection (a holomorphic map onto the x-axis $(= \mathbb{P}^1)$) has ramification one at each of these points. The number of such points is $n(n-1)$.



Using the Riemann-Hurwitz formula we have

$$2p - 2 = -2n + n(n-1) \; ; \; 2p - 2 = n(n-3)$$

$$\text{or} \quad p = \frac{n^2 - 3n + 2}{2} .$$

We can use the same method if C_n has singularities. Suppose C_n has an ordinary singularity of multiplicity k at $(0,0,1)$. Dehomogenizing $f(x,y,z)$ to $f(x,y,1)$ we have

$$f = \prod_{i=1}^{k} (\alpha_i x + \beta_i y) + \text{higher order terms}$$

where the k tangents lines $\alpha_i x + \beta_i y = 0$ are all distinct. Thus

$f_y = \prod_{i=1}^{k-1} (\gamma_i x + \delta_i y) + \text{higher order}$ and so $f = 0$ and $f_y = 0$ have $k(k-1)$ intersections at $(0,0,1)$. Now assume C_n has s ordinary singularities of multiplicities $k_1, \ldots, k_s$. Then the total ramification for the projection of C_n onto the x-axis is $n(n-1) - \sum k_j(k_j - 1)$ or

$$p = \frac{(n-1)(n-2)}{2} - \sum \frac{k_j \cdot (k_j - 1)}{2}.$$

Let us look more closely at singularities of multiplicity two at, say, $(0,0,1)$.

$f = ax^2 + bxy + cy^2 + \text{higher order}.$

$f_y = bx + 2cy + \text{higher order}.$

Local parametric equations for $f_y = 0$ are

$$x = 2ct + c_2 t^2 + \cdots$$

$$y = -bt + b_2 t^2 + \cdots .$$

Thus $f(x,y) = a(4c^2t^2 + \cdots) + b(-2bct^2 + \cdots) + c(b^2t^2 + \cdots) + o(t^3)$

$= c(4ac - b^2)t^2 + o(t^3)$.

If $4ac - b^2 = 0$, $f = 0$ and f_y have two intersections (node).

If $4ac - b^2 = 0$, $f = 0$ and $f_y = 0$ have at least three intersections (cusp).

If $f = (ax + by)^2 + (ax + by)Q(x,y) + 4^{\text{th}}$ degree terms and higher order, then $f = 0$ and $f_y = 0$ have 4 intersections (tacnode).

Nodes add nothing to the ramification of the propection map.

Cusps add one to this ramification and tacnodes add nothing.

If δ = # of nodes, κ = # of simple cusps, η = # of simple tacnodes for C_n and there are no further singularities then

$$2p - 2 = -2n + n(n-1) - 2\delta - 3\kappa - 4\eta + \kappa$$

$$p = \frac{(n-1)(n-2)}{2} - \delta - \kappa - 2\eta .$$

For a general plane curve of degree n we have

$$p = \frac{(n-1)(n-2)}{2} - \delta$$

where δ will be called the <u>δ-invariant</u> (sometimes called <u>the number of nodes suitable counted</u>).

Ordinary k-fold singularities add $\frac{k(k-1)}{2}$ to the δ-invariant.

Simple cusps add 1 to the δ-invariant.

Simple tacnodes add 2 to the δ-invariant.

<u>1.4 Divisors on Riemann surfaces</u>. A <u>divisor</u> D on a Riemann surface X is a singular zero-chain, written $D = \sum n_i z_i$ for $n_i \in Z$ and $z_i \in X$. The <u>degree</u> of D, deg D, is $\sum n_i$, an integer. If $n_i \geq 0$ for all i then D is said to be <u>positive</u> (or <u>integral</u>) and we write $D \geq 0$. The zero divisor is positive.

If $D = \sum n_i z_i$ and $D' = \sum n'_i z_i$ are two divisors then by the <u>greatest common divisor</u> of D and D', written (D,D'), we mean the divisor $\sum \min(n_i, n'_i) z_i$. Thus to write $(E,z) = 0$ is to say that the coefficient of z in E is zero. If $(E,F) = E$ we will write $E \subseteq F$.

If f is a meromorphic function on X, the divisor of f is written (f) where (f) = (zeros of f) - (poles of f)

$$= (f)_0 - (f)_p .$$

Note that $\deg(f) = 0$.

Two divisors D,E are said to be (linearly) <u>equivalent</u> if there exist an $f \in M(X)$ and $(f) = D - E$ (an equivalence relation).

<u>Notation:</u> $D \equiv E$.

<u>Definition</u>. If D is a divisor, $L(D) = \left\{ f \in M(X) \,\middle|\, (f) + D \geq 0 \right\}$.

Theorem. If X is compact then $L(D)$ is a finite dimensional vector space over $\mathbb{C}$.

Definition. If D a divisor on X, $\ell(D) = \dim_{\mathbb{C}} L(D)$, the affine dimension of $L(D)$. $r(D) = \ell(D) - 1$, the projective dimension of $L(D)$.

A meromorphic (or abelian) differential ω is a one-form that locally can be written

$$\omega = f(z)dz$$

where f is meromorphic. The value of a meromorphic differential at a point of X is not well defined; however, zeros and poles are well defined so we can consider the divisor of ω,

$$(\omega) = (\text{zeros of } \omega) - (\text{poles of } \omega).$$

If ω_1 and ω_2 are two meromorphic differentials then ω_1/ω_2 is a well defined meromorphic function on X. Since $\deg(\omega_1/\omega_2) = 0$ we see that $\deg(\omega_1) = \deg(\omega_2)$.

Theorem. If ω is a meromorphic differential on X_p then

$$\deg(\omega) = 2p - 2.$$

Definition. If D is a divisor

$$\Omega(D) = \left\{\text{meromorphic differentials } \omega \,\middle|\, (\omega) - D \geq 0\right\}$$

$i(D) =$ dimension of $\Omega(D)$ as a vector space over $\mathbb{C}$.

Lemma. If K is the divisor of a meromorphic differential then $i(D) = \ell(K - D)$.

Now we list some of the basic theorems concerning compact Riemann surfaces.

Riemann-Roch theorem. If D is a divisor on a compact Riemann surface of genus p, then

$$r(D) = \deg D - p + i(D).$$

Brill-Noether formulation of the Riemann-Roch theorem. If D and D' are two divisors so that $D + D' \equiv (\omega)$ where ω is a meromorphic differential, then $\deg D - 2r(D) = \deg D' - 2r(D')$.

Notice that the divisors in the Brill-Noether theorem need not be positive.

Definition. A meromorphic differential without poles is called holomorphic (or analytic, or regular, or an abelian differential of the first kind).

Definition. If $D \geq 0$ and $i(D) > 0$ then D is said to be special and $i(D)$ is often called the index (of speciality) of D.

Clifford's theorem. If D is special then $\deg D - 2r(D) \geq 0$.

Definition. If D is a divisor, the Clifford index of D, denoted $c(D)$, is:

$$c(D) = \deg D - 2r(D) = p - r(D) - i(D) .$$

Strong form of Clifford's theorem. If X_p admits a special divisor D where $\deg D \neq 0$, $\deg D \neq 2p-2$ and $c(D) = 0$, then X_p is hyperelliptic.

Definition. $\Omega(X_p)$ = {holomorphic differentials on X_p}.

Corollary. $\dim_{\mathbb{C}} \Omega(X_p) = p$. If $\omega \in \Omega(X_p)$ then $i\big((\omega)\big) = 1$.

1.5 Linear series. Let D be a divisor on X_p. Then the complete linear series determinant by D, denoted $|D|$, is

$$|D| = \left\{ D' \geq 0 \,\middle|\, D' \equiv D \right\} .$$

$|D|$ can be empty. If $D \geq 0$ then $|D|$ is a set of divisors parametrized by the projectification of $L(D)$.

$$|D| = \left\{ D' \,\middle|\, D' - D = (f) , f \in L(D) \right\} .$$

Let S be a linear subspace of $L(D)$ of dimension $r+1$, $r \leq r(D)$. The set of divisors parametried by S

$$\left\{ D' \,\middle|\, D' - D = (f) , f \in S \right\}$$

is called a linear series and denoted g^r_n where $n = \deg$ and $r + 1 = \dim S$.

If $r < r(D)$, g^r_n is said to be incomplete.

Example: K = divisors of holomorphic differentials = g^{p-1}_{2p-2}. K is the unique linear series of dimension $p-1$ and degree $2p-2$.

A linear series g^r_n can have base (or fixed) points, that is, a divisor common to all divisors in g^r_n. If F of degree f is such a divisor then $g^r_n - F = g^r_{n-f}$ stands for the linear series without base points obtained by subtracting F from each divisor in g^r_n.

Let g^r_n be a linear series without base points. Then we can map X into $\mathbb{P}^r$ as follows. Suppose

$$g^r_n = \left\{ D' \,\middle|\, D' - D = (f) , f \in S \right\} , S \subset L(D) .$$

Let $f_0, \cdots, f_r$ be a basis for S. If $x \in X$ let

$$\Theta(x) = \big(f_0(x), \cdots, f_r(x)\big) \in \mathbb{P}^r(\mathbb{C}) .$$

If we pick a local parameter z at x_0, $z(x_0) = 0$ then

$$f_i = z^{n_i} g_i(z) \text{ where } g_i(0) \neq 0 .$$

Locally z is mapped by Θ into $\left(\cdots, z^{n_i} g_i(z), \cdots\right) \in \mathbb{P}^r$.

Let $n = \min\{n_i\}$. Then $z \overset{\Theta}{\mapsto} \left(\cdots, z^{n_i - n} g_i(z), \cdots\right)$ and so

$x \overset{\Theta}{\mapsto} \left(\cdots, z^{n_i - n} g_i(z)\Big|_{z=0}, \cdots\right)$ and at least one component is non-zero. That is, Θ is well defined on all of X .

Let $C = \Theta(X) \subset \mathbb{P}^r$ be the image of X under Θ . Then the hyperplane sections of C pull back via Θ to the divisors of g^r_n . For if $\sum_{i=0}^{r} a_i y_i = 0$ is a hyperplane H in $\mathbb{P}^r$ then

$\Theta(x) \in H \cap C \Leftrightarrow \sum a_i f_i(x) = 0 \Leftrightarrow x$ is a zero of $\sum a_i f_i$. If $f_r \equiv 1$ then D is the pull back of the hyperphane (at ∞) $y_r = 0$.

If the map Θ is one-to-one in general, then g^r_n is said to be <u>simple</u>. In this case X is (conformally equivalent to) the Riemann surface (or normalization) of C . (C can, of course, have singularities.) Each hyperplane in $\mathbb{P}^r$ cuts C in n points (counting multiplicities) so we write C_n for C , a curve of degree n in $\mathbb{P}^r$.

If the map Θ is not one-to-one in general then g^r_n is said to be <u>composite</u> . Then X is a t-sheeted covering ($t \geq 2$) of (the normalization of) C and C has degree n/t , since g^r_n has no base points. (Thus a linear series of dimension $r \geq 2$ of prime degree greater than r without base points is simple.) In this case, if Y_q is the normalization of C then Y_q admits a $g^r_{n/t}$. If $\pi : X_p \to Y_q$ is the t-sheeted covering then $g^r_{n/t}$ on Y_q lifts to g^r_n on X_p . The fibers of this map π are called an <u>involution of genus q</u>

and denoted γ^1_n. In this case we say that g^r_n is compounded of the involution γ^1_t. (In the sequel, we shall be careful not to confuse the two meanings of the word "involution".)

Given g^r_n and $\Theta : X \to \mathbb{P}^r$, Θ depends on the choice of bases $\{f_0, \cdots, f_r\}$ in $S \subset L(D)$. A second choice of basis gives another curve in $\mathbb{P}^r$, but the two curves are related by a projective transformation whose matrix is that which relates the two bases of S.

If g^2_n is simple, then $\Theta(X) \subset \mathbb{P}^2$ is called a plane model for X.

For a g^1_n without base points $\Theta : X \to \mathbb{P}^1$ corresponds to an n-sheeted covering of $\mathbb{P}^1$. If X_p $(p \geq 2)$ admits a g^1_2 then X_p is hyperelliptic, and conversely.

If X_p admits a g^r_r then $p = 0$. If X admits a complete g^r_{r+1}, $r \geq 2$, then $p = 1$.

The canonical curve is the image of X_p by Θ in $\mathbb{P}^{p-1}$ given by the canonical series g^{p-1}_{2p-2}. g^{p-1}_{2p-2} is base point free. It is composite if and only if X_p is hyperelliptic in which case it is compounded of the g^1_2.

For a non-hyperelliptic X_3, the canonical curve is a non-singular plane quartic, C_4 where the lines in $\mathbb{P}^2$ cut out the canonical series.

If g^r_n is a complete linear series and $D \in g^r_n$ then $r = n - p + i(D)$. Since $i(D)$ does not depend on the divisor chosen, we will occasionally write $i(g^r_n)$ in place of $i(D)$.

If g^r_n is a complete linear series and D is an integral divisor, by $|g^r_n - D|$ we mean the divisors E of g^r_n containing D, minus D.

$$\left|g^r_n - D\right| = \left\{E - D \;\middle|\; E \in g^r_n, D \subseteq E\right\}.$$

If the degree of D is m then $(g^r_n - D) = g^{r-t}_{n-m}$ and we say

D imposes t (independent, linear) conditions on g^r_n. If $|D| = g^s_m$ then we would write

$$g^r_n = g^s_m + g^{r-t}_{n-m}$$

since any divisor in g^s_m will impose t conditions on g^r_n. (Of course, we are assuming some divisor in g^r_n contains D.)

As a final remark, when considering linear series one should always keep in mind the three possibilities: simple or composite; complete or incomplete; base points or no base points.

1.6 Line bundles. We give now a very brief discussion of line bundles. We refer to [13] and [14] for a more complete discussion.

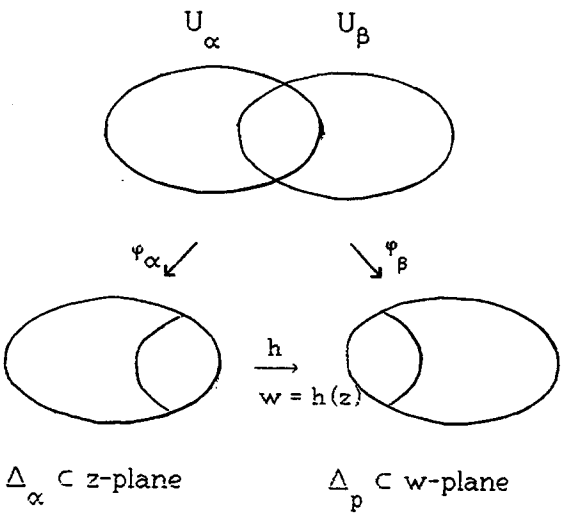
Let $\{(U_\alpha, \varphi) \mid \alpha \in A\}$ be a set of charts where $X_p = \bigcup U_\alpha$ and $\varphi_\alpha : U_\alpha \to \Delta_\alpha$ where Δ_α is a simply connected open set (disk) in $\mathbb{C}$. A (holomorphic) line bundle, η, is a collection of functions called transition functions $\{g_{\alpha\beta} \mid \alpha, \beta \in A, U_\alpha \cap U_\beta \neq \emptyset\}$ so that

(1) $g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathbb{C} - \{0\}$ is holomorphic

(2) $g_{\alpha\alpha} \equiv 1$

(3) $g_{\alpha\beta} g_{\beta\gamma} g_{\gamma\alpha} \equiv 1$ or $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$.

Example: The canonical line bundle. If for α and β we have the following diagram



then $g_{\alpha\beta} = \frac{dh}{dz} \circ \varphi_\alpha$. The canonical line bundle will also be denoted by K.

A holomorphic (meromorphic) section of a line bundle η is a set of holomorphic (meromorphic) functions $\{s_\alpha \mid \alpha \in A\}$ $s_\alpha : U_\alpha \to \mathbb{C}$ (or $\mathbb{P}^1$) so that

$$s_\alpha = g_{\alpha\beta} s_\beta \text{ in } U_\alpha \cap U_\beta .$$

The set of sections of a lie bundle form a vector space over $\mathbb{C}$. For example, the sections of the canonical bundle are differentials.

Theorem. On a compact Riemann surface the vector space of holomorphic sections of a holomorphic line bundle is finite dimensional.

Two line bundles $\eta = \{g_{\alpha\beta}\}$ and $\nu = \{h_{\alpha\beta}\}$ are said to be equivalent if there exists a set of non-zero holomorphic functions $\{t_\alpha \mid \alpha \in A\}$ and

$$t_\alpha g_{\alpha\beta} = h_{\alpha\beta} t_\beta .$$

Thus if $\{s_\alpha\}$ is a section of η then $\{t_\alpha s_\alpha\}$ is a section of ν.

Thus the sections of equivalent line bundles can appear to be quite different. What is invariant?

(a) If $\{s_\alpha\}$ is a section of a line bundle then the divisor of $\{s_\alpha\}$ makes sense since the transition functions are non-zero holomorphic functions. Let $(\{s_\alpha\})$ denote the divisor of the section $\{s_\alpha\}$. Then the set of divisors of holomorphic sections of a line bundle is independent of the equivalence class of the line bundle.

(b) If $\{s_\alpha\}$ and $\{s'_\alpha\}$ are two holomorphic sections of a line bundle η then $\{s_\alpha / s'_\alpha\}$ defines a global meromorphic function. Therefore $(\{s_\alpha\}) \equiv (\{s'_\alpha\})$. So the divisors of holomorphic sections of a line bundle form a linear series which is easily seen to be complete.

(c) If $s_0, \cdots, s_r$ are $r+1$ linearly independent sections of η then we can map $X \to \mathbb{P}^r$ by

$$x \to (s_0(x), \cdots, s_r(x)) .$$

(The curve in $\mathbb{P}^r$ is determined up to a projective transformation of $\mathbb{P}^r$ corresponding to different choices of bases in $\langle s_0, \cdots, s_r \rangle$, the span of those sections.) The curve is independent of the equivalence class of η.

(d) If D is a divisor on X we form a line bundle $[D]$ as follows. In

U_α let f_α be a meromorphic function whose divisor is $D \cap U_\alpha$. Let $g_{\alpha\beta} = f_\alpha / f_\beta$ in $U_\alpha \cap U_\beta$, a non-zero holomorphic function. (If $D \cap U_\alpha = 0$ let $f_\alpha = 1$.) Then clearly $\{f_\alpha\}$ is a meromorphic section of $\{g_{\alpha\beta}\}$ $\left(= [D]\right)$, whose divisor is D. Any holomorphic section of $[D]$ will have a positive divisor equivalent to D. Thus vector spaces of holomorphic sections of $[D]$ will parametrize linear series of divisors equivalent to D.

(e) There is a one-to-one correspondence between complete linear series of projective dimension greater than or equal to zero and holomorphic line bundles admitting holomorphic sections, given by $|D| \longleftrightarrow [D]$.

If $\eta = \{g_{\alpha\beta}\}$ and $\nu = \{h_{\alpha\beta}\}$ are line bundles then $\{g_{\alpha\beta} h_{\alpha\beta}\}$ and $\{g^{-1}_{\alpha\beta}\}$ are also line bundles, the former denoted $\eta\nu$ (or $\eta \otimes \nu$) called the (tensor) <u>product</u> of η and ν, and the latter denoted η^{-1} called the <u>inverse</u> of η. If K is the canonical line bundle then $K \otimes K \otimes \cdots \otimes K$ (k times) or K^k, will be called the <u>k-canonical line bundle</u> (or written additively, kK) whose sections are called k-canonical differentials. With respect to a local parameter z a k-canonical differential is written $f(z)(dz)^k$ and the transition functions for kK are $\left(\frac{dh}{dz}\right)^k \circ \varphi_\alpha$, referring to the diagram earlier in this section.

Chapter 2. Some exceptional points on Riemann surfaces.

2.1. On a Riemann surface, X_p, the points where a holomorphic differential has a zero of order greater than $p-1$ are called Weierstrass points. There are only a finite number of such points, at most $(p-1)\,p\,(p+1)$. If there are fewer then to each Weierstrass point can be associated a weight and the sum of all the weights is again $(p-1)\,p\,(p+1)$. The maximum weight at any point is $p(p-1)/2$, and if the maximum weight is attained, then the Riemann surface is hyperelliptic. In this case the Weierstrass point is a fixed point for the hyperelliptic involution, that is, a branch point for the unique g^1_2 on X_p. The purpose of this chapter is to extend to more general linear series these results of Hurwitz [15].

2.2. Let g^r_n be a linear series without base points but not necessarily simple or complete. Let $\langle s_0, s_1, \cdots, s_r \rangle$ be a basis for the sections of the line bundle L determined by g^r_n which generates the divisors in g^r_n. Fix x_0 in X. In a local parameter z at x_0, $z(x_0) = 0$, consider the Wronskian $W(s_0, \cdots, s_r) =$

$$\begin{vmatrix} s_0(z) & s_1(z) & \cdots & s_r(z) \\ s'_0(s) & s'_1(z) & \cdots & s'_r(z) \\ \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdots & \cdot \\ s^{(r)}_0(z) & s^{(r)}_1(z) & \cdots & s^{(r)}_r(z) \end{vmatrix}$$

where $s^{(k)}(z) = \frac{d^k s}{dz^k}(z)$, the kth derivative of s with respect to the local parameter z. If we choose a different basis $\langle t_0, \cdots, t_r \rangle$ the two bases are related by a constant matrix whose determinant is A and

$$W(s_0, \cdots, s_r) = A\, W(t_0, \cdots, t_r).$$

Thus the order of vanishing of $W(s_0, \cdots, s_r)$ at x_0 is independent of the basis chosen for g^r_n.

2.3. We now choose a basis for the sections adapted to x_0 as follows. Using the local parameter z, $z(x_0) = 0$, we will find integers

$$0 = n_0 < n_1 < n_2 < \cdots < n_r$$

and a basis of sections $\langle t_0, \cdots, t_r \rangle$ so that t_j has order of vanishing at x_0 equal to n_j. $\langle t_0, \cdots, t_r \rangle$ will be a basis adapted to x_0 and the order of vanishing of $W(t_0, \cdots, t_r)$ at x_0 will be

$\sum_{j=0}^{r} (n_j - j)$. This last sum will be seen to be independent of the various choices made.

Let $n_0 = 0$, and let t_0 be a section not vanishing at x_0 (since g^r_n is without base points). Let $E_0 = (t_0)$ a divisor disjoint from x_0.

Let $S_1 = \left\{\text{sections } s \,\middle|\, s(0) = 0\right\}$. Let n_1 be the smallest order of vanishing at 0 of sections in S_1. $n_1 \geq 1$. Choose a section t_1 which vanishes precisely to order n_1 at 0.

Then $\quad (t_1) = n_1 x_0 + E_1, (x_0, E_1) = 0$

and $\quad E_1 \in (g^r_n - n_1 x_0) \ (= g^{r-1}_{n-n_1})$.

Let $S_2 = \left\{\text{sections } s \,\middle|\, s^{(n_1)}(0) = 0\right\}$. Let n_2 be the smallest order of vanishing at 0 for $s \in S_2$. Choose $t_2 \in S_2$ which vanishes to order n_2 at $z = 0$. $n_2 > n_1$.

Then $\quad (t_2) = n_2 x_0 + E_2, (x_0, E_2) = 0$

and $\quad E_2 \in (g^r_n - n_2 x_0) \ (= g^{r-2}_{n-n_2})$.

Continue. We obtain sections $\langle t_0, \cdots, t_r \rangle$ and integers $0 = n_0 < n_1 < n_2 < \cdots < n_r$ where

(i) $\quad (t_j) = n_j x_0 + E_j, (x_0, E_j) = 0$

and $\quad E_j \in (g^r_n - n_j x_0) \ (= g^{r-j}_{n-n_j})$.

(ii) in the local parameter z

$$t_j(z) = z^{n_j} g_j(z), \ g_j(0) \neq 0.$$

2.4. We now show the order of vanishng of $W(t_0, \cdots, t_r)$ at x_0.

$(z = 0)$, is $\sum_{j=0}^{r} (n_j - j)$. We follow Farkas-Kra [11, p. 78].

First we note that if g is holomorphic in a neighborhood of x_0 then $W(gt_0, gt_1, \cdots, gt_r) = g^{r+1} W(t_0, t_1, \cdots, t_r)$. For a typical column of the matrix whose determinant is $W(gt_0, \cdots, gt_r)$ is

$$^t(gt_i, g't_i + gt'_i, g''t_i + 2g't'_i + gt''_i, \cdots).$$

Using elementary row operations this reduces to

$$^t(gt_i, gt'_i, gt''_i, \cdots)$$

and the result follows. It now follows that the order of vanishing of $W(s_0, \cdots, s_r)$ at x_0 does not depend on the local parameter.

Now to prove the order of vanishing of $W(t_0, \cdots, t_r)$ is $\sum (n_j - j)$ we proceed by induction on r. For $r = 0$ the result is immediate. Suppose it is true for integers less than r.

$$W\left(t_0, t_1, \cdots, t_r\right) = t_0^{r+1} \; W\left(1, \frac{t_1}{t_0}, \cdots, \frac{t_r}{t_0}\right)$$

$$= t_0^{r+1} \; W\left(\left(\frac{t_1}{t_0}\right)', \cdots, \left(\frac{t_r}{t_0}\right)'\right).$$

Now $\frac{t_j}{t_0} = z^{n_j} h_j(z)$ where $h_j(0) \neq 0$

$$\left(\frac{t_j}{t_0}\right)' = z^{n_j} h'_j + n_j z^{n_j - 1} h_j$$

$$= z^{n_1 - 1} (z^{n_j - n_1})(z h'_j + n_j h_j).$$

Therefore $W(t_0, \cdots, t_r) = (z^{n_1 - 1})^r \, W(f_1, \cdots, f_r)$ where

$f_j = z^{n_j - n_1} g_j$ and $g_j(0) \neq 0$. By induction $W(f_1, \cdots, f_r)$ vanishes to order

$$\sum_{j=1}^{r} (n_j - n_1) - (j-1) .$$

Thus $W(t_0,\cdots,t_r)$ vanishes to order

$$r(n_1 - 1) + \sum_{j=1}^{r} (n_j - n_1 - j + 1)$$

$$= \sum_{j=0}^{r} n_j - j \quad (n_0 = 0) .$$

2.5. We now show that $W(s_0,\cdots,s_r)$ is a section of the line bundle $L^{r+1}K^{r(r+1)/2}$ where L is the line bundle corresponding to the completion of g^r_n and K is the canonical line bundle.

In $U_z \cap V_w$, the intersection of two parametric neighborhoods of x_0 with local parameters z and w let g be the transition function for L. If s_i is a section, let $f_i(z)$ and $h_i(w)$ be the representatives of s_i in U_z and V_w respectively where $w = w(z)$ is biholomorphic. Then

$$f_i(z) = g(z)h_i\big(w(z)\big)$$

$$W(f_0,\cdots,f_r) = W(gh_0,\cdots,gh_r) = g^{r+1} W\Big(h_0\big(w(z)\big),\cdots,h_r\big(w(z)\big)\Big) .$$

Now

$$\frac{d}{dz} h_i\big(w(z)\big) = \frac{dh_i}{dw} \cdot w'(z)$$

$$\frac{d^2}{dz^2} h_i\big(w(z)\big) = \frac{d^2h_i}{dw^2} \cdot (w')^2 + \frac{dh_i}{dw} \cdot w''$$

$$\frac{d^3}{dz^3} h_i\big(w(z)\big) = \frac{d^3h_i}{dw^3} \cdot (w')^3 + 3\frac{d^2h_i}{dw^2} \cdot w'w'' + \frac{dh_i}{dw} \cdot w''' .$$

Again by elementary row operations we see that the ith column in the determinant becomes

$${}^t\Big(h_i, \frac{dh_i}{dw} \cdot w', \frac{d^2h_i}{dw^2} \cdot (w')^2, \frac{d^3h_i}{dw^3}, (w')^3, \cdots\Big) .$$

Thus we see that

$$W(f_0,\cdots,f_r) = g^{r+1}(w')^{r(r+1)/2} W(h_0,\cdots,h_r) .$$

Since $g^{r+1}(w')^{r(r+1)/2}$ is the transition function for $L^{r+1}K^{r(r+1)/2}$ the proof is complete.

It follows that the degree of $W(s_0,\cdots,s_r)$ is

$$(r+1)n + \big(r(r+1)/2\big)(2p-2)$$
$$= (r+1)\big(n + r(p-1)\big) \qquad [9, \text{p. } 132]\,.$$

2.6. We now make some definitions and summarize the results of the last several sections in a theorem.

Let g^r_n be a linear series (without base points) which is a subseries of the complete linear series corresponding to a line bundle L. A point $x \in X_p$ where a divisor of g^r_n has order greater than r will be called a generalized Weierstrass point for g^r_n The Weierstrass weight of g^r_n at x, denoted $w(g^r_n, x)$, is computed as in Section 2.4.

Theorem 2.6. The total sum $\sum_{x \in X_p} w(g^r_n, x)$ is equal to $(r+1)\big(n + r(p-1)\big)$. The divisor of generalized Weierstrass points for g^r_n (counting multiplicities) is a divisor corresponding to the line bundle $L^{r+1}K^{r(r+1)/2}$.

If g^r_n is complete and non-special this total is $(r+1)^2 p$. In particular, if g^r_n is the k-canonical series, which we denote by kK $(k \geq 2)$, then this total is $(2k-1)^2(p-1)^2 p$. A generalized Weierstrass point for the k-canonical series will be called a higher order Weierstrass point. It is known that the higher order Weierstrass points are dense in X_p [22].

If $g^r_n = |G|$, a complete linear series, we will often write $w(G,x)$ for $w(g^r_n, x)$.

For a linear series g^r_n and $x \in X$, the integers $n_0, n_1, \cdots, n_r$ will be called the non-gaps for g^r_n at x. The complimentary integers in $[0,n]$ will be called the gaps for g^r_n at x. (For the canonical series

note that this terminology differs from the usual terminology.) Then

$$w(g^r_n, x) = \sum_{j=0}^{r} (n_j - j) = \sum_{j=1}^{n-r} (r + j - g_j)$$

where $g_1, \cdots, g_{n-r}$ are the gaps for g^r_n at x. If we let $\hat{n}_i$ denote a non-gap greater than r (an extraordinary non-gap) and let $\hat{g}_i$ denote a gap less than $r + 1$ (an extraordinary gap) then

$$w(g^r_n, x) = \sum \hat{n}_i - \hat{g}_i$$

and the number of extraordinary gaps equals the number of extraordinary non-gaps.

2.7 Examples. The generalized Weierstrass points for g^1_n are the branch points for the cover $X_p \to \mathbb{P}^1$ given by g^1_n. The number is $2(n - p + 1)$ which agrees with the Riemann-Hurwitz formula.

For a simple g^2_n the inflection points (flexes) of the corresponding plane model, C_n, are among the generalized Weierstrass points. However, one sees that cusps also contribute to the total, in particular, a simple cusp contributes 2 to the total. If the plane curve is non-singular then the total weight is $3\big(n + 2(p - 1)\big) = 3n(n - 2)$ since $p = (n - 1)(n - 2)/2$, and this counts the number of inflection points.

For $p = 3$ and X_3 non-hyperelliptic, the C_4 corresponding to the canonical g^2_4 has 24 flexes (counting multiplicities).

2.8 Theorem. Let g^r_n be a complete linear series without base points. The maximum weight at a point of X_p is $(r = n - p + i)$

$$\frac{(n-r)(n-r+1)}{2} = \frac{(p - i)(p - i + 1)}{2}$$

The maximum is attained only on hyperelliptic Riemann surfaces and only at hyperelliptic Weierstrass points.

Remarks. For $g^r_n = g^{p-1}_{2p-2}$ this gives Hurwitz's result. The method of proof of the theorem is that of Segre [24], [9, p. 292].

Proof. For $x_0 \in X_p$ we find a basis of sections for a line bundle corresponding to g^r_n adapted to x_0 as in Section 2.3. We obtain

$$0 = n_0 < n_1 \cdots < n_r$$

so that we have divisors in g^r_n :

$$n_j x_0 + E_j$$

where $(x_0, E_j) = 0$, $|E_j| = g^{r-j}_{n-n_j}$, $j = 0, 1, \cdots, r$.

Let s be the smallest integer so that $n_s > s$. Then $s \geq 1$ and for $j \geq s$, $g^{r-j}_{n-n_j}$ is special since

$$(n - n_j) - (r - j) = n - r - (n_j - j) < n - r = p - i \leq p .$$

By Clifford's theorem for $j \geq s$

$$n - n_j \geq 2r - 2j$$

or

$$n - 2r + j \geq n_j - j .$$

Now $n_s \geq s + 1$ so $n - 2r + s \geq n_s - s \geq 1$

and $s \geq 2r - n + 1$.

Computing: $w(g^r_n, x_0) = \sum_{j=s}^{r} (n_j - j)$

$$\leq \sum_{j=2r-n+1}^{r} (n - 2r + j)$$

$$= \sum_{k=1}^{n-r} k = \frac{(n-r)(n-r+1)}{2} .$$

If we have equality in the last formula we have equality in all the preceeding inequalities, and so X_p hyperelliptic by the strong form of Clifford's theorem. In particular if we let $j = r - 1$

then $n - n_{r-1} = 2r - 2r + 2 = 2$

and $n - n_r = 0$.

Consequently $|E_{r-1}| = g^1_2$ and $g^r_n = |n x_0|$.

It follows that $2x_0 \in g^1_2$ and so x_0 is a hyperelliptic Weierstrass point.

q.e.d.

Chapter 3. The inequality of Castelnuovo-Severi.

3.1. As motivation for this inequality we consider for hyperelliptic Riemann surfaces some properties which are part of the folklore of the subject. These properties will be generalized by the Castelnuovo-Severi inequality and also by some later results on automorphisms.

3.2 Lemma. Suppose X_p is hyperelliptic and $f \in M(X_p)$. Let $\pi : X_p \to \mathbb{P}^1$ be a 2-sheeted cover. (Then π is a meromorphic function of order 2.) Suppose $o(f) \leq p$. Then there exists a rational function $R(\mathbb{X}) \in \mathbb{C}(\mathbb{X})$ and $f = R \circ \pi$ and so $o(f)$ is even.

Proof. We may assume that no poles of f lie on the branch points of π. For if T is a fractional linear transformation then $T \circ f$ will have the same properties as f except that by choosing T properly the poles of $T \circ f$ can be moved away from any finite set on X_p. If the conclusion of the lemma holds for $T \circ f$ it will hold for f.

Let B be the image of the $2p + 2$ branch points of π. For $z \in \mathbb{P}^1 - B$, let $\pi^{-1}(z) = \{x_1, x_2\}$ and let $g(z) = \left(f(x_1) - f(x_2)\right)^2$. Then g is meromorphic on $\mathbb{P}^1 - B$ and bounded near B and so can be extended to be meromorphic on all of $\mathbb{P}^1$. g, so extended, has at most $2\,o(f)$ poles, and at least $2p + 2$ zeros at the points of B. If $g(z)$ is not identically zero then

$$2\,o(f) \geq \#\text{ of poles of } g = \#\text{ of zeros of } g \geq 2p + 2.$$

But $o(f) \leq p$, so g is identically zero. That is, f identifies the fibers of π, and so f is the lift of a function R from $\mathbb{P}^1$. $f = R \circ \pi$ where R is rational.

q.e.d.

3.3 Lemma. For $p \geq 2$, the g^1_2 on a hyperelliptic X_p is unique.

Proof. A g^1_2 is the set of fibers of a meromorphic function of order 2. If f has order 2 then $f = R \circ \pi$ where π is the original function of order 2. Thus R must have order one in $M(\mathbb{P}')$; that is, R is a fractional linear transformation. Consequently π and f have the same set of fibers as maps from $X_p \to \mathbb{P}^1$.

q.e.d.

3.4. Here is another result that asks to be generalized but which will not be considered further since the methods considered in these notes do not seem entirely appropriate.

Lemma 3.4. Suppose X_p is hyperelliptic and $\pi : X_p \to Y_q$ is a map of

compact Riemann surfaces. If $q \geq 2$ then Y_q is hyperelliptic.

Proof. Let f be a function of order 2 on X_p. If the map $X_p \to Y_q$ is n-sheeted, for almost all y in Y_q let $\pi^{-1}(y) = \{x_1, \cdots, x_n\}$. Define

$$g(y) = f(x_1) f(x_2) \cdots f(x_n)$$

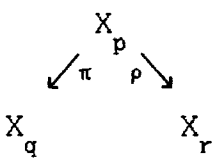
which can be extended to be a well defined meromorphic function on Y. If zeros and poles of f happen to lie in the same fiber of the map π, replace f by $f - \lambda_0$ for some $\lambda_0 \in \mathbb{C}$. Then $g(y)$ will be a function of order 2 on Y_q.

q.e.d.

3.5 Statement of the inequality of Catelnuovo-Serveri.

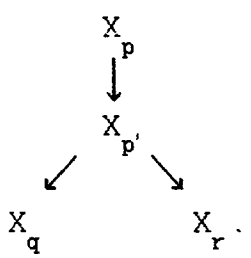
Theorem 3.5. [8].

(i) Suppose we have three compact Riemann surfaces related in the following manner



where the holomorphic maps π and ρ have m and n sheeets respectively.

(ii) Suppose further that the diagram admits no proper factorization; that is, there does not exist an $X_{p'}$ with $p' < p$ so that we have the following diagram



(iii) Then $p \leq mq + nr + (m-1)(n-1)$.

(iv) On X_q we have divisors of degree n

$$\left\{\pi \circ \rho^{-1}(y) \,\middle|\, y \in X_r\right\}$$

called an algebraic series of degree n and index m, again

denoted γ^1_n . (If $m = 1$, γ^1_n is an involution of genus r .)

In general m divisors of γ^1_n contain the same point $x_0 \in X_q$.

(v) If we have equality in (iii) above, then all the divisors in γ^1_n are linearly equivalent (but do not necessarily form a linear series.)

3.6 Applications. Many applications are based on the fact that if the inequality in (iii) does not hold then (ii) is false; that is, the diagram does then admit a proper factorization.

(i) If $q = r = 0$, $m = 2$ and $n \leq p$, then the results of Lemma 3.2 follow.

(ii) A Riemann surface of genus greater than 3 admitting a meromorphic function of order 3 is called trigonal. With $q = r = 0$, $m = 3$ and $2n < p$, we see that any meromorphic function f on a trigonal Riemann surface of order less than $p/2$ is a rational function of a function of order 3 and also $3 \mid o(f)$. On a trigonal Riemann surface of genus 5 or more, the g^1_3 is unique.

(iii) If n is prime and $p > (n-1)^2$ then any g^1_n without base points is unique on X_p .

(iv) Suppose $p > 9$ and X_p admits two complete g^1_4's without base points. Then with $q = r = 0$ and $m = n = 4$ we see that the diagram admits a proper factorization and $X_{p'}$ admits two complete g^1_2's . Consequently $p' = 1$. Thus X_p is elliptic-hyperelliptic; that is, a 2-sheeted covering of a Riemann surface of genus one .

3.7 Some remarks. An equivalent way of stating hypothesis (ii) of Theorem 3.5 is the following.

$$\left[\pi^*\left(M(X_q)\right), \rho^*\left(M(X_r)\right)\right] = M(X_p)$$

where the bracket indicates the field generated by the two fields inside.

The theorem of the primitive element then assures us that

$$M(X_p) = \rho^*\left(M(X_r)\right)[g]$$

where g is some function in $\pi^*\left(M(X_q)\right)$. It follows that for almost all $y \in X_r$, $\rho^{-1}(y)$ is a set of n distinct points on which g takes n distinct values. From this it follows that almost all divisors in γ^1_n are composed of n distinct points.

3.8 Proof of Theorem 3.5.

Let g^{n-1}_{n-1+q} be a complete, non-special linear series on X_q, without base points. If $|D| = g^{n-1}_{n-1+q}$ let $f_1, \cdots, f_n$ be a basis of $L(D)$. g^{n-1}_{n-1+q} lifts to X_p to become a $g^{n-1}_{m(n-1+q)}$ (no longer necessarily complete). Let $\hat{f}_i = f_i \circ \pi$. If $y \in X_r$ let $\rho^{-1}(y) = \{x_1, \cdots, x_n\}$ and define $h(y)$ as follows.

$$h(y) = \begin{vmatrix} \hat{f}_1(x_1) & \cdots & \hat{f}_1(x_n) \\ \vdots & & \vdots \\ \hat{f}_n(x_1) & \cdots & \hat{f}_n(x_n) \end{vmatrix}^2 = \begin{vmatrix} f_1(\pi x_1) & \cdots & f_1(\pi x_n) \\ \vdots & & \vdots \\ f_n(\pi x_1) & \cdots & f_n(\pi x_n) \end{vmatrix}^2$$

We prove the inequality in 3 steps:

(i) We show g^{n-1}_{n-1+q} can be chosen so $h(y)$ is not identically zero.

(ii) We then show that the number of poles of $h(y)$ is less than or equal to $2m(n - 1 + q)$.

(iii) We show the numbers of zeros of $h(y)$ is greater than or equal to $\text{ram}\,\rho = 2p - 2 - n(2r - 2)$.

Since the number of poles of h equals the number of zeros we have

$$2m(n - 1 + q) \geq 2p - 2 - n(2r - 2)$$

which is the inequality of Castelnuovo-Severi.

(i) Suppose that $h(y)$ is identically zero $\big(h(y) \equiv 0\big)$. Then every divisor in g^1_n must lie in a divisor of g^{n-1}_{n-1+q}. For if $h(y) = 0$, $\rho^{-1}(y) = \{x_1, \cdots, x_n\}$, then there exist $\lambda_1, \cdots, \lambda_n \in \mathbb{C}$, not all zero, and $\sum_{j=1}^{n} \lambda_j f_j(\pi x_i) = 0$ for all i. Thus $\{\pi x_1, \cdots, \pi x_n\}$ is part of the divisor

$$D_\lambda = \left\{ z \in X_q \;\middle|\; \sum \lambda_j f_j(z) = 0 \right\} .$$

Now fix $y_0 \in X_r$ so that $\pi \circ \rho^{-1}(y_0)$ $\left(= \{\pi x_1, \cdots, \pi x_n\}\right)$ is a set of n distinct points, and no x_i is a branch point of ρ. Let D_q be a non-special integral divisor on X_q of degree q, and $\pi x_n \notin D_q$; $|D_q| = g^0_q$. Let $|D_q + \pi x_1 + \cdots + \pi x_{n-1}| = g^r_{n-1+q}$. Since this linear series is non-special $r = n - 1$. Moreover, $g^r_{n-1+q} - \pi x_1 - \cdots - \pi x_{n-1} = D_q$. Thus no divisor of g^r_{n-1+q} contains $\pi x_1 + \cdots + \pi x_{n-1} + \pi x_n$.

(ii) and (iii) We will combine these proofs in a lemma.

Lemma 3.8.1. Let $\rho : X_p \to X_r$ be an n-sheeted covering of compact Riemann surfaces $(n \geq 2)$. Let D be a divisor on X_p so that $|D| = g^R_N$ where $R \geq n - 1$. Let $f_1, \ldots, f_n$ be n independent functions in $L(D)$. For $y \in X_r$, let $\rho^{-1}(y) = \{x_1, \cdots, x_n\}$. Define $h(y)$ by

$$h(y) = \begin{vmatrix} f_1(x_1) & & f_1(x_n) \\ \cdot & & \cdot \\ \cdot & & \cdot \\ \cdot & & \cdot \\ f_n(x_1) & & f_n(x_n) \end{vmatrix}^2$$

If $h(y)$ is not identically zero then $N \geq p - 1 - n(r - 1)$.

Proof. Since $R \geq 1$ we can assume that D is a divisor of N distinct points which are disjoint from the set of branch points of ρ. Then if k points of D are in $\rho^{-1}(y)$ one sees that h has a pole of order at most $2k$ at y. Consequently h has at most $2N$ poles.

Now we show that if $x_0 \in X_p$ and $\text{ram}_{x_0} \rho = s$ then x_0 contributes at least $s - 1$ to the zeros of $h(y)$.

Choose a basis of $\langle f_1, \cdots, f_n \rangle$ adapted to x_0 (Section 2.3). Call this new basis $\langle h_1, \cdots, h_n \rangle$. In a parametric neighborhood of x_0, U, with local parameter t $\left(t(x_0) = 0\right)$ the map ρ is equivalent to $t \to t^s = z$,

where z is a local parameter at $\rho(x_0) \in X_r$ $\left(z\left(\rho(x_0)\right) = 0\right)$. Using these local parmeters we have

$$\rho^{-1}(z) \cap U = \{t, t\tau, \cdots, t\tau^{s-1}\}$$

where $\tau = e^{2\pi i/s}$ and $t^s = z$. Then

$$\begin{aligned} h_1(t) &= t^{\ell_1} k_1(t) \\ h_2(t) &= t^{\ell_2} k_2(t) \\ &\vdots \\ h_n(t) &= t^{\ell_n} k_n(t) \end{aligned}$$

where $0 \le \ell_1 < \ell_2 < \cdots < \ell_n$, $j \le \ell_j$ and $k_j(0) \neq 0$. Using these parameters we have

$$h(z) = A^2 \begin{vmatrix} h_1(t) & h_1(\tau t) & h_1(\tau^{s-1} t) \\ h_2(t) & h_2(\tau t) & h_2(\tau^{s-1}\tau) \\ \vdots & & \\ h_s(t) & h_s(\tau t) & h_s(\tau^{s-1} t) \\ h_{s+1}(t) & h_{s+1}(\tau t) & h_{s+1}(\tau^{s-1} t) \\ \vdots & & \\ h_n(t) & h_n(\tau t) & h_n(\tau^{s-1} t) \end{vmatrix}^2 \qquad (A \neq 0)$$

Each $s \times s$ determinant from the first s rows has a zero (as a function of t) of order at least $0 + 1 + 2 + \cdots + (s-1) = s(s-1)/2$. Thus as a function of t the square of the determinant is divisible by $t^{s(s-1)}$. But $t^s = z$, so h has a zero of order at least $s - 1$ coming from the first s rows of the determinant.

Thus $h(y)$ has at least $\operatorname{ram}\rho$ zeros and $\operatorname{ram}\rho = 2p - 2 - n(2r - 2)$. Since $h(y)$ has an equal number of zeros and poles the result follows.

q.e.d.

This completes the proof of the Castelnuovo-Severi inequality.

Now suppose we have equality. Then every inequality in the proof becomes an equality. In this case the zeros of $h(y)$ arise only from the ramification points of ρ. Any g^{n-1}_{n-1+q} for which the corresponding $h(y)$ is not identically zero has no divisor containing a divisor of g^1_n for that would add another zero to $h(y)$. If g^{n-1}_{n-1+q} is chosen so that one divisor of g^1_n is in one of its divisors then <u>every</u> divisor of g^1_n lies in some divisor of g^{n-1}_{n-1+q} since $h(y) \equiv 0$.

Now pick $D_n \in g^1_n$ so that D_n has distinct points. Choose G to be a non-special divisor of distinct points of degree q.

Then $|G + D_n| = g^n_{n+q}$ and we can assume it is without base points.

Since $|G| = g^0_q$, the n points of D_n impose independent conditions on g^n_{n+q}; that is, if

$$D_n = x_1 + x_2 + \cdots + x_n \text{ then}$$

$$g^n_{n+q} - x_{i_1} - x_{i_2} - \cdots - x_{i_t} = g^{n-t}_{n+q-t}, \quad 1 \le t \le n.$$

Now g^n_{n+q} gives a map $\Theta : X_p \to \mathbb{P}^n$. The statement that D_n imposes independent conditions on g^n_{n+q} translates into the statement: $\Theta(x_1), \cdots, \Theta(x_n)$ span a linear space of dimension $n-1$ in $\mathbb{P}^n$. This is an open condition; that is, for E close to D ($\deg E = n$, $E \ge 0$) E will also impose independent conditions on g^n_{n+q}.

<u>Lemma 3.8.2</u>. (Castelnuovo.) If $E_n \in g^1_n$, then E_n imposes independent conditions on g^n_{n+q} if and only if $E_n \equiv D_n$.

<u>Proof</u>. Let $G = P_1 + \cdots + P_q$, q distinct points. Consider

$g^n_{n+q} - P_1 = |D_n + P_2 + \cdots + P_q| = g^{n-1}_{n+q-1}$, which contains D_n.

Consequently g^{n-1}_{n+q-1} contains all divisors of g^1_n. Suppose

$E_n \in \gamma^1_n$ imposes independent conditions on g^n_{n+q}.

$$g^{n-1}_{n+q-1} = |E_n + H_{q-1}|$$

Thus

$$g^n_{n+q} = |E_n + H_{q-1} + P_1|$$

$$g^n_{n+q} - E_n = |H_{q-1} + P_1| = g^0_q$$

since E_n imposes independent conditions on g^n_{n+q}. Therefore $P_1 \in g^n_{n+q} - E_n$. But there was nothing unique about P_1. Thus $P_j \in g^n_{n+q} - E_n$ for all j.

$$g^n_{n+q} - E_n = P_1 + \cdots + P_q = G$$

$$g^n_{n+q} = |D_n + G| = |E_n + G|$$

or $D_n \equiv E_n$.

Conversely, if $D_n \equiv E_n$ then $g^n_{n+q} - E_n = |G| = g^0_q$, so E_n imposes independent conditions on g^n_{n+q}.

q.e.d.

We complete the proof that equality in the Castelnuovo-Severi inequality implies all divisors in γ^1_n are linearly equivalent by a connectivity argument.

Let $X^{(n)}_q$ = {all integral divisors of degree n on X_q} and let $\tilde{\gamma}^1_n$ = {divisors in γ^1_n consisting of n distinct points}. Then $\tilde{\gamma}^1_n$ is a connected subset of $X^{(n)}_q$ when the latter is given the natural topology. Fix $D_n \in \tilde{\gamma}^1_n$. Let $S = \left\{E_n \in \tilde{\gamma}^1_n \,\middle|\, E_n \equiv D_n\right\}$. Then S is easily seen to be closed in $\tilde{\gamma}^1_n$. But the lemma of Castelnuovo shows that S is open, that is, $S = \tilde{\gamma}^1_n$. Since $\tilde{\gamma}^1_n = \gamma^1_n$ - {finite set of divisors} the proof is complete.

3.9 Castelnuovo's Riemann-Roch theorem.

Theorem 3.9. [9, p. 324], [6]. Suppose γ^1_n is an algebraic series of

index one and genus q on X_p ; that is, γ^1_n is the set of fibers of an n-sheeted map $X_p \to X_q$. Let g^R_N be a complete linear series on X_p where $R \geq n - 1$. Suppose

$$N - R < p - nq .$$

Then any divisor in γ^1_n will not impose independent conditions on g^R_N ; that is, if $D \in \gamma^1_n$ then

$$|g^R_N - D| = g^{R-n+1+\varepsilon}_{N-n}$$

where $\varepsilon \geq 0$.

Remarks. Notice that g^R_N may be composite and that we may assume g^R_N is without base points without loss of generality. Clearly g^R_N must be special. For $q = 0$ the theorem is a consequence of the Riemann-Roch theorem.

Proof. For the proof we will assume g^R_N is without base points. Also for technical convenience we will prove the theorem only for divisors $D \in \gamma^1_n$ made up of n distinct points.

Let $D_n = x_1 + \cdots + x_n$ be a divisor in γ^1_n made up of n distinct points. Suppose D_n imposes n independent conditions on g^R_N . We will arrive at a contradiction.

Let $D_{n-1} = x_1 + \cdots + x_{n-1}$. If $R \geq n$ we can find a divisor Y of $R - n + 1$ distinct points so that $D_{n-1} + Y$ imposes R conditions on g^R_N , and the unique divisor in g^R_N containing $Y + D_{n-1}$ does not contain x_n .

Now $|g^R_N - Y| = g^{n-1}_{N-R+n-1}$. Since

$$N - R + n - 1 < p - 1 - n(q - 1)$$

every divisor in γ^1_n lies in some divisor of $g^{n-1}_{N-R+n-1}$ (Lemma 3.8.1) . Let $D_{n-1} + E$ be a divisor in $g^{n-1}_{N-R+n-1}$ containing D_n . Then $x_n \in E$. But $Y + D_{n-1} + E$ is the unique divisor in g^R_N containing $Y + D_{n-1}$, and so $x_n \notin E$. This contradiction

proves the theorem for divisors in g^1_n of n distinct points.

q.e.d.

3.10 An application of Castelnuovo's Riemann-Roch Theorem.

Theorem 3.10. Let C_N be a plane model for a Riemann surface X_p where

$$p = (N-1)(N-2)/2 - \delta .$$

Suppose we have integers $m \geq 2$ and $q \geq 0$ so that

$$\delta + mq < (N-m)(N-m-1)/2 .$$

Then X_p does not admit a g^1_m of index one and genus q.

Proof. If g^2_N is the linear series cut out on C_N by the lines in $\mathbb{P}^2$, then curves of degree $m-1$ cut out a g^ρ_ν,

$$|(m-1)\, g^2_N| = g^\rho_\nu$$

where $\nu = (m-1)N$ and $\rho = (m-1)(m+2)/2 + \varepsilon$, $\varepsilon \geq 0$.

Suppose X_p admits such a g^1_m.

Now, $\nu - \rho = (m-1)N - (m-1)(m+2)/2 - \varepsilon$

$$\leq (m-1)N - (m-1)(m+2)/2 .$$

Since $p = (N-1)(N-2)/2 - \delta$, elementary algebra shows that

$$\nu - \rho < p - mq .$$

Thus a divisor of m distinct points in g^1_m, $x_1 + \cdots + x_m$, imposes at most $m-1$ conditions on g^ρ_ν. Suppose that any divisor in g^ρ_ν containing $x_1 + \cdots + x_{m-1}$ must contain x_m. We can find lines in $\mathbb{P}^2$, $\ell_i = 0$ so that $\ell_i = 0$ contains x_i but not x_m, $i = 1, 2, \cdots, m-1$. $\ell_1 \ell_2 \cdots \ell_{m-1} = 0$ cuts out a divisor in g^ρ_ν containing $x_1 + \cdots + x_{m-1}$ but not containing x_m. This contradiction shows that g^1_m does not exist.

q.e.d.

This generalizes the fact that a non-singular plane curve of degree 4 or more is not hyperelliptic. ($\delta = 0$, $m = 2$, $q = 0$.)

Chapter 4. Smooth and branched coverings of Riemann surfaces.

4.1. In the following discussion many standard facts will be stated without proof. (Reference: Ahlfors and Sario [4].)

Definition. A continuous map $f : Y \to X$ between two surfaces will be called a (smooth) covering if f is a local homeomorphism and for every $x \in X$, there is a connected open neighborhood Δ of x so that $f^{-1}(\Delta) = \bigcup_{i=1}^{n \text{ or } \infty} \Delta_i$ where the sets $\{\Delta_i\}$ are the components of $f^{-1}(\Delta)$ and $f : \Delta_i \to \Delta$ is a homeomorphism.

Definition: Suppose $f : Y \to X$ is a local homeomorphism between surfaces. Let $\gamma : [0,1] \to X$ be a continuous map (path), and suppose $\gamma(0) = x_0$. Suppose $\tilde{\gamma} : [0,1] \to Y$ is a path so that $\tilde{\gamma}(0) = y_0$, $f(y_0) = x_0$, and $f \circ \tilde{\gamma} = \gamma$. Then $\tilde{\gamma}$ will be called a lift of γ starting at y_0.

Lemma. If $\tilde{\gamma}$ and $\tilde{\gamma}'$ are lifts of γ starting at y_0 then $\tilde{\gamma} = \tilde{\gamma}'$ (uniqueness of lifts).

Lemma. If $f : Y \to X$ is a local homeomorphism between surfaces, then the following are equivalent:

(a) f is a covering;

(b) If $\gamma : [0,1] \to X$ is a path and $f(y_0) = \gamma(0)$, then there exist a lift of γ starting at y_0.

Monodromy Theorem. Let $f : Y \to X$ be a covering. Let γ_0, γ_1 be paths in X so that $\gamma_0(0) = \gamma_1(0)$, $\gamma_0(1) = \gamma_1(1)$, and γ_0 is homotopic to γ_1. Let $\tilde{\gamma}_0$ and $\tilde{\gamma}_1$ be lifts of γ_0 and γ_1 starting at the same point above $\gamma_0(0)$. Then $\tilde{\gamma}_0$ is homotopic to $\tilde{\gamma}_1$, and in particular $\tilde{\gamma}_0(1) = \tilde{\gamma}_1(1)$.

Theorem. Let $f : Y \to X$ be a covering. Then $\text{card}\{f^{-1}(x)\}$ is a constant function on X, possibly infinite. This integer (or ∞) is called the number of sheets of the covering.

4.2 Classification of smooth coverings.

Let X be a surface, $x_0 \in X$. A triple (Y, f, y_0) will be called a covering of (X, x_0) if $f : Y \to X$ is a covering of surfaces, $y_0 \in Y$, and $f(y_0) = x_0$. If (Y, f, y_0) and (Z, g, z_0) are coverings of (X, x_0), they will be said to be equivalent if there exists a homeomorphism $h : Y \to Z$ so that $h(y_0) = z_0$ and $g \circ h = f$.

Suppose (Y,f,y_0) is a covering of (X,x_0). Let $[\gamma]$ be an equivalence class of paths in $\pi_1(X,x_0)$. Let $D \subset \pi_1(X,x_0)$ be the paths $[\gamma]$ so that the lift of γ starting at y_0 is a closed path. By the monodromy theorem D is a well defined subgroup of $\pi_1(X,x_0)$. In fact if we define $f_* = \pi_1(Y,y_0) \to \pi_1(X,x_0)$ by $f_*[\gamma] = [f \circ \gamma]$ then $D = f_*\left(\pi_1(Y,y_0)\right)$.

Existence Theorem. If D is a subgroup of $\pi_1(X,x_0)$, there exists a covering (Y,f,y_0) of (X,x_0) so that $D = f_*\left(\pi_1(Y,y_0)\right)$.

Theorem. Suppose $D \subset D' \subset \pi_1(X,x_0)$ and let (Y,f,y_0) and (Y',f',y'_0) be coverings as in the existence theorem. Then there exist a covering $g : Y \to Y'$ so that (Y,g,y_0) is a covering of (Y',y'_0) and $f = f' \circ g$.

Corollary. (Y,f,y_0) and (Y',f',y'_0) are equivalent coverings of (X,x_0) if and only if $D = D'$.

Suppose for each subgroup $D \subset \pi_1(X,x_0)$ we pick one covering (Y_D,f_D,y_D) as in the existence theorem. We define a partial ordering of these covers as follows: $(Y_D,f_D,y_D) > (Y_{D'},f_{D'},y_{D'})$ if there exist a covering $g : Y_D \to Y_{D'}$ so that (Y_D,g,y_D) is a covering of $(Y_{D'},y_{D'})$ and $f_D = f_{D'} \circ g$. Then $(Y_D,f_D,y_D) > (Y_{D'},f_{D'},y_{D'})$ if and only if $D \subsetneq D'$. Thus the map $D \to (Y_D,f_D,y_D)$ is an order reversing lattice isomorphism. A covering (Y_D,f_D,y_D) corresponding to $D = \langle e \rangle \subset \pi_1(X,x_0)$ is called a (the) universal covering of X and $Y_{\langle e \rangle}$ is simply connected.

4.3. Lifting conformal structures. If $f : Y \to X$ is a covering of surfaces and X is a Riemann surface then there is a unique conformal structure on Y so that f is holomorphic. For if $x \in X$ and Δ is a small simply connected open neighborhood which is mapped by h biholomorphically onto $\{|z| < 1\}$ then on $f^{-1}(\Delta) = \bigcup \Delta_i$, $h \circ f$ restricted to each Δ_i gives the conformal structure for Y. If $\tilde{X}$ is the universal covering of X then $\tilde{X}$ is a simply connected Riemann surface.

Uniformization Theorem. Every simply connected Riemann surface is conformally equivalent to $\mathbb{P}^1$, $\mathbb{C}$, or $\mathbb{D}$ (the unit disk).

4.4. Cover transformations.

Notational convention for Sections 4.4, 4.5, 4.6, and 4.7.

We will write all functional notation with the argument on the left; e.g. $x \mapsto (x)f$. Then the composition of f followed by g will be written

$f \circ g$ or fg. This notational convention will help distinguish between isomorphisms and anti-isomorphisms which occur naturally in discussing cover transformations.

Suppose $f : Y \to X$ is a covering. A homeomorphism $g : Y \to Y$ will be called a cover transformation if $g \circ f = f$. Under composition the set of cover transformations is a group denoted $G(Y/X)$ and is called the Galois group of the covering. If X and Y are Riemann surfaces and f is holomorphic the cover transformations are biholomorphic.

Lemma 4.4. A cover transformation with a fixed point is the identity.

Proof. Suppose $g \in G(Y/X)$ and $(y_0)g = y_0$. If y_1 is any point on Y let $y_2 = (y_1)g$. Let $\tilde{\gamma}$ be a path from y_0 to y_1. Then $\tilde{\gamma} \circ g$ is a path from y_0 to y_2. But $(\tilde{\gamma})f = (\tilde{\gamma} \circ g)f = \gamma$, a path in X starting at x_0. $\tilde{\gamma}$ and $\tilde{\gamma} \circ g$ are lifts of γ starting at y_0. By the uniqueness of lifts $\tilde{\gamma} = \tilde{\gamma} \circ g$ and so $y_1 = y_2$.

q.e.d.

Theorem 4.4. Let (Y,f,y_0) be a covering of (X,x_0). Let $D = f_*\left(\pi_1(Y,y_0)\right) \subset \pi_1(X,x_0)$. Then $G(Y/X) \cong N(D)/D$ where $N(D)$ is the normalizer of D in $\pi_1(X,x_0)$.

Proof. Fix $g \in G(Y/X)$. Let $\tilde{\sigma}$ be a path in Y connecting y_0 to $(y_0)g$. Let $\sigma = \tilde{\sigma} \circ f$. Now the covering $\left(Y,f,(y_0)g\right)$ is equivalent to (Y,f,y_0). Since $g \circ f = f$ we see that f_* maps both $\pi_1(Y,y_0)$ and $\pi_1\left(Y,(y_0)g\right)$ isomorphically onto D. But

$$\pi_1\left(Y,(y_0)g\right) = \tilde{\sigma}^{-1}\pi_1(Y,y_0)\tilde{\sigma}.$$

Thus $$D = \sigma^{-1} D \sigma.$$

Now consider the map from $G(Y/X) \to N(D)/D$

$$g \to D\sigma_g$$

where $\tilde{\sigma}_g$ connects y_0 to $(y_0)g$. We must check several things about this map.

(i) The map is well defined. If $\tilde{\tau}_g$ connects y_0 to $(y_0)g$ then $\tilde{\tau}_g = \tilde{\tau}_g \tilde{\sigma}_g^{-1} \tilde{\sigma}_g$. But $\tilde{\tau}_g \tilde{\sigma}_g^{-1}$ is a closed path starting at y_0 and so maps onto D by f_*. Therefore $\tau_g \in D\sigma_g$.

(ii) The map is injective. If $D\sigma_g = D\sigma_h$ then $\sigma_h \sigma_g^{-1} \in D$.

$\sigma_h \tilde{\sigma}_g^{-1}$ is a closed path on Y. Thus $(1)\tilde{\sigma}_h = (1)\tilde{\sigma}_g$ or $(y_0)h = (y_0)g$. Consequently $h = g$.

(iii) The map is surjective. If $\sigma^{-1} D \sigma = D$ let $\tilde{\sigma}$ be the lift of σ starting at y_0. Let $y_1 = (1)\tilde{\sigma}$. Then $\pi_1(Y,y_1) = \tilde{\sigma}^{-1} \pi_1(Y,y_0) \tilde{\sigma}$. Since f_* maps both of these groups onto D we see that (Y,f,y_0) is equivalent to (Y,f,y_1). The homeomorphism $g : Y \to Y$ so that $y_1 = (y_0)g$ is a cover transformation and $\sigma = \sigma_g$.

(iv) The map is an anti-isomorphism. A path connecting y_0 to $(y_0)gh$ is seen to be $\tilde{\sigma}_h(\tilde{\sigma}_g)h$ which projects by f onto $\sigma_h \sigma_g$. Thus $\sigma_{gh} = \sigma_h \sigma_g$.

q.e.d.

Remark. If $\varphi : G \to H$ is an anti-isomorphism then $\psi : G \to H$, $(g)\psi = (g^{-1})\varphi$ is an isomorphism.

4.5. Some remarks on permutation groups.

Let $\sum$ be a set and let $\mathcal{S}(\sum)$ be the group of permutations of $\sum$. If $\sum = \{1,2,\cdots,n\} \subset \mathbb{Z}$ then $\mathcal{S}(\sum)$ will be denoted $\mathcal{S}_n$. A subgroup $H \subset \mathcal{S}(\sum)$ is said to be transitive if for all $x,y \in \sum$ there is a $h \in H$ so that $(x)h = y$. A subgroup H is said to be regular if the only element in H which fixes any element of $\sum$ is the identity. If H is regular and $h \in H$, $h \neq e$, then when we write h as a product of disjoint cycles, all cycles have the same length.

Examples. The right and left Cayley representations. If G is a group the right Cayley representation $T : G \to \mathcal{S}(G)$ is given by $g \to T_g$ where $(h)T_g = hg$. Then T is an injective homomorphism and $(G)T \subset \mathcal{S}(G)$ is a regular transitive subgroup. The left Cayley representation $\tau : G \to S(G)$ is given by $g \to \tau_g$ where $(h)\tau_g = gh$. τ_g is an injective anti-homomorphism and again $(G)\tau$ is a regular transitive subgroup of $\mathcal{S}(G)$. The two representations are the same if and only if G is abelian.

4.6. Suppose $(Y,f,y_0) \to (X,x_0)$ is a covering of Riemann surfaces. Then $G(Y/X)$ acting on the fiber of f over x_0 is a regular group of permutations since only the identity cover transformation has a fixed point. Let this fiber, $(x_0)f^{-1}$ be denoted by I.

From the proof of Theorem 4.4 we see that $G(Y/X)\,|\,I$ is transitive if and only if D is normal in $\pi_1(X,x_0)$ $\left(D = \left(\pi_1(Y,y_0)\right)f_*\right)$. In this case the covering is said to be Galois or normal. Assuming the covering is Galois, for $x \in X$, $y \in (x)f^{-1}$ we have $(x)f^{-1} = \left\{(y)g \,\middle|\, g \in G(Y/X)\right\}$. Denote this set, the $G(Y/X)$-orbit of y, by $[y]$. Let $\left\{[y] \,\middle|\, y \in Y\right\}$ be denoted by $Y/G(Y/X)$. Since each $G(Y/X)$-orbit $[y]$ can be identified with $(y)f$, $Y/G(Y/X)$ can be given the structure of a Riemann surface and $Y/G(Y/X) \cong X$.

Lemma 4.6. Suppose (Y,f,y_0) is a Galois covering of (X,x_0) and H is a subgroup of $G(Y/X)$. Then there is a factoring of the covering $f: Y \to X$, $Y \to Z \to X$, where $Y \to Z$ is a Galois covering and $G(Y/Z) = H$. H is normal in $G(Y/X)$ if and only if the covering $Z \to X$ is a Galois covering. In this case

$$G(Z/X) \cong G(Y/X) \,/\, G(Y/Z).$$

Proof: Let D be the image of $\pi_1(Y,y_0)$ under f_*. Then D is normal in $\pi_1(X,x_0)$ and $\pi_1(X,x_0)/D \cong G(Y/X)$. Let φ be the quotient map of $\pi_1(X,x_0)$ onto $G(Y/X)$. Then $D \subset (H)\varphi^{-1} \subset \pi_1(X,x_0)$. Let Z be the Riemann surface which is the covering of X corresponding to $(H)\varphi^{-1}$. Since D is normal in $(H)\varphi^{-1}$ the covering $Y \to Z$ is Galois with Galois group isomorphic to $(H)\varphi^{-1}/D$, which is isomorphic to H.

If H is normal in $G(Y/X)$ then $(H)\varphi^{-1}$ is normal in $\pi_1(X,x_0)$ and $G(Z/X) \cong \pi_1(X,x_0)/(H)\varphi^{-1} \cong G(Y/X)/H$.

q.e.d.

If $D = \langle e \rangle$ then $\tilde{X}$, the universal covering surface of X, is a Galois cover with $G(\tilde{X}/X) \cong \pi_1(X,x_0)$. If $\tilde{X}$ is isomorphic to the unit disk $\mathbb{D}$, then $G(\tilde{X}/X)$ is a group of fixed-point-free conformal self-maps of $\mathbb{D}$ (fractional linear transformations).

4.7. The monodromy group of the cover $(Y,f,y_0) \to (X,x_0)$ [25, p. 198].

Let $I = \{x_0\}f^{-1} = \{y_0, y_1, y_2, \cdots\}$. Suppose $[\gamma] \in \pi_1(X,x_0)$. (Hereafter we will write γ instead of $[\gamma]$ since all statements will depend only on the homotopy class of γ.) We obtain a permutation $(\gamma)\mu$ in $\mathcal{S}(I)$ as follows. Let $\tilde{\gamma}_i$ be the lift of γ starting at y_i. Then define $(y_i)(\gamma)\mu = (1)\,\tilde{\gamma}_i$.

(i) By the monodromy theorem μ is well defined as a function from $\pi_1(X,x_0)$ into $\mathcal{S}(I)$.

(ii) μ is seen to be a homomorphism.

(iii) Since Y is connected the image of μ is a transitive subgroup of $\mathcal{S}(I)$. It is called the monodromy group of the covering and will be denoted $M(Y/X)$.

(iv) $\left(\pi_1(Y,y_0)\right)f_* = \left\{\gamma \in \pi_1(X,x_0) \,\middle|\, (y_0)(\gamma)\mu = y_0\right\}$.

(v) $\left(\pi_1(Y,y_j)\right)f_* = \left\{\gamma \,\middle|\, (y_j)(\gamma)\mu = y_j\right\}$. Call this group D_j.

(vi) Let γ_j be a path whose lift, $\tilde{\gamma}_j$, to Y starting at y_0 ends at y_j. Then $\tilde{\gamma}_j^{-1}\pi_1(Y,y_0)\tilde{\gamma}_j = \pi_1(Y,y_j)$. Consequently $\gamma_j^{-1} D_0 \gamma_j = D_j$.

(vii) $\pi_1(X,x_0) = D_0\gamma_0 \cup D_0\gamma_1 \cup D_0\gamma_2 \cup \cdots$, a right coset decomposition of $\pi_1(X,x_0)$. $D_0\gamma_j$ is the set of paths which when lifted to start at y_0 end at y_j.

(viii) $\ker\mu = \bigcap_j D_j = \bigcap_{\sigma\in\pi_1(X,x_0)} \sigma^{-1} D_0 \sigma$. This is the largest normal subgroup of $\pi_1(X,x_0)$ lying in D_0. Now if the cover $Y \to X$ is Galois then $D_0 = \ker\mu$ and $M(Y/X)$ is a regular subgroup of $\mathcal{S}(I)$.

(ix) Thus if $Y \to X$ is Galois we have two regular transitive subgroups of $\mathcal{S}(I)$ namely $M(Y/X)$ and $G(Y/X) \,\middle|\, I$. Both groups are isomorphic to $\pi_1(X,x_0)/\left(\pi_1(Y,y_0)\right)f_*$.

(x) How are these two subgroups of $\mathcal{S}(I)$ related?
Let $G = \pi_1(X,x_0)/D_0 = \{D_0\gamma_0, D_0\gamma_1, \cdots\}$. Identify y_j with $D\gamma_j$. If μ is the monodromy map then $(y_j)(\gamma)\mu = (1)(\tilde{\gamma}_j\gamma)$. That is, $(\gamma)\mu$ acting on $D\gamma_j$ is $D\gamma_j\gamma = D\gamma_j D\gamma$. The monodromy map is thus the right Cayley representation. If $g_j \in G(Y/X)$ where $(y_0)g_j = y_j$ then $(y_i)g_j = (1)\tilde{\gamma}_j\tilde{\gamma}_i$. (See the proof of Theorem 4.4.) That is, g_j acting on $D\gamma_i$ is $D\gamma_j\gamma_i = D\gamma_j D\gamma_i$. Thus $G(Y/X) \,\middle|\, I$ is the left Cayley representation.

Theorem 4.7. Suppose $(Y,f,y_0) \to (X,x_0)$ is a Galois covering. Then the

Galois group acting on $(x_0)f^{-1}$ $(= I)$ and the monodromy group are regular transitive subgroups of $\mathcal{S}(I)$. If we identify I with the right cosets $\{D\gamma_i\}$ of D in $\pi_1(X,x_0)$ as above $\left(D = \left(\pi_1(Y,y_0)\right)f_*\right)$, then $M(Y/X)$ acting on I is the right Cayley representation and $G(Y/X)$ is the left Cayley representation.

Corollary. The two representatons are the same if and only if $G(Y/X)$ is abelian.

4.8. We now abandon the notational convention adopted at the beginning of Section 4.4.

Theorem. (Hurwitz). Let G be a finite group. Then there exists a compact Riemann surface Y which admits a group of automorphisms isomorphic to G.

Proof. Let $g_1,\cdots,g_s$ be generators for G. Let X be a Riemann surface of genus s. Let $A_1,\cdots,A_s,B_1,\cdots,B_s$ be a canonical basis for $\pi_1(X,x_0)$. Then $\Pi[A_i,B_i] = e$, and this is the only defining relation. Then $\mu : \pi_1(X,x_0) \to G$ can be defined where $\mu(A_i) = g_i$ and $\mu(B_i) = e$ and μ is a homomorphism from $\pi_1(X,x_0)$ onto G. The covering Riemann surface Y corresponding to the kernal of μ has $G(Y/X) \cong G$.

q.e.d.

4.9 Galois (or normal) closure.

Let $(Y,f,y_0) \to (X,x_0)$ be a covering of surfaces where $D = f_*\left(\pi_1(Y,y_0)\right)$. Let $N = \bigcap_{\sigma\in\pi_1(X,x_0)} \sigma^{-1}D\sigma$ which is the kernel of the monodromy map. The covering Y_N corresponding to N is called the Galois (or normal) closure for the covering $Y \to X$. Note that $G(Y_N/X) = M(Y/X)$.

4.10. Branched coverings.

The coverings of surfaces we have considered so far in this chapter will often be called smooth coverings in contradistinction to branched coverings. The latter will be continuous maps between surfaces which locally are equivalent to $z \to z^n$. An example of a branched covering is a non-constant holomorphic map between two compact Riemann surfaces.

Consider the map from $f : \mathbb{D} - \{0\} \to \mathbb{D} - \{0\}$, $f(z) = z^n$, a smooth covering of Riemann surfaces. Any finite sheeted smooth covering of

$\mathbb{D} - \{0\}$ is of this type. f can obviously be extended to a holomorphic map of $\mathbb{D} \to \mathbb{D}$ by sending $0 \to 0$. f so extended is now a branched covering.

(i) Let X be a Riemann surface. Let B be a discrete set in X. (If X is compact B is finite.) Let $X^* = X - B$ and let $f : Y^* \to X^*$ be a finite sheeted smooth covering. If $b_0 \in B$, let $\Delta \subset X$ be a parametric disk centered at b_0 so that $\Delta \cap B = \{b_0\}$. $f^{-1}\left(\Delta - \{b_0\}\right)$ is a collection of finite sheeted coverings of $\Delta - \{b_0\}$. To each component of $f^{-1}\left(\Delta - \{b_0\}\right)$ we add a point to Y^* above b_0 and extend f to these points to map onto b_0. We do this for each point $b_0 \in B$. Let $Y = Y^* \cup \{\text{points added above } B\}$ and $f^e : Y \to X$ is f extended. Then $f^e : Y \to X$ is a finite-sheeted holomorphic map branched (if at all) only above B.

Again consider $b_0 \in \Delta$, $b_0 \in B$ as above. A path $\gamma \in \pi_1(X^*, x_0)$, $x_0 \in X^*$, will be said to "circle" b_0 if $\gamma = \sigma\tau\sigma^{-1}$ where σ is a path from x_0 to $x_1 \in \Delta - \{b_0\}$ and τ is a path in $\Delta - \{b_0\}$ beginning and ending at x_1 and winding once (positively) around b_0. If μ is the monodromy representation for the cover $Y^* \to X^*$ then $\mu(\gamma)$ is a permutation, each cycle of which corresponds to a component of $f^{-1}\left(\Delta - \{b_0\}\right)$; that is, each cyle of $\mu(\gamma)$ corresponds to a point of Y above b_0. The length of a cycle in $\mu(\gamma)$ will be the multiplicity of f^e at the corresponding point of Y.

(ii) Also any cover transformation g of $G(Y^*/X^*)$ extends to a biholomorphic map g^e of $Y \to Y$, and since $f \circ g = f$ we have $f^e \circ g^e = f^e$. Such a map of $Y \to Y$ will be called a cover transformation of Y and the group of such cover transformations will be denoted $G(Y/X)$.

(iii) Let X be a compact Riemann surface, B a finite set $\{b_1, \cdots, b_s\}$, $X^* = X - B$. Let D be a normal subgroup of $\pi_1(X^*, x_0)$, $x_0 \in X^*$, and let Y^* be the corresponding Galois cover of X^*, $f : Y^* \to X^*$. We then extend Y^* to Y, f to f^e, and $G(Y^*/X^*)$ to $G(Y/X)$. Then $G(Y/X)$ is isomorphic to $\pi_1(X^*, x_0)/D$. Now for each $b_j \in B_0$

let γ_j be a path circling b_j. Then $\mu(\gamma_j)$ is a regular permutation of length, say, ν_j. The ramification of f^e above b_j is then seen to be $(n/\nu_j)(\nu_j - 1)$ where n is the number of sheets in the covering. We obtain The Riemann-Hurwitz formula for Galois coverings

$$2p_Y - 2 = n(2p_X - 2) + \sum_{j=1}^{s} n(1 - \frac{1}{\nu_j}) .$$

Lemma 4.10. Let G be a finite group of automorphisms on a Riemann surface X. Suppose there is an $x_0 \in X$ so that $g(x_0) = x_0$ for all $g \in G$. Then G is a cyclic group.

Proof. Let Δ be a parametric disk centered at x_0. Let Δ_0 be the component of $\bigcap_{g \in G} g\Delta$ containing x_0. Then Δ_0 is simply connected and invariant under G. By the uniformization theorem we may assume that G acting on Δ_0 is a finite group of fractional linear transformation of $\mathbb{D} \to \mathbb{D}$ which leaves zero fixed. It now follows that G is cyclic.

(iv) Let Y be a Riemann surface with a finite group automorphisms G. Let Y/G be the set of orbits of G. If the only element of G with a fixed point is the identity then Y/G is made a Riemann surface X by giving it the quotient topology and requiring that the natural map $f : Y \to X$ be holomorphic. In fact, f is a smooth covering.

Suppose G admits non-identity elements with fixed points. For $y \in Y$ let $St(y) = \{g \in G \mid g(y) = y\}$, the stabilizer of y. If $h \in G$ then $St(h(y)) = h\, St(y) h^{-1}$. Thus the fixed points of non-identity elements of G are orbits of G.

Let $Y^* = \{y \mid St(y) = e\}$. Then $G - \{e\}$ is fixed-point-free on Y^*. Let $X^* = Y^*/G$. By the previous discussion $f : Y^* \to X^*$ can be extended to $f^e : Y \to X$ where we add to X (points corresponding to) the orbits of points in Y with stabilizers larger than $\langle e \rangle$. f^e is now holomorphic and branched precisely over $X - X^*$.

We now summarize this discussion in a theorem.

Theorem 4.10. If G is a finite group of automorphisms on a Riemann surface Y, then the orbit space Y/G is naturally a Riemann surface so that the projection $f : Y \to Y/G$ is a holomorphic map. The branch-

ing of f occurs precisely at the fixed points of non-identity elements of G. The number of sheets in the covering $Y \to Y/G$ is the order of G.

(With a slight modification of the above discussion one sees that this theorem is true for countable groups of automorphisms, G.)

4.11. We saw in Lemma 4.6 that if $Y \to Z$ is a Galois covering and H is normal in $G(Y/Z)$ then there exist a factoring $Y \to X \to Z$ where both intermediate coverings are normal. We want to reverse this procedure. If $Y \to X$ and $X \to Z$ are Galois coverings, when do we know that $Y \to Z$ is Galois? This then is the Riemann surface (or function field) version of a basic problem in group extensions.

Theorem 4.11. (A. M. Macbeath [18]). Let X be a surface. Let D be a normal subgroup of $\pi_1(X,x_0)$ and let $(Y,f,y_0) \to (X,x_0)$ be the corresponding smooth Galois covering. Let φ be a homeomorphism of X (possibly with fixed points). Let α be a path in X joining x_0 to $\varphi(x_0)$. Define $\varphi_* : \pi_1(X,x_0) \to \pi_1(X,x_0)$ by

$$\varphi_*(\gamma) = \alpha\,\varphi(\gamma)\,\alpha^{-1}.$$

Suppose $\varphi_*(D) = D$.

(Since D is normal this criterion is independent of α.)

Let y_1 be any point above $\varphi(x_0)$.

Then there exists a homeomorphism $\tilde{\varphi}$ of Y onto Y so that $\tilde{\varphi}(y_0) = y_1$, and $f \circ \tilde{\varphi} = \varphi \circ f$. That is, $\tilde{\varphi}$ preserves the fibers of f.

Consequently $\tilde{\varphi}$ normalizes $G(Y/X)$.

Proof. Let $\tilde{\alpha}$ be a path in Y connecting y_0 to y_1. Let $\alpha = f(\tilde{\alpha})$, connecting x_0 to $\varphi(x_0)$. If $y \in Y$ let $\tilde{\sigma}$ connect y_0 to y. If $\sigma = f(\tilde{\sigma})$ let $x = \sigma(1)$. $\varphi(\sigma)$ connects $\varphi(x_0)$ to $\varphi(x)$. Let $\tilde{s}$ be the lift of $\varphi(\sigma)$ starting at y_1. Define $\tilde{\varphi}(y) = \tilde{s}(1)$.

To show that $\tilde{\varphi}$ is well defined, suppose $\tilde{\tau}$ connects y_0 to y. Then $\tau = f(\tilde{\tau})$ connects x_0 to x. Now $\tilde{\sigma}\tilde{\tau}^{-1}$ is a closed path in Y so $\sigma\tau^{-1}$ is a closed path in D. Let $\tilde{t}$ be the lift of $\varphi(\tau)$ starting at y_1. Now $\varphi_*(\sigma\tau^{-1})$ is in D. Therefore, the lifts of $\alpha\,\varphi(\sigma)$ and $\alpha\,\varphi(\tau)$ have the same endpoint in Y; that is, $\tilde{t}(1) = \tilde{s}(1)$. This shows that $\tilde{\varphi}$ is well defined.

That $\tilde{\varphi}$ is locally $f^{-1} \circ \varphi \circ f$ shows that $\tilde{\varphi}$ is a local homeomorphism. That $\tilde{\varphi}$ has an inverse sending y_1 to y_0 is seen by inverting the steps in the proof.

4.12. If φ is an element in a group, let $o(\varphi)$ denote its order.

If $o(\varphi)$ is finite in the previous theorem, it will always be the case that $o(\varphi)$ divides $o(\tilde{\varphi})$, but equality is not necessary. We now give a simple criterion for $o(\varphi) = o(\tilde{\varphi})$. (In the language of group extensions this means that the extension

$$0 \to K \to \langle K, \tilde{\varphi} \rangle \to \langle \varphi \rangle \to 0$$

splits, where $K = G(Y/X)$).

Corollary 4.12. Suppose we have the hypotheses of Theorem 4.11 where X and Y are Riemann surfaces and φ and $\tilde{\varphi}$ are biholomorphic. Suppose φ has a fixed point. Then $\tilde{\varphi}$ can be choosen so that $o(\varphi) = o(\tilde{\varphi})$.

Proof. Let x_0 be a fixed point of φ. Suppose $f(y_0) = x_0$ Then we can find $\tilde{\varphi}$ so that $\tilde{\varphi}(y_0) = y_0$. Now $(\tilde{\varphi})^{o(\varphi)}$ is in $G(Y/X)$ and has a fixed point y_0; consequently $(\tilde{\varphi})^{o(\varphi)} = e$. Therefore $o(\tilde{\varphi})$ divides $o(\varphi)$.

q.e.d.

4.13. When $G(Y/X)$ is abelian the criterion for lifting automorphisms can be stated in terms of the first homology group of X, $H_1(X,Z) \cong \pi_1(X,x_0)/[\pi_1(X,x_0), \pi_1(X,x_0)]$.

Theorem 4.13. Suppose $f : Y \to X$ is a smooth Galois covering of surfaces where $G(Y/X)$ is abelian. Let D be the subgroup of $\pi_1(X,x_0)$ corresponding to this covering. Let D' be the image of D under the quotient map of $\pi_1(X,x_0)$ onto $H_1(X,Z)$. A homeomorphism φ of X acts on $H_1(X,Z)$ in a natural way. If such an aciton of φ on $H_1(X,Z)$ leave D' invariant, then φ lifts to Y as in Theorem 4.11.

Proof. If $\gamma \in D$ then $\varphi_*(\gamma) \in D \cdot [\pi_1(X,x_0), \pi_1(X,x_0)]$. Therefore $\varphi_*(\gamma) \in D$ since the commutator subgroup is a subgroup of D.

Corollary 4.13. Let $Y \to X$ be a smooth Galois cover of Riemann surfaces where $G(Y/X)$ is abelian. Suppose X is hyperelliptic. Then the hyperelliptic involution lifts to Y.

Proof. In $H_1(X,Z)$ the hyperelliptic involution sends every cycle into its inverse and, therefore, preserves all subgroups of $H_1(X,Z)$.

4.14. Conjugate automorphism groups.

Lemma 4.14. Let X be a Riemann surface with a finite group of automorphisms G. Let φ be another automorphism, $\varphi \notin G$. Then

X/G is conformally equivalent to $X/\varphi G\varphi^{-1}$.

Proof. Let $[x]$ be a G-orbit. Then $\left[\varphi(x)\right]$ is a $\varphi G\varphi^{-1}$-orbit. For if $y = g(x)$ then $\varphi(y) = \varphi g\varphi^{-1}\left(\varphi(x)\right)$. Thus the map $[x] \to \left[\varphi(x)\right]$ takes X/G onto $X/\varphi G\varphi^{-1}$.

4.15. We now include some consequences of the previous paragraphs which will be of use later.

4.15.1. A Riemann surface of genus 2 does not admit an automorphism of order 7. A Riemann surface of genus 3 does not admit an automorphism of order 5.

4.15.2. A Riemann surface of even genus does not admit an automorphism group isomorphic to $Z_2 \times Z_2 \times Z_2$ where Z_2 is the cyclic group of order 2.

4.15.3. Let $Y \to X$ be a Galois covering of compact Riemann surfaces where $G(Y/X)$ is abelian. If the covering is not smooth, then it must be branched above at least 2 points of X.

The first three results follow from the Riemann-Hurwitz formula for Galois covers. The fourth follows from the observation that the commutator subgroup of an abelian group is the identity.

Chapter 5. Automorphisms of Riemann surfaces, I.

5.1. If X is a Riemann surface let $\mathrm{Aut}(X)$ stand for the group of automorphisms of X.

We now reproduce, with slight modifications, Hurwitz's proof that a group of automorphisms on a compact Riemann surface of genus greater than one is finite [15].

Lemma 5.1. Let X be a compact Riemann surface of genus p. Let φ be an automorphism of X. Then φ has at most $2p+2$ fixed points.

Proof. Suppose φ has s fixed points. Let f be a meromorphic function with precisely $p+1$ poles at $p+1$ distinct points, none of which are fixed points for φ and so that none of the poles of f and $f \circ \varphi$ coincide. Then $f - f \circ \varphi$ has $2p+2$ poles and at least s zeros. Therefore $2p+2 \geq s$.

q.e.d.

Theorem 5.1. (H. A. Schwarz [23]). If X_p is a compact Riemann surface with genus $p \geq 2$, then $\mathrm{Aut}(X_p)$ is a finite group.

Proof. For $p \geq 2$ the total weight of the $3K$ points (higher order Weierstrass points for the tri-canonical series) is $25(p-1)^2 p$. The maximum weight at any point is $p(p+1)/2$. Thus there are at least $25(p-1)^2 p / \big(p(p+1)/2\big)$ distinct $3K$ points. For $p \geq 2$ this number is greater than $2p+2$.

Any automorphism permutes the $3K$-points. We thus have a map from $\mathrm{Aut}(X_p) \to \mathcal{S}(3K\text{-points})$. The kernel of this map fixes the $3K$-points, and so by Lemma 5.1 the kernel is the identity. Thus $\mathrm{Aut}(X_p)$ is isomorphic to a subgroup fo a finite permutation group.

q.e.d.

Definition. For $p \geq 2$ let $N(p)$ be the largest order of an automorphism group on a Riemann surface of genus p.

Many applications of what follows will give information about the nature of this function, $N(p)$.

5.2. Let $f : Y \to \mathbb{P}^1$ be a Galois covering of n-sheets, Y compact. Suppose the ramification occurs above $B = \{z_1, \dots, z_s\} \subset \mathbb{P}^1$. Let γ_i be a path circling z_i so that

$$\pi_1(\mathbb{P}^1 - B, z_0) = \langle \gamma_1, \gamma_2, \cdots, \gamma_s \mid \gamma_1 \gamma_2 \cdots \gamma_s = e \rangle .$$

Let ν_i be the ramification of a single point over z_i. Then $\gamma_i^{\nu_i} = e$. If

$\mu : \pi_1(\mathbb{P}^1 - B, z_0) \to \mathcal{S}_n$ is the monodromy map then $M(Y/\mathbb{P}^1)$ is generated by $\mu(\gamma_1), \cdots, \mu(\gamma_s)$. If $\mu(\gamma_i) = a_i \in \mathcal{S}_n$ then the a_i's satisfy the following: $a_i^{\nu_i} = e$, $i = 1, 2, \cdots, s$, and $a_1 a_2 \cdots a_s = e$. These generators obviously can be expected to satisfy other relations.

Definition. A finite group generated by elements $a_1, \cdots, a_s$ so that $a_1^{\nu_1} = \cdots = a_s^{\nu_s} = a_1 a_2 \cdots a_s = e$ will be denoted $(\nu_1, \cdots, \nu_s)$.

In the discussion preceeding the definition both $G(Y/\mathbb{P}^1)$ and $M(Y/\mathbb{P}^1)$ are $(\nu_1, \cdots, \nu_s)$ groups since the two groups are isomorphic. If p is the genus of Y then the Riemann-Hurwitz formula is:

$$(*) \quad 2p - 2 = -2n + n \sum_{k=1}^{s} \left(1 - \frac{1}{\nu_k}\right) = n\left(s - 2 - \sum_{k=1}^{s} \frac{1}{\nu_k}\right).$$

When $G(Y/\mathbb{P}^1)$ is abelian, in the identification of $G(Y/\mathbb{P}^1)$ with $M(Y/\mathbb{P}^1)$ the permutation $\mu(\gamma_i)$ is equal to the stabilizer of one (and therefore all) of the points in Y above z_i in $\mathbb{P}^1$. This will, in fact, be the case for arbitrary $G(Y/\mathbb{P}^1)$ whenever $\mu(\gamma_i)$ is central in $M(Y/\mathbb{P}^1)$. The observations of this paragraph are valid for any Galois cover $Y \to X$ and do not depend on the genus of X being zero. All these remarks follow from Theorem 4.7.

5.3. Groups of genus zero.

In 5.3 assume that $p = 0$, that is $Y = \mathbb{P}^1$.

Definition. A finite group fo automorphism of $\mathbb{P}^1$ will be called a group of genus zero.

In the fomula $(*)$ of 5.2 we see that $s - 2 - \sum_{k=1}^{s} \frac{1}{\nu_k}$ must be negative, equal to $-\frac{2}{n}$, in fact. Since $\nu_k \geq 2$ we see that s must be less than or equal to 3.

If $s = 2$ $G(\mathbb{P}^1/\mathbb{P}^1)$ is an (n, n), a cyclic group of order n.

If $s = 3$ we list the possibilities in the following table:

$G(\mathbb{P}^1/\mathbb{P}^1)$	order	description
(2,2,n)	2n	Dihedral group D_n
(2,3,3)	12	Alternating group A_4
(2,3,4)	24	Symmetric group $\mathcal{S}_4$
(2,3,5)	60	Alternating group A_5

This allows us to completely describe hyperelliptic automorphism groups.

Theorem 5.3. Let X_p be a compact hyperelliptic Riemann surface, $p \geq 2$. Let H be the hyperelliptic involution. Then the subgroup of order 2, $\langle H \rangle$, is normal (and central) in $\mathrm{Aut}(X_p)$ and $\mathrm{Aut}(X_p)/\langle H \rangle$ is a group of genus zero.

Proof. Since the g^1_2 is unique, the hyperelliptic involution is unique. But for $g \in \mathrm{Aut}(X_p)$, $g^{-1}Hg$ is also a hyperelliptic involution (Lemma 4.14). Consequently $g^{-1}Hg = H$. The result now follows from Lemma 4.6.

5.4. Groups of genus one.

If X_1 is a Riemann surface of genus one, any finite group of X_1 will be called a group of genus one provided the quotient is $\mathbb{P}^1$. Since $\pi_1(X_1, \cdot)$ is isomorphic to $Z \times Z$, we can find a fixed point free finite group of automorphism isomorphic to $Z_m \times Z_n$ for any two positive integers m, n.

For a Galois cover $X_1 \to \mathbb{P}^1$ the Riemann-Hurwitz formula is

$$\left(s - 2 - \sum_{k=1}^{s} \frac{1}{\nu_k}\right) = 0$$

with no obvious condition on n. The four possibilities are

$s = 4$	(2,2,2,2)
$s = 3$	(2,3,6)
	(2,4,4)
	(3,3,3)

In analogy with hyperelliptic Riemann surfaces we define a Riemann surface to be elliptic-hyperelliptic if it is a 2-sheeted branched covering of a Riemann surface of genus one. An involution whose orbits are the

fibers of this 2-sheeted covering will be called an elliptic-hyperelliptic involution. Later in this chapter we will show that an elliptic-hyperelliptic Riemann surface of genus greater than 5 has a unique elliptic-hyperelliptic involution. (This also follows from the Castelnuovo-Severi inequality.) Then we can repeat the proof of Theorem 5.3 to conclude that for genus greater than 5 , an elliptic-hyperelliptic automorphism group has a normal (central) subgroup of order two , and the quotient group is a group of genus one .

5.5. Maximal automorphism groups for $p \geq 2$.

We now reproduce Hurwitz's proof that for $p \geq 2$

$$N(p) \leq 84(p-1) \qquad [15].$$

Lemma 5.5. Let $Y \to X$ be a Galois covering of n sheets of Riemann surfaces. Assume p_y , the genus of Y , is greater than one . If p_x , the genus of X , is positive then

$$o\big(G(Y/X)\big) \leq 4(p_y - 1) .$$

Proof. The Riemann-Hurwitz formula for the covering is

$$2p_y - 2 = n(2p_x - 2) + n \sum_{i=1}^{s} \left(1 - \frac{1}{\nu_i}\right) .$$

If $p_x \geq 2$ then $2p_y - 2 \geq n(2p_x - 2) \geq 2n$, or $n \leq p_y - 1$.

If $p_x = 1$ then $2p_y - 2 = n \sum_{i=1}^{s} \left(1 - \frac{1}{\nu_i}\right) \geq \frac{1}{2}n$.

q.e.d.

Thus to obtain large automorphism groups we must have $p_x = 0$.

Then $2p_y - 2 = n\left(s - 2 - \sum_{i=1}^{s} \frac{1}{\nu_i}\right)$.

If $s \geq 5$ then $2p_y - 2 \geq n\left(3 - \frac{5}{2}\right) = \frac{1}{2}n$.

If $s = 4$ then $2p_y - 2 = n\left(2 - \sum_{i=1}^{4} \frac{1}{\nu_i}\right)$.

Since $p_y > 1$, not all the ν_i can equal 2 . The order of $G(Y/X)$, n , attains its maximum when $(\nu_1, \nu_2, \nu_3, \nu_4) = (2,2,2,3)$ and $n = 12(p_y - 1)$.

If $s = 3$ then $2p_y - 2 = n\left(1 - \sum_{i=1}^{3} \frac{1}{\nu_i}\right)$.

In order to assure that $p_y > 1$, we must exclude those values of (ν_1, ν_2, ν_3) which occur for groups of genus zero and one. Thus we choose (ν_1, ν_2, ν_3) so that $\left(1 - \sum \frac{1}{\nu}\right)$ is positive but as small as possible. The appropriate values are $(2,3,7)$ in which case $n = 84(p_y - 1)$.

Theorem 5.5. (Hurwitz). For $p \geq 2$, $N(p) \leq 84(p - 1)$. If a Riemann surface of genus p admits a group G of automorphisms of order $84(p - 1)$ then G is a $(2,3,7)$ and the quotient map $X_p \to \mathbb{P}^1$ is branched over 3 points of $\mathbb{P}^1$ above which the multiplicities of the branch points are $2, 3$, and 7.

Conversely, if G is an abstract $(2,3,7)$, then there exists a Riemann surface X_p admitting an automorphism group isomorphic to G of order $84(p - 1)$.

Proof. To prove the statement in the second paragraph, let G be an abstract $(2,3,7)$, $G = \langle a, b \rangle$ where $a^2 = b^3 = (ab)^7 = e$. Define $\mu : \pi_1\left(\mathbb{P}^1 - \{0, 1, \infty\}, 2\right) \to G$ by $\mu(\gamma_0) = a$, $\mu(\gamma_1) = b$ and $\mu(\gamma_\infty) = (ab)^{-1}$. The kernel of μ corresponds to a Galois covering Y^* of $\mathbb{P}^* \left(= \mathbb{P}^1 - \{0, 1, \infty\}\right)$ with $G(Y^*/\mathbb{P}^*) \cong G$. Filling in the branch points over $0, 1$, and ∞ gives a branched Galois covering $Y \to \mathbb{P}^1$ and $N(p_y) = 84(p_y - 1)$.

q.e.d.

Thus the problem of finding Riemann surfaces admitting automorphism groups of maximum order is the purely group theoretic problem of finding finite $(2,3,7)$'s.

To show that $N(p) = 84(p - 1)$ for an infinite number of p's, one must find an infinite number of $(2,3,7)$'s. This was done by G. A. Miller [21] in 1902, about 10 years after Hurwitz's original work.

5.6. An infinite number of $(2,3,7)$'s.

We shall follow A. M. Macbeath's discussion of $(2,3,7)$'s [18]. We first show the existence of a famous group.

Lemma 5.6. There exist a $(2,3,7)$ of order 168.

Proof. Let $G = GL(3,2)$, the group of all 3×3 non-singular matrices

over the field K_2 of two elements. By counting the number of ordered bases in the 3-dimensional vector space over K_2 we see that the order of G is $168 = 7 \cdot 6 \cdot 4$. Now consider G as the group of automorphisms of the elementary abelian group of order 8, G_8. Then $G_8 = \langle a,b,c \rangle$ or $G_8 = \{e, a, b, c, ab, bc, ca, abc\}$

$= \{e, g_1, g_2, g_3, g_4, g_5, g_6, g_7\}$

where we relable the group elements g_i in the manner indicated; e.g. $g_5 = bc$. Let A be the automorphism $a \to b$, $b \to a$, $c \to abc$. Using the subscripts of the g_j's we can describe A by the permutation $(12)(37)$. Let $B = (145)(267)$. Then $C = AB = (1673245)$. If we let $\langle A,B \rangle = H$ then H is a $(2,3,7)$ and we want to show that the order of H is 168. Since 3 and 7 divide the order of H, we show H contains a subgroup of order 8. $\langle A, B^{-1}AB \rangle$ is such a subgroup since $AB^{-1}AB = (1372)(46)$, that is $\langle A, B^{-1}AB \rangle = D_4$, a $(2,2,4)$.

q.e.d.

With a little more work one shows that G is simple.

Applying Theorem 5.5 gives us Klein's Riemann surface of genus 3 admitting 168 automorphisms.

Theorem 5.7. (Macbeath [18,19]. There exist a sequence $p_n \to \infty$ and $N(p_n) = 84(p_n - 1)$.

Proof. Let X_3 be Klein's Riemann surface. Let H_n be the characteristic subgroup of $\pi_1(X_3, x_0)$ containing the commutator subgroup and all n^{th} powers. H_n is the kernel of the surjective homomorphism

$$\pi_1(X_3, x_0) \to Z_n \times Z_n \times Z_n \times Z_n \times Z_n \times Z_n .$$

Since H_n is characteristic every element of $\mathrm{Aut}(X_3)$ leaves H_n invariant. The Riemann surface Y_n corresponding to H_n is a smooth, n^6-sheeted Galois covering of X_3 and all of $\mathrm{Aut}(X_3)$ lifts to Y_n. The genus of Y_n is $2n^6 + 1$, and the order of $\mathrm{Aut}(Y_n)$ is $168n^6$.

q.e.d.

This method of Macbeath's, that is, lifting automorphisms to Galois coverings corresonding to characteristic subgroups of $\pi_1(X, x_0)$ is very useful.

5.7 Smooth cyclic coverings of prime order.

Let n be a prime integer. We need a criterion which assures that an automorphism of order n on a Riemann surface is fixed point free.

Lemma 5.7. Let S_n be a cyclic group of prime order n on a Riemann surface X_p. Suppose $p - 1 = sn$ and $n > 2s + 3$. Then the covering

$$X_p \to X_p/S_n (= Y)$$

is smooth.

Proof. Suppose not. The Riemann-Hurwitz formula for the covering is $2p_X - 2 = n(2p_Y - 2) + t(n - 1)$ where t is the number of branch points and $t > 0$. Then

$$t(n - 1) = 2p_X - 2 - n(2p_Y - 2) = 2n(s + 1 - p_Y).$$

Therefore $n - 1 \mid 2(s + 1 - p_Y)$ or $n - 1 \leq 2s + 2$. This is the desired contradiction.

q.e.d.

5.8. $N(p) < 84(p - 1)$ infinitely often.

Theorem. (Macbeath [19]). Let n be a prime greater than 84. Let $p = n + 1$. Then $N(p) < 84(p - 1)$.

Proof. Suppose X_p admits a group G of order $84(p - 1) = 84n$. Let S_n be an n-Sylow subgroup of G, a cyclic group of order n. Since $n > 84$, S_n is a normal subgroup of G (Sylow's theorem). By Lemma 5.7 $X_p \to X_p/S_n$ is a smooth covering and X_p/S_n is a Riemann surface of genus 2 admitting a group of automorphisms of order 84 (Lemma 4.6). But a Riemann surface of genus 2 does not admit an automorphism of order 7 (4.15.1).

q.e.d.

We shall return to this function $N(p)$ in Chapter 7.

5.9. Groups with partitions [2].

We now generalize the hyperelliptic situation.

Definition. A finite group G_0 is said to admit a partition if there is a collection of subgroups G_i, $i = 1, 2, \cdots, s$, $s \geq 3$, so that

(1) $G_0 = \bigcup_{i=1}^{s} G_i$ and (2) if $0 < i < j$ then $G_i \cap G_j = \langle e \rangle$. The examples of such groups that will interest us are groups in which every element has order 2, and dihedral groups.

Theorem 5.9. Let X be a compact Riemann surface of genus p. Suppose X admits a finite group of automorphisms, G_0, where G_0 admits a partition. Let the pertinent subgroups be $G_1, G_2, \cdots, G_s$.

Let the order of G_i be n_i, let $X_i = X/G_i$ and let p_i be the genus of X/G_i for $i = 0,1,2,\cdots,s$. Then:

$$(s-1)p + n_0 p_0 = \sum_{i=1}^{s} n_i p_i .$$

Proof. Let the Riemann-Hurwitz formula for $X \to X/G_i$ be

$$(*) \qquad 2p - 2 = n_i(2p_i - 2) + ram_i .$$

We show $\sum_{i=1}^{s} ram_i = ram_0$. For $x \in X$ the elements in G_0 which leave x fixed form a cyclic group whose generator lies in one of the G_i's. It follows that $St(x)$ lies completely in this G_i. Consequently, the contribution of $St(x)$ to ram_0 contributes to one and only one of the ram_i's.

The theorem now follows by summing equation $(*)$ as i runs from 1 to s and comparing to equation $(*)$ with $i = 0$, noting that

$$n_0 = \sum_{i=1}^{s} n_i + 1 - s .$$

q.e.d.

The theorem is of interest because the ramifications do not enter into the formula. For a beautiful generalization of the material in this section see the paper of Ernst Kani "Relations between the genera and between the Hasse-Witt invariants of Galois coverings of curves" Canadian Mathematical Bulletin Vol. 28 (3), 1985, pp. 321-327.

5.10. $Z_2 \times Z_2$.

We first apply Theorem 5.9 to the non-cyclic group of order 4 (four group). There are 3 subgroups of order 2 so in this case $s = 3$. The formula becomes

$$p + 2p_0 = p_1 + p_2 + p_3 .$$

We use this formula to deduce a folk theorem that appears to be due to Enriques [10].

Lemma 5.10. Let X_3 be a covering of an X_2. Then X_3 is hyperelliptic.

Proof. $X_3 \to X_2$ is smooth and 2-sheeted. Let T be the involution in $Aut(X_3)$ so that $X_3/\langle T \rangle = X_2$. The hyperellipic involution on X_2 lifts to an automorphism S on X_3 of order two with some fixed points. S

normalizes $\langle T\rangle$; consequently, $\langle S,T\rangle$ is a four group on X_3 with $X_3/\langle S,T\rangle = \mathbb{P}^1$. Apply the formula above with $p = 3$, $p_0 = 0$ and one of the p_i's equal to 2 . The other p_i's must be 0 and 1 .

q.e.d.

Similarly, one shows that an X_5 (or X_7) which is a two-sheeted covering of a hyperelliptic X_3 (or X_4) must be hyperelliptic or elliptic-hyperelliptic.

5.11. $Z_2 \times Z_2 \times Z_2$.

There are 7 subgroups of $Z_2 \times Z_2 \times Z_2$ of order 2 . Theorem 5.9 gives

$$3p + 4p_0 = \sum_{i=1}^{7} p_i .$$

Lemma 5.11. A Riemann surface of genus 3 admitting a $Z_2 \times Z_2 \times Z_2$ is hyperelliptic.

Proof. If X_3 is not hyperelliptic then all p_i's above $(i \geq 1)$ must be one , whereas the left-hand side is at least 9 .

q.e.d.

In fact, by relabeling, if necessary, it can be arranged that $p_1 = 0$, $p_2 = p_3 = p_4 = 1$, and $p_5 = p_6 = p_7 = 2$.

A hyperelliptic Riemann surface cannot admit $Z_2 \times Z_2 \times Z_2 \times Z_2$ as a group of automorphisms because $Z_2 \times Z_2 \times Z_2$ is not a group of genus zero (Theorem 5.3).

5.12. Dihedral groups of automorphisms.

Let D_n be a dihedral group of order $2n$. Let R generate the cyclic subgroup of order n . If V is an element of order 2 not in $\langle R\rangle$ then $V_i = R^iV$ $(i = 1,2,\cdots,n)$ are the elements in D_n not in $\langle R\rangle$. If D_n is realized as a group of automorphisms on a Riemann surface X of genus p , let $X_R = X/\langle R\rangle$, $X_i = X/\langle V_i\rangle$, $i = 1,\cdots,n$ and let $X_0 = X/D_n$ with genera $p_R, p_1, \cdots, p_n, p_0$ respectively. Theorem 5.9 now gives

$$np + 2np_0 = np_R + 2\sum_{i=1}^{n} p_i .$$

If n is odd then all the groups $\langle V_i\rangle$ are conjugate and so $p_i = p_1$,

for $i = 2, \cdots, R$ (Lemma 4.14). Thus we obtain

$$p + 2p_0 = p_R + 2p_1 .$$

If n is even then $\langle V_i \rangle$ is conjugate to $\langle V_j \rangle$ if and only if $i \equiv j \pmod 2$.

Let p_1 and p_2 be the two genera in this case. Then

$$(*) \qquad p + 2p_0 = p_R + p_1 + p_2 .$$

This last formula holds for n odd or even with n odd implying $p_1 = p_2$.

5.13. Commuting involutions.

By the Castelnuovo-Severi inequality we know that if $p > 4p_0 + 1$ and X_p admits an involution T so that the genus of $X_p/\langle T \rangle$ is p_0 then this property makes T unique. If, however, $p \leq 4p_0 + 1$ then T need not be unique. The next application is an extension of these ideas.

Lemma 5.13. Let p_1 and p_2 be non-negative integers. Let X be a Riemann surface of genus p so that

$$2p \geq 3p_1 + 3p_2 + 3 .$$

Let X admit two distinct involutions S_1 and S_2 so that $X/\langle S_i \rangle$ has genus p_i, $i = 1, 2$. Then S_1 and S_2 commute.

Proof. Let $G_0 = \langle S_1, S_2 \rangle$ a dihedral group of order $2n$. Set $S_1 = V_1$ and $S_2 = V_2$. Then

$$p + 2p_0 = p_1 + p_2 + p_R .$$

We wish to show $n = 2$. Suppose $n \geq 3$. The Riemann-Hurwitz formula for $X \to X/\langle R \rangle$ is

$$2p - 2 = n(2p_R - 2) + \text{ram} \geq 3(2p_R - 2)$$

or

$$3p_R \leq p + 2 .$$

Since $p_0 \geq 0$ we have

$$3p \leq 3p + 6p_0 = 3p_1 + 3p_2 + 3p_R \leq 3p_1 + 3p_2 + p + 2 .$$

This contradiction proves the lemma.

q.e.d.

If $p_1 \neq p_2$ the above result can be strengthened since $n \neq 2$ implies $n \geq 4$; for in this case S_1 and S_2 cannot be conjugate.

Chapter 6. When are fixed points of automorphisms exceptional in some other sense?

6.1. Lewittes' Theorem.

The following theorem is due to J. Lewittes [17].

Theorem 6.1. Let X_p be a Riemann surface of genus $p \geq 2$. Let T be an automorphism with 5 or more fixed points. Then each fixed point is an ordinary Weierstrass point.

This theorem will follow from the following lemma.

Lemma 6.1. Let $f : X_p \to X_q$ be an n-sheeted branched covering of Riemann surfaces. Suppose the total ramification of this covering is greater than $4(n-1)$. If z is a branch point of f of multiplicity n then z is an ordinary Weierstrass point.

Proof. We show that X_p admits a meromorphic function whose only pole is at z and whose order is less than or equal to p.

Let $y = f(z)$. Let g be a meromorphic function on X_q whose only pole is at y and whose order is less than or equal to $q+1$. The Riemann-Hurwitz formula for the cover is:

$$2p - 2 = n(2q-2) + \text{ram} > 2nq - 2n + 4n - 4$$

or

$$p + 1 > n(q+1).$$

Consequently, $g \circ f$ is meromorphic on X_p; its only pole is at z and its order is less than or equal to $n(q+1)$. Consequently its order is less than or equal to p.

q.e.d.

Proof of Lewittes' Theorem. Let T be the automorphism of order n with 5 or more fixed points. Since the ramification of the cover $X_p \to X_p/\langle T \rangle$ is $5(n-1)$ or more, the lemma shows every fixed point to be a Weierstrass point.

6.2. G_{168} on X_3. (Some folklore.)

Let X_3 be a Riemann surface of genus 3 admitting the finite (2,3,7) group G_{168} of automorphisms of order 168. Consider the Galois covering $W_3 \to \mathbb{P}^1$ of 168 sheets. Let $z_2 \in \mathbb{P}^1$ be the point over which there are 84 branch points of multiplicity 2; let $z_3 \in \mathbb{P}^1$ be the point over which there are 56 branch points of multiplicity 3; let $z_7 \in \mathbb{P}^1$ be the point over which there are 24 branch points of multiplicity 7.

We claim that the 24 points above z_7 are the 24 ordinary

Weierstrass points in X_3. For the weights of points in a single G-orbit must all be the same. But the total Weierstrass weight is 24. The only place the Weierstrass points can fit is over z_7, all other orbits having more than 24 points.

Consider the points over z_3. The stabilizer of one of these points is an automorphism of order 3; call it T. If the genus of $W_3/\langle T\rangle$ is zero, there would be 5 fixed points, by the Riemann-Hurwitz formula, and so all would be ordinary Weierstrass points, by Lewittes' theorem, a contradiction to the previous paragraph. Consequently, the genus of $X_3/\langle T\rangle$ $(= X_1)$ must be one, and the Galois covering of 3 sheets must have 2 fixed points. Let $2x + 2y$ be the divisor of branch points in the covering $X_3 \to X_1$. Then this is a canonical divisor on X_3, and corresponds to the intersection of a line with $\varphi(X_3)$ where $\varphi : X_3 \to \mathbb{P}^2$ is the map given by the canonical series onto a non-singular plane quartic C_4. Thus $2x + 2y$ corresponds to the two points of contact of a bitangent to C_4. Since there are 56 points above z_3, these points are the 56 points of contact of the 28 bitangent to C_4.

We now locate the 2nd order Weierstrass points (which we call 2K points).

The total weight of the 2K points is 108. Let w_2, w_3, and w_7 be the 2K weight of a single point above z_2, z_3, and z_7 respectively. (Since all other G-orbits have order 168, none of these points can have positve 2K weight.) Thus

$$(*) \qquad 84w_2 + 56w_3 + 24w_7 = 108 .$$

The only possible solution in positive integers to this equation is

$$w_2 = 1 , w_3 = 0 , \text{and } w_7 = 1 .$$

The total weight of the kK points is $(2k - 1)^2 \cdot 12$. Let w_0 be the kK weight above points of $\mathbb{P}^1 - \{z_2, z_3, z_7\}$. Thus if w_2, w_3, and w_7 are the kK weights above z_2, z_3, and z_7 we have $168w_0 + 84w_2 + 56w_3 + 24w_7 = (2k - 1)^2 \cdot 12$. If we reduce this equation module 7 we have:

$$3w_7 \equiv 5(2k - 1)^2 \pmod 7$$

or

$$w_7 \equiv 4(2k - 1)^2 \pmod 7 .$$

Since $0 \le w_7 \le 5$ we see that w_7 is the smallest non-negative residue of $4(2k-1)^2 \pmod 7$. If we let $w_7(k)$ $(k \ge 2)$ be the kK weight of a point above z_7, we see that $w_7(2) = 1$, $w_7(3) = 2$, $w_7(4) = 0$, $w_7(5) = 2$, $w_7(6) = 1$, $w_7(7) = 4$, $w_7(8) = 4$, and $w_7(k)$ is periodic in k with period 7 for $k \ge 2$.

6.3. Cyclic coverings of $\mathbb{P}^1$ of prime order: Examples. (More folklore.)[3].

The title of this section refers to the Riemann surface for the polynomial $P(x,y) = y^n - \prod_{j=0}^{s} (x - a_j)^{f_j}$ where n is prime, $0 < f_j < n$ and (for convenience) we assume $\sum_1^s f_j \equiv 0 \pmod n$.

Then the n-sheeted covering $\varphi : X_p \to \mathbb{P}^1$ for this polynomial has branching at points above the a_j's of multiplicity n and is smooth over ∞. Thus the Riemann-Hurwitz formula gives $2p = (s-2)(n-1)$. Let A_j be the point on W_p above a_j. Our problem is to discover to what extent the A_j's are higher order Weierstrass points. In this section we shall work out some examples which will motivate the more general discussion that will follow.

On X_p, the Riemann surface for $P(x,y)$ above, there are functions y and $\varphi^*(x)$, which we will call simply x, where $M(X_p) = \mathbb{C}(x,y)$. Also we have a meromorphic differential, the lift of dx on $\mathbb{P}^1$, which we shall also denote by dx. We wish to find the divisors of these objects on X_p.

Since $\sum_{j=1}^{s} f_j \equiv 0 \pmod n$ the covering $\varphi : X_p \to \mathbb{P}^1$ is smooth over ∞. Call the n points on X_p over $\infty, \infty_1, \cdots, \infty_n$. Thus the function $x - x_0$ on $\mathbb{P}^1$ lifts to a function with n zero above x_0 and simple poles at $\infty_1, \cdots, \infty_n$. dx lifted has a zero of order $n-1$ at each A_j and n poles of order 2 at each point

above ∞. Near $\varphi^{-1}(\infty)$, $y = \sqrt[n]{\prod (x-a_j)^{f_j}}$ so that at each ∞_i, y has a pole of order $\sum f_j / n$. Thus over ∞, y has a total of $\sum f_j$ poles. Near A_j, $y = (x - a_j)^{(f_j/n)} k(x)$ where $k(a_j) \neq 0$. A local parameter t at A_j can be choosen so that $t^n = x - a_j$; so we see that y has a zero of order f_j at A_j.

We can summarize this information about x, y and dx on X_p by the following table:

	A_1	A_2	$\cdots$	A_s	total over ∞	total over $\varphi^{-1}(x_0)$
dx	$n-1$	$n-1$		$n-1$	$-2n$	
y	f_1	f_2	$\cdots$	f_s	$-\sum f_j$	
$x-x_0$	0	0		0	$-n$	n

As an example consider the polynomial

$$P(x,y) = y^5 - (x-a_1)(x-a_2)(x-a_3)(x-a_4)^2 \qquad (p = 4)$$

$K = g^3_6$	A_1	A_2	A_3	A_4	total over ∞
dx	4	4	4	4	-10
y	1	1	1	2	-5
$x-a_1$	5	0	0	0	-5
$\omega_1 : dx/y^2$	2	2	2	0	0
$\omega_2 : (x-a_4)\,dx/y^3$	1	1	1	3	0
$\omega_3 : (x-a_4)\,dx/y^4$	0	0	0	1	5
$\omega_4 : (x-a_4)^2\,dx/y^4$	0	0	0	6	0
$\omega_5 : (x-a_1)(x-a_4)\,dx/y^4$	5	0	0	1	0

The last 5 rows correspond to holomorphic one-forms on X_4, $\omega_1, \omega_2, \cdots, \omega_5$. We see that all A_j's are ordinary Weierstrass points with A_1, A_2, and A_3 having weight 2 and A_4 having weight 4.

By multiplying together two holomorphic one-forms we can compute

non-gaps for quadratic differentials as follows:

$2K = g^8_{12}$	A_1	A_2	A_3	A_4	total over ∞
ω_1^2	4	4	4	0	0
ω_2^2	2	2	2	6	0
ω_3^2	0	0	0	2	10
ω_4^2	0	0	0	12	0
ω_5^2	10	0	0	2	0
$[(x-a_2)/(x-a_1)]\,\omega_5^2$	5	5	0	2	0
$\omega_1\omega_2$	3	3	3	3	0
$\omega_1\omega_5$	7	2	2	1	0
$\omega_2\omega_5$	6	1	1	4	0
$\omega_3\omega_4$	0	0	0	7	5
$\omega_2\omega_4$	1	1	1	9	0

The 2K non-gaps at A_1 are $0,1,2,3,4,5,6,7,10$; weight 2 .

The 2K non-gaps at A_4 are $0,1,2,3,4,6,7,9,12$; weight 8 .

By multiplying one-forms with quadratic differentials we can obtain the 3K non-gaps at the A_j's . And so forth.

The table on the following page for a slightly more complicated case is worked out. It is readily seen that the kK weights at each A_j are periodic with period 7 for $k \geq 2$. Several features of this table will later be shown to be special cases of more general phenomena.
A final useful observation is the following.

Let $y^n = \prod_{j=1}^{s} (x - a_j)^{f_j}$ define a Riemann surface whose function field is $\mathbb{C}(x,y)$. Then letting

$$z = y^{\delta} \prod_{j=1}^{s} (x - a_j)^{\alpha_j} \,, \quad \delta \not\equiv 0 \pmod{n}$$

we see that $\mathbb{C}(x,y) = \mathbb{C}(x,z)$ and

kK-nongaps for $y^7 = (x-a_1)(x-a_2)(x-a_3)^2(x-a_4)^3$ (p=6)

		nongaps																				weight
k = 1	a_1			0	1	2	3	4				7										2
	a_3			0	1	2	3		5				8									4
	a_4			0	1	2		4	5						9							6
k = 2	a_1	0-8	9	10	11		13	14	15													3
	a_3	0-8	9	10	11		13			16			18									8
	a_4	0-8	9	10	11		13	14					18									6
k = 3	a_1	0-18	19	20	21	22	23	24														0
	a_3	0-18	19	20	21		23	24		26												4
	a_4	0-18	19	20		22	23			26	27											8
k = 4	a_1	0-28	29	30	31		33		35		37											6
	a_3	0-28	29		31	32		34		36				39								12
	a_4	0-28	29	30	31	32			35	36												4
k = 5	a_1	0-38	39	40	41	42	43	44														0
	a_3	0-38	39	40	41	42		44			47											4
	a_4	0-38	39	40	41			44	45			48										8
k = 6	a_1	0-48	49	50	51	52	53				57											3
	a_3	0-48	49	50		52		54	55		57											8
	a_4	0-48	49	50		52	53	54			57											6
k = 7	a_1	0-58	59	60	61		63	64											70	8		
	a_3	0-58	59	60		62	63		65										70	10		
	a_4	0-58	59		61	62	63			66									70	12		
k = 8	a_1	0-68		70	71	72	73	74			77											8
	a_3	0-68		70	71	72	73		75			78										10
	a_4	0-68		70	71	72		74	75				79									12

$$z^n = \prod_{j=1}^{s} (x - a_j)^{\delta f_j + n\alpha_j}$$

defines the same covering of $\mathbb{P}^1$. Thus

$$y^7 = (x - a_1)(x - a_2)(x - a_3)^5$$
$$y^7 = (x - a_1)^2(x - a_2)^2(x - a_3)^3$$
$$y^7 = (x - a_1)^3(x - a_2)^3(x - a_3)$$
$$y^7 = (x - a_1)^4(x - a_2)^4(x - a_3)^2 \quad \text{etc.}$$

all define the same covering $x : X_3 \to \mathbb{P}^1$.

6.4. Lemma. Let $y^n = \prod_{j=1}^{s} (x - a_j)^{f_j}$ where n is prime, $0 < f_j < n$, and $\sum f_j \equiv 0 \pmod{n}$. Let A_j be the point on the Riemann surface X_p above a_j. Then $n(2p - 2)A_j \equiv nK$ for all j and so $w(nK, A_j) > 0$ for all j.

Proof. Let g^1_n be the fibers of the cover $X_p \to \mathbb{P}^1$. Then by Theorem 2.6 we have (B is the branched locus)

$$K \equiv -2g^1_n + B$$

where we write the result additively. Now $g^1_n \equiv nA_1$, and

$$B = \sum_{j=1}^{s} (n - 1)A_j.$$

$$\text{Thus } nK \equiv -2ng^1_n + \sum_{j=1}^{s} (n - 1)nA_j \equiv -2n^2 A_1 + s(n - 1)nA_1$$
$$= n\big(-2n + s(n - 1)\big)A_1 = n(2p - 2)A_1.$$

q.e.d.

This lemma motivates most of the work in this chapter.

6.5. We first review some material form Chapter 2, with slightly different notation. Let $|G| = g^R_N$ be a complete linear series without

base point where G is an integral divisor on X_p. For $z \in X_p$ there are $R+1$ G-non-gaps for g^R_N at z, $0 = n_0 < n_1 < \cdots < n_R \leq N$. The $N-R$ $(=T)$ G-gaps at z are $g_1 < g_2 < \cdots < g_T \leq N$. $T = N - R = p - i_0$ where i_0 is the index of speciality of g^R_N. For all $j = 0,1,\cdots,R$

$$G \equiv n_j z + E_j$$

where $|E_j| = g^{R-j}_{N-n_j}$ and $(E_j, z) = 0$. Then the generalized Weierstrass weight of $|G|$ at z, denoted $w(G,z)$, and called simply the G-weight at z is:

$$w(G,z) = \sum_{j=0}^{R}(n_j - j) = \sum_{j=1}^{T}(R + j - g_j) = \sum \hat{n} - \hat{g}$$

where $\hat{n}$ stands for the extraordinary non-gaps $n_j > R$ and $\hat{g}$ stands for the extraordinary gaps $g_j \leq R$.

6.6. Lemma. The following three conditions are equivalent.

(1) g is a G-gap at z.

(2) $i(G - gz) + 1 = i\big(G - (g+1)z\big)$.

(3) There exists an integral divisor E of degree $g + 2p - N - 1$ so that $\big(|E|, z\big) = 0$ and

$$G + E \equiv (g+1)z + K.$$

(Remark: It is not required that $G - gz$ be integral.)

Proof. (1) $\Leftrightarrow$ (2): g is a G-gap at $z \Leftrightarrow r(G - gz) = r\big(G - (g+1)z\big)$ $\Leftrightarrow N - g - p + i(G - gz) = N - (g+1) - p + i\big(G - (g+1)z\big)$.

(2) $\Leftrightarrow$ (3): $i\big(G - (g+1)z\big) > 0$. Therefore there exists an integral divisor E so that $G - (g+1)z + E \equiv K$ or $G + E \equiv (g+1)z + K$. Now z is a fixed point of $|E| \Leftrightarrow r(E - z) = r(E) \Leftrightarrow i(G - gz) = i\big(G - (g+1)z\big)$. Therefore z is not a fixed point of $|E|$. Now reverse the argument.

q.e.d.

(Note: If $G = K$ we get the usual results about ordinary Weierstrass points.)

6.7. Lemma. Let $i_0 = i(G)$. Then $i_0 = i(G - g_1 z)$ and $i\big(G - (g_j + 1)z\big) = i_0 + j$, $j = 1,2,\cdots,T$.

Proof. Because G_1 is the first gap we have $r(G - g_1 z) = R - g_1$.

Therefore $|G - g_1 z| = g^{R-g_1}_{N-g_1}$, and it is complete; thus it has the same index as $|G|$.

By the previous lemma, if n is a non-gap then $i(G - nz) = i\big(G - (n+1)z\big)$. Thus the index of $G - nz$ changes only at the gaps, and at each gap the change is one.

q.e.d.

6.8. Lemma. If $|G|$ is not special then for m a positive integer $w(G + mz, z) = w(G, z)$.

Proof. It suffices to prove the lemma for $m = 1$ (and then use induction).

For any j $\quad G \equiv n_j z + E_j \quad (E_j, z) = 0$

$$G + z \equiv (n_j + 1)z + E_j .$$

Now $|G + z| = g^{R+1}_{N+1}$, so the $(G+z)$-non-gaps are

$$0, n_0 + 1, n_1 + 1, \cdots, n_R + 1$$

and the $(G+z)$-gaps are

$$g_1 + 1, \cdots, g_T + 1 .$$

Thus the sum $\sum \hat{n} - \hat{g}$ is unchanged.

6.9. Lemma. The smallest G-gap at z, g_1, satisfies:

$$g_1 \geq N - 2p + 1 .$$

Equality occurs $\Leftrightarrow G \equiv (g_1 + 1)z + K$

(in which case $g_1 + 1$ is a non-gap).

Proof. $i_0 = i(G - g_1 z)$. If $i_0 > 0$ then $N \leq 2p - 2$ and there is nothing to prove. Therefore suppose $i_0 = 0$. Then $i\big(G - (g_1 + 1)z\big) = 1$ so $\deg\big(G - (g_1 + 1)z\big) \leq 2p - 2$ or $N - (g_1 + 1) \leq 2p - 2$.

Equality occurs $\Leftrightarrow G - (g_1 + 1)z \equiv K$ since K is characterized by having degree $2p - 2$ and index 1.

6.10. Lemma. If $m \geq 2$ then $w(K + mz, z) = w(K, z) + p$.

Proof. Let $G = mz + K$. Then $|G| = g^R_N$ where $N = m + 2p - 2$, $N - R = p$. By Lemma 6.9 $g_1 = m - 1$. Let $n'_0, \cdots, n'_{p-1}$ be the K-non-gaps at z. Since K is fixed-point-free there are p G-non-gaps

$$m + n'_j \qquad j = 0,1,\cdots,p-1 .$$

Thus the $m + p - 1$ G-non-gaps are

$$0,1,\cdots,m-2 , m + n'_0,\cdots,m + n'_{p-1} .$$

Thus $w(G,z) = \sum_{j=0}^{p-1} (m + n'_j) - (m - 1 + j)$

$$= p + \sum_{j=0}^{p-1} (n'_j - j) = p + w(K,z) .$$

6.11. Lemma. Suppose $G = Nz$ where $|G| = g^R_N$, $N - R = p$. Then (1) $g_p = N - 1$ and (2) $w(G,z) = p + w(K,z)$.

Proof. (1) Since $i_0 = 0$ Lemma 6.7 gives, for $j = p$ $i\big((N - g_p - 1)z\big) = p$. Since K is fixed-point-free this implies $N - g_p - 1 = 0$.

(2) By Lemma 6.7 $i\big((N - g_j - 1)z\big) = j$.

Thus $\big\{N - g_j - 1 \,\big|\, j = 1,2,\cdots,p\big\}$ are the K-non-gaps at z.

$$w(K,z) = \sum_0^{p-1} n'_j - j = \sum_1^p (N - g_j - 1) - (j - 1)$$

$$= \sum (N - p + j) - g_j - 1 \quad \Big(\sum_1^p j = \sum_1^p p + 1 - j\Big)$$

$$= w(G,z) - p . \qquad (N - p = R)$$

6.12. Lemma. Let $G = Nz$, $\deg H = M$, $M \geq 2$, and $|G - H|$ is not special. Then $w(G - H,z) = w(H + K,z) = w(G + H + K,z)$.

Proof. Since $G = Nz$ the second equality follows form Lemma 6.8. For the first equality, let g_j be a $(G - H)$-gap. By Lemma 6.6 there exists an integral divisor E_j, $(E_j,z) = 0$ and

$$G - H + E_j \equiv (g_j + 1)z + K$$

or

$$(N - g_j - 1)z + E_j \equiv H + K .$$

Thus $N - g_j + 1$ is an $(H + K)$-non-gap. Now $g_j \leq N - M$

so $M - 1 \le N - g_j - 1$.

$|H + K| = g^{M+p-2}_{M+2p-2}$. There are $M + p - 1$ $(H + K)$-non-gaps of which the first $M - 1$ are $0,1,\cdots,M-2$ (Lemma 6.9). The other $(H + K)$-non-gaps are $N - g_j - 1$, $j = 1,2,\cdots,p$. Compute $w(H + K,z)$ for non-gaps $\ge M - 1$.

$$w(H + K,z) = \sum_{j=1}^{p} \left[(N - g_j - 1) - \Big((M - 1 + (j - 1)\Big)\right]$$

$$= \sum_{j=1}^{p} \Big(N - M - (j - 1) - g_j\Big) .$$

But $w(G - H,z) = \sum_{j=1}^{p} (N - M - p + j - g_j)$.

q.e.d.

6.13. In the light of Lemma 6.4 the following theorem explains many features of the results tabulated in Section 6.3.

Theorem 6.13. Suppose for $z \in X_p$ we have $m(2p - 2)z \equiv mK$.

(1) If $m = 1$, then for all $\ell = 2,3,4,\cdots$ we have
$$w(\ell K,z) = w(K,z) + p .$$

(2) If $k,\ell \ge 2$ and $k \equiv \ell \pmod m$ then
$$w(kK,z) = w(\ell K,z) .$$

(3) If $m \ge 2$, $k + \ell \equiv 1 \pmod m$, and $k,\ell \ge 2$ then
$$w(kK,z) = w(\ell K,z) .$$

(4) If $m \ge 2$, $\ell = 1,2,\cdots$ then
$$w(\ell mK,z) = w\Big((\ell m + 1)K,z\Big) = w(K,z) + p .$$

Proof. (1) follows from Lemma 6.11.

(2) follows from Lemma 6.8.

(3) follows from Lemma 6.12. For any $t \ge 1$ and k so that $2 \le k \le tm - 2$ let $G = tmK$ and $H = kK$ in Lemma 6.12.

(4) follows from Lemma 6.11 and (3) above.

6.14. Cyclic converings in general.

We now show how to generalize the results of Theorem 6.13 to cyclic coverings of prime order where the quotient is no longer required to be the Riemann sphere. Thus we consider a Riemann surface X_p of genus p with an automorphism T of prime order n and $X_q = X_p/\langle T\rangle$. The branched covering $\varphi : X_p \to X_q$ will have n-sheets . If $M(X_q)$ is the

field of meromorphic functions on X_q then we will denote $\varphi^*(M(X_q))$ by $M^*(X_q)$ a subfield of $M(X_q)$ of index n.

Lemma 6.14. There exists $y \in M(X_p) - M^*(X_q)$, with $y^n \in M^*(X_q)$ and $M(X_p) = M^*(X_q)[y]$.

Proof. Pick $y \in M(X_p) - M^*(X_q)$. Since n is prime, $M(X_p) = M^*(X_q)[y]$. Let $\tau = e^{2\pi i/n}$. Then $\sum_{m=0}^{n-1} \tau^{km} = 0$ if $k = 1,\cdots,n-1$. For $k = 0,1,\cdots,n-1$ let $y_k = \sum_{m=0}^{n-1} \tau^{-km}(y \circ T^m)$.

Thus $y_k \circ T = \tau^k \sum_{m=0}^{n-1} \tau^{-k(m+1)}(y \circ T^{m+1}) = \tau^k y_k$.

Also $\sum_{k=0}^{n-1} y_k = \sum_{m=0}^{n-1} \left(\sum_{k=0}^{n-1} \tau^{-km} \right) y \circ T^m = ny$.

Therefore, there is a y_k, not in $M^*(X_q)$, so that $y_k^n \in M^*(X_q)$ since $(y_k \circ T)^n = y_k^n \circ T = y_k$.

q.e.d.

We can and will assume that y is chosen as in this lemma so that $y \circ T = \tau y$. Let $y^n = F \in M^*(X_q)$. Then $F = \tilde{F} \circ \varphi$ for a function $\tilde{F} \in M(X_q)$. The divisor of $\tilde{F}$ can be written

$(\tilde{F}) = \sum_{j=1}^{s} f_j a_j + nE_0$ where E_0 is a divisor on W_q and

$0 < f_j < n-1$. Because $\deg(\tilde{F}) = 0$, we have $\sum f_j \equiv 0 \pmod{n}$. (Some of the a_j's can also occur in E_0.) Since X_p is the Riemann surface for the polynomial $X^n - \tilde{F} \in M(X_q)[X]$ we see that the covering $\varphi : X_p \to X_q$ is branched over each a_j and only there. (If $(\tilde{F}) = nE_0$,

then the covering is smooth.)

The ramification over each a_j is $n-1$ so the Riemann-Hurwitz formula for the covering is

$$2p - 2 = n(2q - 2) + s(n - 1) .$$

Let A_j be the point on X_p above a_j on X_q. Then the divisor of y is

$$(y) = \sum_{j=1}^{s} f_j A_j + \varphi^{-1}(E_0)$$

and $\sum f_j \equiv 0 \pmod{n}$.

6.15. Lemma. Suppose $g \in M(X_p)$ where $\varphi : X_p \to X_q$ is an n-sheeted covering of compact Riemann surfaces. Suppose there is a divisor of degree zero on X_q, D_0, so that $(g) = \varphi^{-1}(D_0)$. Then $g^n \in \varphi^* M(X_q)$. (We do not assume that n is prime or that the covering is Galois.)

Proof. For $y \in X_q$, let $\varphi^{-1}(y) = \{x_1, \cdots, x_n\}$. Define

$G(y) = \prod_{j=1}^{n} g(x_j)$, a well defined function on X_q. Then $(G) = n D_0$.

So $G \circ \varphi$ and g^n have the same divisor.

q.e.d.

6.16. Invariant divisors.

Definition. Let H be a group of automorphisms on a Riemann surface X. A linear series g^R_N will be said to be invariant under H if for all $h \in H$ and all divisors D in g^R_N, we have $hD \in g^R_N$.

For example the k-canonical series will be invariant under any subgroup of $\mathrm{Aut}(X)$.

Lemma 6.16. Let g^R_N be a complete linear series invariant under a cyclic group $\langle T \rangle$ of prime order. Then there exists a divisor G_0 in g^R_N so that $TG_0 = G_0$.

The proof of this lemma will depend on the following general fact.

Proposition 6.16. Let $|G| = g^R_N$ be a complete linear series invariant under a group $H \in \mathrm{Aut}(X)$. Consider a map

$$\theta : X \to \mathbb{P}^R$$

given by $R+1$ independent functions in $L(G)$. Then there is a group of projective transformations in $\mathbb{P}^R$, $\hat{H}$, isomorphic to H and $\theta(X)$ is invariant under $\hat{H}$. Moreover, $\hat{H}$ acts on the hyperplanes of $\mathbb{P}^R$ in the same manner that H acts on the divisors in $|G|$.

Proof. Let $\langle f_0, \cdots, f_R \rangle$ be a basis of $L(G) = \{ f \in M(X) \mid (f) + G > 0 \}$. Then $\langle f_0 \circ T, \cdots, f_R \circ T \rangle$ is a basis of $L(TG)$ where $T \in H$. Since g^R_N is invariant under T, there exists $F \in M(X)$ and

$$(F) = G - TG.$$

Thus $(f_k \circ T)F \in L(G)$ and so

$$(f_k \circ T)F = \sum_{j=0}^{R} a_{k_j} f_j \qquad (a_{k_j})^{(R+1)\times(R+1)}.$$

Define $\theta : X \to \mathbb{P}^R$ by $x \to (f_0(x), \cdots, f_R(x))$. Then Tx maps into

$$\begin{aligned}
&(f_0 \circ T(x), \cdots, f_R \circ T(x)) \\
&= \big((f_0 \circ T(x))F(x), \cdots, (f_R \circ T(x))F(x)\big) \\
&= \Big(\sum_{j=0}^{R} a_{0j} f_j, \cdots, \sum_{j=0}^{R} a_{Rj} f_j\Big).
\end{aligned}$$

Thus T induces on $\mathbb{P}^R$ a projective transformation T given with respect to this basis by the matrix (a_{kj}). Everything now follows.

q.e.d.

Proof of Lemma 6.16. Suppose g^R_N is invariant under $\langle T \rangle$. Then $\hat{T}$ acting on $\mathbb{P}^R$ also acts on $\mathbb{P}^{R*}$, the hyperplanes in $\mathbb{P}^R$. Suppose the action of $\hat{T}$ on $\mathbb{P}^{R*}$ is given by the matrix $\tilde{M}$. Thus $\tilde{M}^n = \lambda I$, $\lambda \in \mathbb{C}$. If $M = \frac{1}{\sqrt[n]{\lambda}} \tilde{M}$ then $M^n = I$; that is, we can lift the cyclic group to $\mathbb{C}^{R+1} - \{0\}$ over $\mathbb{P}^R$. Now we proceed as before. For $v \in \mathbb{C}^{R+1}$ define

$$v_k = \sum_{m=0}^{n-1} \tau^{-mk} M^m v$$

and so $Mv_k = \tau^k v_k$ and $nv = \sum v_k$.

Therefore there is a $v_k \neq 0$. The line through the origin and v_k in $\mathbb{C}^{R+1}$ corresponds to a point in $\mathbb{P}^{R*}$, which in turn corresponds to an invariant hyperplane H in $\mathbb{P}^R$. The pull back to X of the hyperplane section $\theta(X) \cap H$ will be a divisor in $|G|$ invarant under T (but not necessarily pointwise invariant).

q.e.d.

6.17. We continue the previous notation, some of which we now recapitulate. T is an automorphism of X_p of prime order n, and $\varphi : X_p \to X_q$ is the quotient map. y generates $M(X_p)$ over $M^*(X_q)$ where $y^n \in M^*(X_q)$

$$(y) = \sum_{j=1}^{s} f_j A_j + \varphi^{-1}(E_0)$$

where E_0 is a divisor on X_q. We will not assume $s > 0$, although most of the complications which follow are the result of assuming T has fixed points.

Let g^R_N be a complete linear series invariant under $\langle T \rangle$. We will assume for convenience that g^R_N is without base points; although this assumption is not necessary. Let G_0 be an invariant divisor in G^R_N; that is $TG_0 = G_0$. Then

$$G_0 = \sum_{j=1}^{s} g_j A_j + \varphi^{-1}(D_0)$$

where $0 \leq g_j < n$ and $N \equiv \sum_{j=1}^{s} g_j \pmod{n}$.

For example if $g^R_N = K_p$, the canonical series on X_p, then

$$G_0 = \sum_{j=1}^{s} (n-1) A_j + \varphi^{-1}(K_q).$$

If $g^R_N = |kK_p|$ then for $1 \le k \le n$

$$G_0 = \sum_{j=1}^{s} (n-k)A_j + \varphi^{-1}\left(kK_q + (k-1)(a_1 + \cdots + a_s)\right).$$

$a_j = \varphi(A_j)$.

Let $L = L(G_0) = \left\{F \in M(X_p) \,\middle|\, (F) + G_0 \ge 0\right\}$. Then $L(G_0)$ is invariant under T and $\dim L = R + 1$. Let $L_m = \left\{F \in L \,\middle|\, F \circ T = \tau^m F\right\}$, $m = 0, 1, \cdots, n$. Let $d_m = \dim L_m$.

Then $L = \sum_{m=0}^{n-1} L_m$ and $R + 1 = \sum_{m=0}^{n-1} d_m$.

Let $\mathcal{S}_m = \left\{(F) + G_0 \,\middle|\, F \in L_m\right\}$ where $\mathcal{S}_m = \varnothing$ if $d_m = 0$.

We want a more explicit description of the divisors in $\mathcal{S}_m$.

Note that $(y^m) = \sum_{j=1}^{s} mf_jA_j + \varphi^{-1}(mE_0)$.

If $F \in L_m$ then $(F/y^m) \circ T = F/y^m$; that is $F/y^m \in M^*(X_q)$. Suppose $(F) = G_m - G_0$ where $G_m \in \mathcal{S}_m$.

$$(F/y^m) = \varphi^{-1}(D) \text{ for some divisor } D \text{ in } X_q$$

thus $G_m - G_0 = (y^m) + \varphi^{-1}(D)$

$$G_m = G_0 + (y^m) + \varphi^{-1}(D).$$

$$G_m = \sum g_jA_j + \varphi^{-1}(D_0) + \sum mf_jA_j + \varphi^{-1}(mE_0) + \varphi^{-1}(D)$$

$$= \sum (mf_j + g_j)A_j + \varphi^{-1}(D')$$

$$= \sum t_{mj}A_j + \varphi^{-1}(D'')$$

where $t_{mj} = \overline{mf_j + g_j}$, the smallest non-negative residue of $mf_j + g_j \pmod n$, D' and D'' are divisors on X_q, and $D'' > 0$.

Lemma 6.17. If $\mathcal{S}_m$ is non-empty then $\mathcal{S}_m$ is the set of all divisors

of the form $\sum t_{mj} A_j + \varphi^{-1}(D_m)$ where a) the divisor is linearly equivalent to G_0, and b) D_m is integral.

Proof. That any divisor with these proeprties is in δ_m follows by reversing the above argument.

Theorem 6.17. For each $m = 0, 1, \cdots, n-1$ there exists on X_q a divisor D_m, not nessarily integral, so that

a) $G_0 \equiv \sum_{j=1}^{s} t_{mj} A_j + \varphi^{-1}(D_m)$ and

b) $d_m = \dim |D_m|$.

Proof. For part a) we unwind the previous calculation.

$$G_0 - \sum_{j=1}^{s} t_{mj} A_j = \sum g_j A_j + \varphi^{-1}(D_0) - \sum (g_j + m f_j) A_j - \varphi^{-1}(B)$$

$$= - \sum_{j=1}^{s} m f_j A_j - m\varphi^{-1}(E_0) + \varphi^{-1}(D_m)$$

$$= -(y^m) + \varphi^{-1}(D_m)$$

where $D_m = mE_0 + D_0 - B$ and B is a divisor supported in $\{a_1, \cdots, a_s\}$. Thus

$$G_0 \equiv \sum_{j} t_{mj} A_j + \varphi^{-1}(D_m).$$

Now $\dim |D_m| = 0$ if and only if there is no integral divisor equivalent to D_m. By the lemma this is equivalent to $\delta_m = \varnothing$, which in turn is equivalent to $d_m = 0$.

If $\dim |D_m| \geq 1$, let D_m^0 be a fixed divisor in $|D_m|$. Then fix $F^0 \in L_m$ where

$$(F^0) = \sum t_{mj} A_j + \varphi^{-1}(D_m^0) - G_0.$$

We define an isomorphism of $L(D_m^0)$ onto L_m as follows.

If $(f) + D^0_m > 0$ let $D'_m = (f) + D^0_m$. Then

$$(F^0\varphi^* f) = \sum_{j=1}^{s} t_{mj} A_j + \varphi^{-1}(D'_m) - G_0 ;$$

that is, $F^0\varphi^* f$ is in L_m. The map $f \to F^0\varphi^* f$ is the required isomorphism. For if $F \in L_m$ then F/F^0 is seen to be $\varphi^* f$ for some $f \in L(D^0_m)$.

q.e.d.

6.18. Lemma. Let $r = s(n-1)$, the ramification of the covering $\varphi : X_p \to X_q$. Then $\sum_{m=0}^{n-1} \sum_{j=1}^{s} t_{mj} = n\frac{r}{2}$.

Proof. As m runs through the integers $0, 1, \cdots, n-1$ $t_{mj} = \overline{g_j + mf_j}$ does also since $f_j \not\equiv 0 \pmod n$. Thus

$$\sum_{j=1}^{s} \sum_{m=0}^{n-1} t_{mj} = \sum_{j=1}^{s} \frac{n(n-1)}{2} = n\frac{s(n-1)}{2} = n\frac{r}{2} .$$

q.e.d.

In Theorem 6.17 $\deg D_m = \left(N - \sum_{j=1}^{s} t_{mj}\right)/n$. Applying the Riemann-Roch theorem to $|D_m|$ on X_q, we have

$$\dim |D_m| = \deg D_m - q + 1 + i_m$$

where i_m is the index of $|D_m|$. Summing this over m gives

$$R + 1 = \sum_m d_m = N - \left(\sum_m \sum_j t_{mj}\right)/n - n(q-1) + \sum_m i_m$$

$$= N - \frac{r}{2} - n(q-1) + \sum_m i_m$$

$$= N - p + 1 + \sum_{m=0}^{n-1} i_m$$

by the Riemann-Hurwitz formula. Another application of the Riemann-Roch theorem on X_p gives this

Theorem 6.18. $i(g^R_N) = \sum_{m=0}^{n-1} i_m$

Corollary 6.18. If g^R_N is not special then

$$d_m = \left(N - \sum_{j=1}^{s} t_{mj}\right)/n - q + 1$$

a non-negative number.

6.19. We now assume that g^R_N is non-special and we fix A_ℓ. Assuming $\mathcal{S}_m \neq \emptyset$ we ask how the divisors in $\mathcal{S}_m$ contribute to the g^R_N non-gaps at A_ℓ. Fix a G_m in $\mathcal{S}_m$ where

$$G_m = \sum_j t_{mj} A_j + \varphi^{-1}(D_m)$$

and $|D_m| = g^{d_m - 1}_{d_m - 1 + q}$. If $D_m = n'a_\ell + D'$, $D' \geq 0$, $(a_\ell, D') = 0$, then $t_{m\ell} + nn'$ is a g^R_N non-gap at A_ℓ. Thus the divisors in $\mathcal{S}_m$ contribute d_m non-gaps at A_ℓ and all of these d_m non-gaps are congruent to $t_{m\ell} \pmod n$. Since there are $R + 1 = \sum d_m$ non-gaps at A_ℓ we see that the only non-gaps at A_ℓ congruent to $t_{m\ell}$ come from divisors in $\mathcal{S}_m$ If a_ℓ is not a D_m-point (to be expected in general) then the non-gaps congruent to $t_{m\ell}$ at A_ℓ are

$$t_{m\ell}, t_{m\ell} + n, t_{m\ell} + 2n, \cdots, t_{m\ell} + (d_m - 1)n.$$

Definition. Let $d_{m\ell}$ be the number of integers in $0, 1, \cdots, R$ congruent to $t_{m\ell} \pmod n$.

(We remark that $\sum_{m=0}^{n-1} d_{m\ell} = R + 1 = \sum_{m=0}^{n-1} d_m$.)

Writing $w(G,A_\ell)$ as $\sum n_j - j$ we obtain:

$$w(G,A_\ell) = \sum_{j=0}^{R} n_j - \sum_{j=0}^{R} j$$

$$= \sum_{m=0}^{n-1} \sum_{j=0}^{d_m-1} (t_{m\ell} + jn) - \sum_{m=0}^{n-1} \sum_{j=0}^{d_{m\ell}-1} (t_{m\ell} + jn)$$

$$= \sum_{m=0}^{n-1} \sum_{j=d_{m\ell}}^{d_m-1} (t_{m\ell} + jn)$$

$$= \sum_{m=0}^{n-1} \tfrac{1}{2}\left(t_{m\ell} + d_{m\ell}n + t_{m\ell} + (d_m - 1)n\right)\left(d_m - d_{m\ell}\right) .$$

Since $\sum_m (d_m - d_{m\ell}) = 0$ we obtain the following.

Theorem 6.19. If $w(D_m, a_\ell) = 0$ for all m then

$$w(G,A_\ell) = \tfrac{1}{2} \sum_{m=0}^{n-1} \left[2t_{m\ell} + n(d_m + d_{m\ell})\right](d_m - d_{m\ell}) .$$

Definition. The right-hand side of the equation in this theorem will be called the expected weight and denoted $w_e(G,A_\ell)$.

Corollary 6.19. (1) $w(G,A_\ell) \geq w_e(G,A_\ell)$

(2) $w(G,A_\ell) \equiv w_e(G,A_\ell) \pmod{n}$.

Proof. If a_ℓ is a D_m-point for some m, the contribution of δ_m to $w(G,A_\ell)$ will increase by a multiple of n over that computed in the theorem.

q.e.d.

6.20. Lemma. The formula for $w_e(G,A_\ell)$ in Theorem 6.19 depends only on $\ell, f_1, \cdots, f_s, g_1 \cdots, g_s$ provided g^R_N is non-special.

Proof. We show $w_e(G,A_\ell)$ does not depend on N or q.

Suppose first that q is fixed and $g^{R'}_{N'}$ has the same $g_1,\cdots,g_s$ as g^{R}_{N}. Now $N \equiv \sum g \pmod{n}$ so $N \equiv N' \pmod{n}$. If $N' = N + n$ then $g^{R'}_{N'} = g^{R+n}_{N+n}$. Since $d_m = \left(N - \sum_{j=1}^{s} t_{mj}\right)/n - q + 1$, d_m is increased by one. d_{mj}, the number of integers congruent to $t_{m\ell}$ in $[0,R]$ is also increased by one. Since $\sum_m (d_m - d_{m\ell}) = 0$ the result follows.

Now suppose q is increased by one while N is fixed. Then d_m is decreased by one. p is increased by n, R is decreased by n and so $d_{m\ell}$ is decreased by one.

q.e.d.

We now consider the extent to which fixed points of automorphisms are higher order Weierstrass points, that is, kK-points for $k \geq 2$.

Theorem 6.20. Suppose $p \geq 2$ and T is an automorphism of prime order n with three or more fixed points. Then each fixed point is an ℓK-point for $\ell \geq 2$ and $\ell \equiv 0 \pmod{n}$ or $\ell \equiv 1 \pmod{n}$. (The case $\ell \equiv 1 \pmod{n}$ appears in Farkas-Kra [11]).

Proof. Consider the cover $X_p \to X_q$ $(= X_p/\langle T\rangle)$. By the above lemma we may apply the results of the case $X_{p-nq} \to \mathbb{P}^1$ provided $p - nq \geq 2$. (Since $s \geq 3$, $p - nq \geq 1$.) But then the results of Theorem 6.13 (4) apply.

The two cases when $p - nq = 1$ are (i) $n = 2$, $s = 4$ and (ii) $n = 3$ and $s = 3$. Then the two cases to consider are (i) $W_3 \to W_1$ where $T^2 = 1d$ and (ii) $W_4 \to W_1$ where $T^3 = 1d$. Here we use the formula in Theorem 6.19 directly. We will only work out the second case $W_4 \to W_1$.

In this case $f_1 = f_2 = f_3 = 1$. For $3K_4$ $(= g^{14}_{18})$ we may take $g_1 = g_2 = g_3 = 0$. Then $d_0 = 6$, $d_1 = 5$, and $d_2 = 4$ while $d_{01} = d_{11} = d_{21} = 5$. Then

$$w(3K_4, A_1) = \frac{1}{2}\Big(2 \cdot 0 + 3(6+5)\Big)(6-5) + \frac{1}{2}\Big(2 \cdot 1 + 3(5+5)\Big)(5-5)$$
$$+ \frac{1}{2}\Big(2 \cdot 2 + 3(4+5)\Big)(4-5)$$
$$= 1 .$$

Actually, as soon as we have $d_0 > d_{01}$ we know there is an extraordinary $3K_4$-non-gap at A_1 and so A_1 is a $3K_4$-point .

Similarly, $w_e(4K_4, A_1) = 1$.

q.e.d.

Chapter 7. Automorphisms of Riemann surfaces, II; $N(p)$.

7.1 In section 5.5 through 5.8 we discussed automorphism groups of order $84(p-1)$ on X_p. If X_p admits a group of automorphisms of order $84(p-1)$ then we immediately know that $N(p) = 84(p-1)$. If, however, $N(p) < 84(p-1)$ then determining $N(p)$ involves two problems: the existence problem, to show there is a Riemann surface of genus p admitting a group of the conjectured order; and the elimination problem, to show that no Riemann surface of genus p admits a group of order greater than that conjectured.

The problem of determining the function $N(p)$ for all p appears to be extremely difficult. Here we shall show how to find some infinite sequences of genera, p_n, where $N(p_n)$ can be determined.

7.2 In showing the existence of a group of automorphisms whose order competes for $N(p)$, very often we will be extending an abelian group by a group of genus zero. In using Macbeath's method of lifting automorphisms, Theorem 4.13 simplifies matters considerably.

Theorem 7.2. For $p \geq 2$, $N(p) \geq 8(p+1)$.

Proof. Fix an integer $p \geq 2$. Let $\tau = e^{2\pi i/(2p+2)}$. Let $B = \{\tau^k \mid k = 0,1,\ldots, 2p+1\}$, the vertices of a regular $(2p+2)$-gon inscribed in the unit circle. Let D_{2p+2} be the dihedral group in $\mathbb{P}^1$ generatd by $Vz = z^{-1}$ and $Rz = \tau z$. D_{2p+2} permutes the set B. If γ_0 is a path circling $\tau^0 = 1$, let $\gamma_k = R\gamma_0$, a little path circling γ^k. Let $Z_2 = \langle\varphi\rangle$. Let $\mu : \pi_1(\mathbb{P}^1 - B, 2) \to Z_2$ take each γ_k onto φ. The kernel of μ in $H_1(\mathbb{P}^1 - B, Z)$ is

$$\left\{\sum_{j=0}^{2p+1} c_j\gamma_j \,\middle|\, \sum c_j \equiv 0 \pmod 2\right\}$$

a subgroup of index 2 which is invariant under D_{2p+2}. The corresponding 2-sheeted covering, with points filled in above the $2p+2$ points of B, is a hyperelliptic Riemann surface X_p. D_{2p+2} lifts to X_p which together with the hyperelliptic involution yields a group of automorphisms of order $8(p+1)$.

q.e.d.

7.3. We continue in the same vein.

Theorem 7.3. If 3 divides p then $N(p) \geq 8(p+3)$.

Proof. Consider the symmetric group S_4 acting as the orientation preserving symmetries of the cube inscribed in the Riemann sphere. Lable the 8 vertices $a_1, a_2, \ldots, a_8$ so that any two of the vertices with odd indices lie at opposite ends of a diagonal of one of the square faces of the cube. Let $A_1 = \{a_1, a_3, a_5, a_7\}$ and $A_2 = \{a_2, a_4, a_6, a_8\}$. The even indexed vertices have the same property. Let $A = A_1 \cup A_2$. For $z_0 \notin A$ and any positive integer m define $\mu : \pi_1(\mathbb{P}^1 - A, z_0) \to Z_m (= \langle\varphi\rangle)$ by sending paths circling the odd indexed vertices into φ and paths circling the even indexed vertices into φ^{-1}. The elements of S_4 permute the sets A_1, A_2 so that S_4 lifts to the Riemann surface X_p covering $\mathbb{P}^1$ corresponding to the kernel of μ. By the Riemann-Hurwitz formula

$$2p - 2 = -2m + 8(m - 1) \quad \text{or} \quad p = 3(m - 1).$$

The order of the group on X_p is $o(S_4)o(Z_m) = 24m$. Thus X_p admits a group of order $8(p + 3)$.

q.e.d.

(The results of Theorems 7.2 and 7.3 together with the result that these bounds are sharp infinitely often were obtained independently by Colin Maclachlan [20] and the author [1]).

7.4. We shall soon see that the lower bound $8(p + 1) \leq N(p)$ is sharp infinitely often, so the following definition is justified.

Definition. A group G of automorphism on a Riemann surface X_p will be called big if $o(G) \geq 8\ (p + 1)$.

In this section we shall show the existence of many sequences of big groups whose orders lie in arithmetic progressions. These groups will all be $(2,4,\lambda)$'s. So let us look at a $(2,4,\lambda) = \langle a,b\rangle$ where $a^2 = b^4 = c^\lambda = e$, $c = ab$, and the order of the group is $\mu\lambda$, where μ is the index of the cyclic group $\langle c\rangle$. Let N be the smallest normal subgroup containing b^2. Then $\langle a,b\rangle/N$ can be expected to be a $(2,2,\lambda')$ where $\lambda' \mid \lambda$, a group of genus zero. If N is abelian then it is an elementary abelian group of order 2^ℓ for some ℓ. Thus $\langle a,b\rangle$ is an extension by a dihedral group of an elementary abelian 2-group.

Theorem. (W. T. Kiley [16]). Let ℓ be a positive integer. If $p \equiv (1 - 2^\ell) \pmod{\ell 2^{\ell-1}}$ then $N(p) \geq 8(p - 1 + 2^\ell)$.

Proof. Fix positive integers ℓ and m. Let

$$A = \{\tau^k \mid \tau = e^{2\pi i/2\ell m},\ k = 0,1,\cdots,2\ell m - 1\}$$

the vertices of a regular $2\ell m$-gon inscribed in the unit circle. Let $A_j = \{\tau^k \mid k \equiv j \pmod{\ell},\ j = 0,1,\ldots,\ell - 1$. There are $2m$ points in each set A_j. Let $D_{2\ell m}$ be generated by R and V where $Rz = \tau z$ and $Vz = z^{-1}$. Then $D_{2\ell m}$ permutes the sets $\{A_j\}$. Let γ_k circle τ^k as before. Define $\mu : \pi_1(\mathbb{P}^1 - A, z_0) \to (Z_2)^\ell, (z_0 \notin A)$ where $(Z_2)^\ell = \langle \varphi_0, \varphi_1, \ldots \varphi_{\ell-1} \rangle$ by

$$\mu(\gamma_k) = \varphi_{\bar{k}} \quad \text{where } \bar{k} \equiv k \pmod{\ell}.$$

In $H_1(\mathbb{P}^1 - A, Z)$ $\ker \mu = \left\{ \sum_{j=0}^{\ell-1} \sum_{k\equiv j} c_k \gamma_k \;\middle|\; \sum_{k\equiv j} c_k \equiv 0 \pmod{2} \right\}$,

which is invariant under $D_{2\ell m}$.

The Riemann surface corresponding to $\ker \mu$, X_p, has a group of automorphisms of order $o(D_{2\ell m}) o\left((Z_2)^\ell\right) = 4\ell m \cdot 2^\ell$.

By the Riemann-Hurwitz formula

$$2p - 2 = -2 \cdot 2^\ell + 2\ell m \cdot 2^{\ell-1}.$$

Thus $4 \cdot 2^\ell \cdot \ell m = 8(p - 1 + 2^\ell)$ and $p = 1 - 2^\ell + m \cdot \ell 2^{\ell-1}$ for all $\ell, m \geq 1$.

q.e.d.

For $\ell = 1$ we have the result of Theorem 7.2. For $\ell = 2$ we have: if $p \equiv 1 \pmod{4}$ then $N(p) \geq 8(p + 3)$ a result different from Theorem 7.3.

7.5. We now state a theorem which shows that the lower bounds on $N(p)$ discussed in the last three sections are sharp infinitely often.

Theorem 7.5. (W. T. Kiley [16]. Suppose we have positive integers s and t so that $s \geq 2$ and $4 \mid st$. Then there are an infinite number of genera p so that

1) $N(p) \leq 8(p - 1 + s)$ and
2) $p \equiv (1 - s) \pmod{(st)/4}$.

(In Theorem 7.4 $s = 2^\ell$ and $t = 2\ell$, $\ell = 1,2,\ldots$.)

The proof of this theorem is an elaboration of the basic idea in Theorem 5.8 where Macbeath showed that $N(p) < 84(p - 1)$ for infinitely many p. There $p = n + 1$ where n was a large prime.

Here we shall assume $p - 1 = sn$ where again n is a large prime satisfying further conditions. Moreover, a closer examination of the steps in the proof will allow us to actually compute $N(p)$ where $p = n + 1$, or $2n + 1$, where n is prime. The proof of Theorem 7.5 will occupy the next several sections.

7.6. Some preliminary results.

From now on n will denote a prime integer and p will denote the genus of a Riemann surface X_p where the subscript may be omitted if no confusion follows.

In the sequel we will need to recall Lemmas 5.5 and 5.7.

From elementary group theory we recall that the group of automorphisms of Z_n is Z_{n-1}. Also by using the transfer one proves that if a finite group G admits a central abelian subgroup S of index i, then the transfer homomorphism from G into S is $g \to g^i$. If S is a Sylow subgroup the map is surjective with kernel of order i [27].

Other results from group theory will be proven in the propositions below.

Now we catalogue the big groups. If X admits a big group of automorphism, G, then the genus of X/G is zero (Lemma 5.5). Recalling the discussion of section 5.5 one sees that the possible big groups on a Riemann surface X_p $(p \geq 2)$ are (for convenience we will divide them into 3 types):

Type (I)	$(2,5,\lambda)$	$5 \leq \lambda \leq 19$
	$(2,6,\lambda)$	$6 \leq \lambda \leq 11$
	$(2,7,\lambda)$	$7 \leq \lambda \leq 9$
	$(3,3,\lambda)$	$4 \leq \lambda \leq 11$
	$(3,4,\lambda)$	$4 \leq \lambda \leq 5$
	$(2,2,2,3)$	
Type II	$(2,3,\lambda)$	$\lambda \geq 7$
Type III	$(2,4,\lambda)$	$\lambda \geq 5$

To prove Theorem 7.5 we will put more and more conditions on the genus of a Riemann surface to eliminate big groups of Type I, Type II, and the unwanted big groups of Type III.

7.7. Theorem. Let s be a positive integer and let n be a prime. Let $p = sn + 1$. Let G be a big group on X_p. Suppose

(i) $n > 84s$;

(ii) n divides $o(G)$.

Then G admits a normal subgroup H where G/H is a cyclic group which is one of the following 4 possibilities:

(i) $(2,3,6)$ G/H has order 6 and X_p/H has genus one ;

(ii) $(2,4,4)$ G/H has order 4 and X_p/H has genus one ;

(iii) $(3,3,3)$ G/H has order 3 and X_p/H has genus one ;

(iv) $(2,5,10)$ G/H has order 10 and X_p/H has genus two.

If $3 \mid (n-1)$ then $o(G) \leq 24(p-1)$.

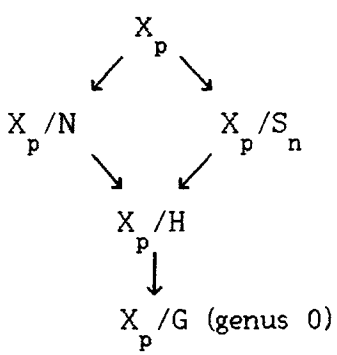
If $3 \nmid (n-1)$ and $4 \mid (n-1)$ then $o(G) \leq 16(p-1)$.

If $(12, n-1) = 2$ then $5 \mid (n-1)$ and $o(G) \leq 10(p-1)$.

In particular, $\left((n-1)/2, 30\right) > 1$.

Proof. Let S_n be the n-Sylow subgroup of G . Now $84(p-1) = 84sn \geq o(G)$. Consequently S_n is a normal cyclic subgroup, by Sylow's theorem. By Lemma 5.7 $X_p \to X_p/S_n$ is a smooth covering.

Consider the homomorphism $\Theta : G \to \mathrm{Aut}(S_n)$ given by $\Theta(g)(h) = ghg^{-1}$. Let H be the kernel of Θ . Then $S_n \subset H$ and H is the centralizer of S_n in G . S_n is thus a central subgroup of H , and since $\mathrm{Aut}(S_n) \cong Z_{n-1}$, G/H is isomorphic to a subgroup of Z_{n-1} . G/H is a cyclic group whose order divides $n-1$. The transfer map $H \to S_n$ is surjective and the kernel of the transfer, N , is a normal complement to S_n in H ; that is, $H = NS_n$ where $N \cap S_n = \langle e \rangle$. We thus have the following diagram.



Now $H/N \cong S_n$ and S_n acting on X_p is without fixed points. Thus the covering $X_p/N \to X_p/H$ is also a smooth covering of n sheets. Consequently, the genus of X_p/H is positive. By Lemma 5.5, $o(H) \leq 4(p-1)$. Since G is big, $o(G/H) > 2$.

Now G is a $(\nu_1, \nu_2, \ldots, \nu_s)$ group ($s = 3$ or 4) and so G/H is a $(\nu_1', \nu_2', \ldots, \nu_s')$ group where $\nu_j' \mid \nu_j$. Since G/H is cyclic and X_p/H has positive genus, 3 of the ν_i''s must be larger than one. By consulting the list of big groups one sees that the 4 possibilities for G/H listed in the statement of the theorem are the only possibilities. Consequently $3, 4$, or 5 must be a factor of $n-1$.

If $3 \mid (n-1)$ then G/H could be a $(2,3,6)$ of order 6. Then X_p/H has genus one so that $o(G) \leq 6 \cdot 4(p-1)$. The remaining parts in the statement of the theorem are proved similarly. Note that if the genus of X_p/H is 2 then $o(H) \leq p-1$. (See the proof of Lemma 5.5.)

q.e.d.

Corollary 7.7. Let s be a positive integer and let n be a prime. Let $p = sn + 1$ and let G be a big group on X_p. Suppose

(i) $n \geq 84s$

(ii) $\left(\frac{n-1}{2}, 30\right) = 1$ $\left(\text{i.e. } n \equiv 23, 47, \text{ or } 59 \pmod{60}\right)$.

Then $n \nmid o(G)$ and G is not a big group of Type I.

Proof. That $n \nmid o(G)$ follows immediately from the theorem.

Now consider the 35 possibilities for the big groups of Type I. The

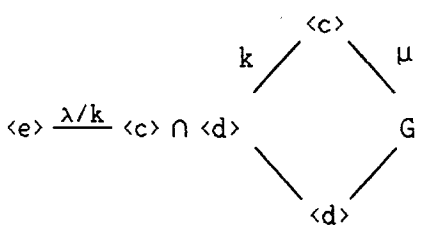
$(2,2,2,3)$ is excluded since it has order $12(p-1)$ and $n \mid p-1$. The other 34 big groups are (α,β,γ)'s. By the Riemann-Hurwitz formula $2p-2 = o(G)(1-\frac{1}{\alpha}-\frac{1}{\beta}-\frac{1}{\gamma})$. We see that $2p-2 = 0(G)\cdot\frac{A}{B}$ where A and B are integers. Since $n \nmid o(G)$ it follows that $n \mid A$. By working out these 34 cases one sees that the largest prime dividing an A is 47. Since $n > 84$, this rules out all Type I big groups.

q.e.d.

7.8. The following group theoretic proposition will eliminate the $(2,3,\lambda)$'s.

Proposition 7.8. Let G be a non-cyclic $(2,3,\lambda)$ of order $\mu\lambda$. Then $\lambda \le \mu^2$ and so $o(G) \le \mu^3$. In fact $\lambda \mid k(\lambda,\mu)$ where $k \le \mu$ and $k \mid \lambda$.

Proof. Suppose $G = \langle a,b\rangle$ where $a^2 = b^3 = c^\lambda = d^\lambda$, $c = ab$ and $d = ba$. Since $dc = b^2$, $G = \langle c,d\rangle$. Now $\langle c\rangle \cap \langle d\rangle$ is normal, in fact, central in G. Consider the following diagram



where $k\mu$ is the index of $\langle c\rangle \cap \langle d\rangle$ in G and $k \mid \lambda$.

Now $k \le \mu$. For in $G' = G/\langle c\rangle \cap \langle d\rangle = \langle c',d'\rangle$ the cyclic

subgroups $\langle c' \rangle$ and $\langle d' \rangle$ intersect only in $\langle e' \rangle$ and so $c'^{\alpha} d'^{\beta}$ are distinct for $0 \le \alpha, \beta \le k-1$, $(\alpha, \beta) \ne (0,0)$. Thus $k^2 \le o(G') = k\mu$ Also $6 \mid k\mu$. If not, say $2 \nmid k\mu$. Then $a \in \langle c \rangle \cap \langle d \rangle \subset \langle c \rangle$ and so G is cyclic. Similarly, $3 \mid k\mu$.

The transfer $t : G \to \langle c \rangle \cap \langle d \rangle$ is the map $g \to g^{k\mu}$. But $6 \mid k\mu$; therefore, a and b map into the identity; that is, $g \to g^{k\mu}$ is the trivial homomorphism. Consequently $c^{k\mu} = e$ or $\lambda \mid k\mu$. Thus $\lambda \le k\mu \le \mu^2$. That $\lambda \mid k(\lambda, \mu)$ is now immediate.

q.e.d.

Theorem 7.8. Assume the hypotheses of Corollary 7.7. Assume, moreover, that $n \ge 216s^2 = 6^3 s^2$. Then G is a $(2,4,\lambda)$ group.

Proof. If G is a $(2,3,\lambda)$ of order $\mu\lambda$, then by the Riemann-Hurwitz formula

$$2ns = 2p - 2 = o(G)\left(1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{\lambda}\right) = \mu\lambda\left(\frac{1}{6} - \frac{1}{\lambda}\right) = \frac{\mu}{6}(\lambda - 6)$$

or

$$12ns = \mu(\lambda - 6).$$

Since $n \nmid \mu$ we see that $n \mid (\lambda - 6)$ or $\lambda - 6 = \tau n$.

Thus

$$12s = \tau\mu$$

and

$$\mu = 12s/\tau \le 12s.$$

By the proposition $o(G) \le (12s)^3$.

But G is a big group. Therefore

$$o(G) \ge 8(p+1) > 8(p-1) = 8ns \ge (12s)^3,$$

a contradiction.

q.e.d.

7.9. Now we prove another group theoretic proposition, very similar to that in 7.8, to help eliminate the unwanted $(2,4,\lambda)$'s .

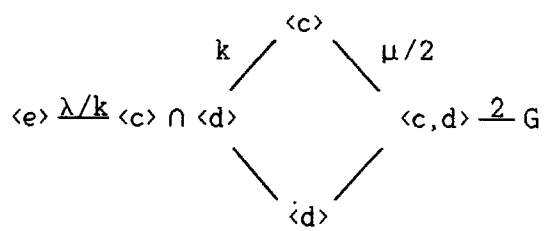
Proposition 7.9. Let G be a $(2,4,\lambda)$ group of order $\mu\lambda$ where $\mu > 4$, $4 \mid \mu$ and $\lambda > \mu^2$ $\left(o(G) > \mu^3\right)$.

Then there is a positive integer k so that $3 \le k \le \mu/2$ and $k \mid \lambda$. If $k = 3$ then $\mu = 8$.

Proof. Let $G = \langle a,b \rangle$ where $a^2 = b^4 = c^\lambda = d^\lambda = e$, $c = ab$, $d = ba$. We show first that $G \neq \langle c,d \rangle$.

Suppose $G = \langle c,d \rangle$. Since $4 \mid \mu$ it follows exactly as in Proposition 7.8 that if the index of $\langle c \rangle \cap \langle d \rangle$ is $k\mu$, the transfer homomorphism $G \to \langle c \rangle \cap \langle d \rangle$, $g \to g^{km}$ is the identity and consequently $\lambda \mid km$. Also $k \le \mu$ as before, so $\lambda \le \mu^2$. But this is excluded by hypothesis.

Now $\langle c,d \rangle$ is normal (since $aca = d$) and must now have index 2 in G . Consider the diagram



Then considering $\langle c,d \rangle / \langle c \rangle \cap \langle d \rangle = \langle c',d' \rangle$ we see as in Proposition 7.8 that $k \le \mu/2$.

If $k = 1$ $\langle c \rangle = \langle d \rangle$ and so $\mu = 2$, a contradiction.

If $k = 2$ $G/\langle c\rangle \cap \langle d\rangle$ is a $(2,4,2)$, of order 8 and so $\mu = 4$, a contradiction.

If $k = 3$ $G/\langle c\rangle \cap \langle d\rangle$, is a $(2,4,3)$ of order 24 and so $\mu = 8$.

q.e.d.

7.10. Proof of Theorem 7.6

We assume s and t are positive integers so that $s \geq 2$ and $4 \mid st$. Let c be a positive integer so that

(a) if $1 \leq k \leq 4s$ then $k \mid c$.

(b) $(st/4) \mid c$.

Let n be a prime integer so that $n \equiv -1 \pmod{c}$ and $n > 216s^2$. Since $60 \mid c$ we see that $n \equiv 59 \pmod{60}$. Also $n > 84s$. Let $p = sn + 1$. Then

$$p = 1 - s + s(n+1) \equiv (1-s) \pmod{c} \equiv (1-s) \left(\mathrm{mod}\left(\frac{st}{4}\right)\right).$$

By Dirichet's theorem on primes in an arithmetic series, there are an infinite number of such genera p. It remains to show that

$$N(p) \leq 8(p - 1 + s).$$

Suppose G is a big group on X_p. Thus by Corollary 7.7 and Theorem 7.8 G is a $(2,4,\lambda)$ group of order, say $\mu\lambda$ and n does not divide $o(G)$. By the Riemann-Hurwitz formula

$$8p - 8 = \mu(\lambda - 4).$$

Now n divides $p - 1$ but n does not divide $o(G)$.

Therefore $n \mid (\lambda - 4)$. Let $\lambda - 4 = \tau n$. Then

$$8ns = 8(p-1) = \mu n \tau \quad \text{or} \quad \mu = 8s/\tau.$$

Since $o(G) = \mu\lambda = 8p - 8 + 4\mu$ it remains to show

$$\mu \leq 2s \quad \text{or} \quad \tau \geq 4.$$

We eliminate the other possibilities for τ . Now

$$\lambda = n\tau + 4 = \tau(n + 1) + (4 - \tau) .$$

Letting $\tau = 1,2$ or 3 we see that $4 - \tau = 3,2$ or 1 .

We apply Proposition 7.9 to produce a k so that $3 \le k \le \mu/2 (= 4s/\tau)$ and $k \mid \lambda$. If $k = 3$ then $\mu = 8$. By the choice of n we also have $k \mid (n + 1)$. Thus k divides $1,2$ or 3 , all cases which are impossible $\left(k = 3 \Rightarrow \tau = 1 \Rightarrow \mu \ge 16\right)$.

Thus $\tau \ge 4$.

q.e.d.

7.11. In order to apply the proof of Theorem 7.5 to produce genera p where $N(p)$ is known, applying condition (a) in Section 7.10 together with $n > 216s^2$ quickly leads to rather large numbers. We conclude this chapter by proving several theorems which allow us to compute $N(p)$ for some genera which are relatively small.

Theorem 7.11. Let n be a prime integer, $n \ge 84$ $(p - 1 = n)$.

(a) If $3 \mid (n - 1)$ then $N(n + 1) = 12n$;
i.e. $N(p) = 12(p - 1)$.

(b) If $3 \nmid (n - 1)$ and $5 \mid (n - 1)$, then $N(p) = 10n$;
i.e. $N(p) = 10(p - 1)$.

(c) If $(n - 1,15) = 1$, then $N(n + 1) = 8(n + 4)$;
i.e. $N(p) = 8(p + 3)$.

The proof of this theorem is derived by examining more closely Theorem 7.7 and 7.8 when $s = 1$.

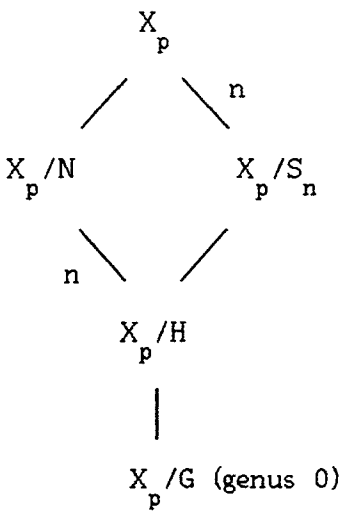
7.12. Theorem. Let n be a prime integer, $n > 84$, $p = n + 1$.

If $3 \mid (n - 1)$, then $N(p) \le 12(p - 1)$.

If $3 \nmid (n-1)$ and $5 \mid (n-1)$, then $N(p) \leq 10(p-1)$.

If $(n-1, 15) = 1$ and G is a big group, then G is not of Type I and $n \nmid o(G)$.

Proof. Again consider the diagram in the proof of Theorem 7.7 when assuming that $n \mid o(G)$ where G is a big group.



Since $X_p \to X_p/S_n$ is smooth we see that the genus of X_p/S_n is 2. The genus of X_p/H is one or 2. A Galois cover $X_2 \to X_1$ must have precisely 2 sheets so the order of H is n or $2n$. If the genus of X_p/H is one then G/H is a cyclic group of genus one and therefore has order bounded by 6, and the order divides $n-1$. Thus if $3 \mid (n-1)$ the maximum order for G is $(6)(2n) = 12n$. (If $o(G) = 12n$ one sees that G must be a $(2,6,6)$).

If $3 \nmid (n-1)$ and the genus of X_p/H is one, the maximum

order for G/H is 4 and $o(G) \leq 8n$ which is not big. So if $3 \nmid n-1$ and G is big then the genus of X_p/H is 2. In this case G/H must be a cyclic group of genus 2, a $(2,5,10)$ which has order 10. Thus if $3 \nmid (n-1)$ and $5 \mid (n-1)$ then the maximum order of G is $10n$. If $3 \nmid (n-1)$ and $5 \nmid (n-1)$ then the situation illustrated in the diagram is impossible and $n \nmid o(G)$. Then we eliminate groups of Type I by the same argument as in the proof of Corollary 7.7.

q.e.d.

7.13. We now complete the proof of Part (c) of Theorem 7.11. We assume $p = n+1$, n is prime, $n \geq 84$, and $(n-1,15) = 1$. X_p admits a big group and $n \nmid o(G)$. By examining the proofs in Section 7.8 we show that without any further restrictions G cannot be a $(2,3,\lambda)$.

If G is a $(2,3,\lambda)$ of order $\mu\lambda$ then there is a k so that $k \mid \lambda$, $k \leq \mu$ and $\lambda \mid k(\lambda,\mu)$. Since $p-1 = n$, the Riemann-Hurwitz formula gives $2p-2 = \mu\lambda(\frac{1}{6} - \frac{1}{\lambda})$ or

$$12n = \mu(\lambda - 6).$$

As before $n \mid \lambda - 6$ so that $\lambda - 6 = \tau n$ and $12 = \mu\tau$. The possibilities for τ are $1,2,3,4$ and 6 and correspondingly $\mu = 12,6,4,3$ and 2.

If $\tau = 1$, $\mu = 12$, $\lambda = n+6$ then $(\mu,\lambda) = 1$. Thus $\lambda \mid k$; that is $\lambda = k \leq \mu = 12$. Consequently $n+6 \leq 12$. This contradicts $n \geq 84$. Thus τ cannot be 1.

If $\tau = 2$, $\mu = 6$, $\lambda = 2n + 6$, then $(\mu, \lambda) = 2$ . . Thus $\lambda \mid 2k$ and so $2n + 6 \leq 12$, again a contradiction .

For $\tau \geq 3$, $\mu \leq 4$ so directly by Proposition 7.8 $o(G) \leq \mu^3 \leq 64$.

Thus our big group is not a $(2,3,\lambda)$.

We conclude that G must be a $(2,4,\lambda)$ group of order $\mu\lambda$. Again the Riemann-Hurwitz formula gives

$$8n = \mu(\lambda - 4) .$$

Again $\lambda - 4 = \tau n$ and $\mu\tau = 8$. Therefore

$$o(G) = \mu\lambda = \frac{8}{\tau}(\tau n + 4) \leq 8(n + 4) = 8(p + 3) .$$

But if $(n - 1, 15) = 1$ then $n = 2, 8$ or $14 \pmod{15}$. Consequently $p = n + 1 \equiv 0 \pmod 3$. By Theorem 7.3 $N(p) = 8(p + 3)$.

The proof of Part (c) of Theorem 7.11 is complete.

7.14. To complete the proof of Parts (a) and (b) of Theorem 7.11 we first exhibit two groups of genus 2 which must act on X_p / S_n in the diagram of Section 7.12.

The $(2,6,6)$ is $Z_2 \times Z_6 = \langle a, b \rangle$ where $a^2 = b^6 = e$.

The $(2,5,10)$ is $Z_{10} = \langle \varphi^5, \varphi^2 \rangle$ where φ generates Z_{10} .

If $3 \mid (n - 1)$ (n prime) we want to extent Z_n by the above $(2,6,6)$ to obtain a $(2,6,6)$ of order $12n$. This will complete the proof of Part (a) of Theorem 7.11.

If $5 \mid (n - 1)$ (n prime) we want to extend Z_n by the above $(2,5,10)$ to obtain a $(2,5,10)$ of order $10n$. This will complete the proof of Part (b) of Theorem 7.11.

Proposition 7.14 Let G be an (α,β,γ) where the l.c.m. $[\alpha,\beta]$ divides γ. Let $G = \langle a,b\rangle$ where $a^\alpha = b^\beta = c^\gamma = e$ $(c = ab)$. Assume $\alpha,\beta,\gamma \geq 2$ and at most one of these integers is 2.

(i) Suppose there exists a surjective homomorphism

$$\Theta : G \to Z_{[\alpha,\beta]}$$

so that $o\big(\Theta(a)\big) = \alpha$ and $o\big(\Theta(b)\big) = \beta$.

(ii) Suppose n is prime and $[\alpha,\beta] \mid (n-1)$ but $n \nmid o(G)$.

Then there exists G', again an (α,β,γ) of order $o(G)n$, and G' satisfies hypothesis (i).

Proof. Θ maps G onto $Z_{[\alpha,\beta]}$ which is a subgroup of $\mathrm{Aut}(Z_n)$. Let G' be the semi-direct product of G and Z_n. That is, $G' = \langle a,b,\varphi\rangle$ where $G = \langle a,b\rangle$, $\langle\varphi\rangle = Z_n$ and $\langle a,b\rangle \cap \langle\varphi\rangle = \langle e\rangle$, together with the following relations:

$$a\varphi a^{-1} = \varphi^x,\ x^\alpha \equiv 1 \pmod n,\ x \not\equiv 1 \pmod n;$$

$$b\varphi b^{-1} = \varphi^y,\ y^\beta \equiv 1 \pmod n,\ y \not\equiv 1 \pmod n.$$

Thus $c\varphi c^{-1} = \varphi^{xy}$ and $(xy)^{[\alpha,\beta]} \equiv 1 \pmod n$.

x and y can be chosen so that $xy \not\equiv 1 \pmod n$ since α and β are not both 2.

Now $(a\varphi)^t = (a\varphi a^{-1})(a^2\varphi a^{-2})\cdots(a^t\varphi a^{-t})a^t = \varphi^{x+x^2+\cdots x^t}a^t$.

Consequently $(a\varphi)^\alpha = \varphi^{x+x^2+\cdots x^{\alpha-1}+1}a^\alpha = e$

$$(b\varphi)^\beta = e$$

and $$(c\varphi)^{[\alpha,\beta]} = c^{[\alpha,\beta]} \quad \big(\text{since } xy \not\equiv 1 \pmod n\big).$$

Thus $$(c\varphi)^\gamma = e.$$

Now $\langle a, b\varphi \rangle$, a subgroup of G', is an (α, β, γ). To show that $G' = \langle a, b\varphi \rangle$ we need only show $\varphi \in \langle a, b\varphi \rangle$. Consider the commutator $[a, b\varphi] \in \langle a, b\varphi \rangle$.

$$[a, b\varphi] = c\varphi a^{-1}(b\varphi)^{-1}$$

$$= \varphi^{xy} c a^{-1} b^{-1} \varphi^{-y}$$

$$= \varphi^{y(x-1)}[a, b]$$

$$[a, b\varphi]^{o(G)} = \varphi^{y(x-1)o(G)}$$

Now $y(x-1)o(G) \not\equiv 0 \pmod{n}$, so $\varphi \in \langle a, b\varphi \rangle$.

Since $\langle \varphi \rangle$ is normal in G' and $G'/\langle \varphi \rangle \cong G$, the proof of the proposition is easily completed.

q.e.d.

The proof of Theorem 7.11 is completed by applying the proposition to the two cases discussed above, $(2,6,6)$ and $(2,5,10)$.

7.15. We now show that the two other bounds given in Theorem 7.7 are sharp infinitely often, by letting $s = 2$ in the theorem.

Theorem 7.15. Suppose n is a prime integer, $n \geq 168$. Let $p = 2n + 1$.

If $3 \mid (n-1)$ then $N(p) = 24(p-1)$.

If $3 \nmid (n-1)$ and $4 \mid (n-1)$ then $N(p) = 16(p-1)$.

Proof. It suffices to show the existence of two groups satisfying the hypotheses of Proposition 7.14, namely:

(i) a $(2,3,12)$ of order 48 and genus 3 with a $(2,3,6)$ of order 6 as a homomorphic image.

(ii) a $(2,4,8)$ of order 32 and genus 3 with a $(2,4,4)$ of order 4 as a homomorphic image.

(i) Let $A_4 = (2,3,3)$ be represented as the symmetries of the regular tetrahedron inscribed in the Riemann sphere. Let

$B = \{b_1, b_2, b_3 b_4\}$ be the 4 vertices which are fixed points for elements of order 3 in A_4. Define $\mu : \pi_1(\mathbb{P}^1 - B, b) \to Z_4$ $(= \langle\varphi\rangle)$ by $\mu(\gamma_i) = \varphi$ for each of the 4 paths circling the b_i's. The kernel of μ is associated to a 3-sheeted cover of $\mathbb{P}^1$ with total ramification 12, that is, a Riemann surface X_3. A_4 lifts to X_3 and so X_3 admits a $(2,3,12)$ of order 48. If the group is $G = \langle a, b\rangle$, $a^2 = b^3 = (ab)^{12} = e$, it is not difficult to see that the commutator subgroup of G is $\langle abab^2, ab^2ab\rangle$ $(= G')$ and G' is isomorphic to the quaternion group of order 8. Thus $G/G' \cong Z_6$ and G/G' is a $(2,3,6)$.

(ii) Let $Z_4 \times Z_4 = \langle\varphi, \psi\rangle$. Let a be the automorphism of $Z_4 \times Z_4$ of order 2 that permutes φ and ψ. Let $G = \langle a, \varphi, \psi\rangle$ be the semi-direct product where $a^2 = \varphi^4 = \psi^4 = e$, $\varphi\psi = \psi\varphi$, $a\varphi a = \psi$ and $a\psi a = \varphi$. It follows that $G = \langle a, \varphi\rangle$, a $(2,4,8)$ of order 32. The subgroup $\langle\varphi\psi^3\rangle$ is seen to be normal and

$$G/\langle\varphi\psi^3\rangle \cong Z_2 \times Z_4, \text{ a } (2,4,4).$$

Thus $G/\langle\varphi\psi^3, a\varphi^2\rangle \cong Z_4$, a $(2,4,4)$.

q.e.d.

All the Riemann surfaces occuring in Theorem 7.15 are elliptic-hyperelliptic. If $G = \langle a, b\rangle$, $c = ab$ then c^6 (resp. c^4)

is the elliptic-hyperelliptic involution in the case of $(2,3,12)$ $\left(\text{resp. } (2,4,8)\right)$.

7.16. We continue this discussion of $N(p)$ by showing that the results of Theorem 7.11 hold for any prime integer $n \geq 11$, $n \neq 13$. Macbeath [19] showed that $N(14) = (13)(84)$.

The following remarks will be useful. If a Riemann surface of genus p admits an automorphism group of order $m(p-1)$ and $m \geq 24$, then the only possible values for m are 24, (132/5), 30, 36, 40, 48, and 84. The reader will see that part of the following discussion depends on looking fairly carefully at the 35 Type I groups. This is slightly tedious but quite easy. Also a useful table of non-cyclic simple groups of low order is appended to this chapter.

Lemma 7.16. Let n be a prime integer, $n \geq 11$, $n \neq 13$ and $n < 84$. Let $p = n + 1$. Let G be a big group on W_p and suppose $n \mid o(G)$. Then $n^2 \nmid o(G)$, G is solvable, and the n-Sylow subgroup is normal in G.

Proof. The only big automorphism group of order $m(p-1)$ where m is prime is a $(2, 3, 78)$ of order $13(p-1)$. Consequently, $n^2 \nmid o(G)$.

Suppose G is not solvable. Then G has a non-cyclic simple factor in any composition series. If $o(G) = mn$, we may assume $m \leq 84$, and so $o(G) < (84)^2$ $(= 7056)$. By considering the non-cyclic simple groups of order less than 7056, we see that the relevant prime factors of the orders of these simple groups (11, 17, 19, 23) have indices (e.g. 60, 144, 180, etc.) inappropriate for automorphism groups. We conclude that G is solvable.

Since G is solvable, in any composition series there is a group H with a subgroup N, normal in H, so that $H/N = Z_n$. Consider the

coverings:

$$W_p \to W_p/N \to W_p/H .$$

The covering $W_p/N \to W_p/H$ is an unramified Z_n covering, and so the genus of W_p/H is positive. Consequently, $o(H) \leq 4(p-1) = 4n$. Therefore, $o(N) \leq 4$ and S_n (the n-Sylow subgroup) is normal in $H (n \geq 11)$. S_n is a characteristic subgroup of H and so is normal in the next subgroup of the composition series and so on until we reach G.

q.e.d.

Now that we know that S_n is normal, the proof of Theorem 7.11 proceeds exactly as before until last sentence, where we consider Type I groups, G, with $n \nmid o(G)$. Such a group has order $(B/A)(2p-2) = (2Bn/A)$ where n must divide A. The possible values for A are 11, 17, 19, 23, 29, 31, 41, and 47, and so these are the values of n to consider. In all but two cases the (α, β, γ) Type I group is perfect $\left((\alpha, \beta) = (\beta, \gamma) = (\gamma, \alpha) = 1\right)$ and so must have a non-cyclic simple group is any composition series. We then see that, with one exception, $o(G)$ is not divisible by the order of a non-cyclic simple group and so cannot occur. The exceptional perfect group is a (2, 5, 9) group of order (180/17). However, Paul Hewitt showed the author that there does not exist a (2, 5, 9) group of order 180. The two non-perfect Type I candidates fail because the group would be too small. For example, a (2, 5, 16) group of order (160/19) on W_{20} would yield a group of order 160 which is less than $8(p+1)$ (= 168). Thus Type I groups are eliminated from the

competition to achieve $N(p)$.

An examination of the proofs in Section 7.13 shows that they work equally well for the primes now under consideration and so Theorem 7.11 does indeed hold for all primes $n \geq 11$, $n \neq 13$. $p = 12$ is the smallest genus for which $N(p) = 8(p + 3)$.

7.17. Now we consider $N(p)$ for $p = 2n + 1$, n prime and $n \geq 11$. We will prove the following.

Theorem 7.17. Let n be a prime integer greater than or equal to 11. Let $p = 2n + 1$.

a) If $3 \mid (n - 1)$ then $N(p) = 24(p - 1)$

b) If $3 \nmid (n - 1)$ and $4 \mid (n - 1)$ then $N(p) = 16(p - 1)$

c) If $\left((n - 1), 12\right) = 1$ then $N(p) = 8(p + 1)$

Remarks. $p = 23$ is the smallest genus for which $N(p) = 8(p + 1)$. Paul Hewitt showed this by finding all previously unknown $N(p)$ for $p \leq 25$. Notice that we have lower bounds in parts a) and b) of the theorem by Theorem 7.15.

The integers n and p will always have the above meaning in this section.

We now adapt Theorem 7.7 to the present situation.

Lemma 7.17. Suppose $o(G) = N(p)$ and $n \mid o(G)$. Then

a) $n^2 \nmid o(G)$.

b) For $n < 168$, G is solvable.

c) For all n, S_n (the n-Sylow subgroup) is normal in G.

Proof. $o(G) = m(p - 1)$. The only case where m is prime is a $(2, 3, 78)$ of order $13(p - 1)$. This is excluded for $p = 23$ since 3

does not divide the order of the group. Thus $n^2 \nmid o(G)$.

If $o(G) = m(2n)$ then $m \leq 84$, and $o(G) \leq 168n$. If $n > 168$ then S_n is normal.

If $n < 168$ then $o(G) < (168)^2$ (= 28, 224). But all non-cyclic simple groups of order less than 28, 224 have the relevant cyclic Sylow subgroups with too great an index to be an automorphism group of a Riemann surface, with the possible exception of a simple group of order 660 (= 22.30)

We consider the possibility of a (2, 3, 10) on W_{23}. S_{11} in this group is unramified and by Sylow's theorem has index 5 in its normalizer. Thus W_{23}/S_{11} admits an automorphism of order 5. We now reach the contradiction by observing that the genus of W_{23}/S_{11} is 3.

Thus G cannot be simple. In fact, it must be solvable since a non-cyclic simple group cannot be fitted into a composition series. We now use the argument in Lemma 7.15 to show that S_n is normal.

q.e.d.

We can now apply the argument of Theorem 7.7 which results from S_n being normal in G. In the figure accompanying the proof of that theorem W_p/S_n has genus 3 . Consequently, possibility (iv) in the statement of the theorem is ruled out because, again, a W_3 does not admit a Z_5 . Thus in the light of Theorem 7.15, the conclusion of Theorem 7.7 for our n and p is:

If $3 \mid (n-1)$ then $o(G) = 24(p-1)$.

If $3 \nmid (n-1)$ and $4 \mid (n-1)$ then $o(G) = 16(p-1)$.

In any case, $(n-1, 12) \neq 1$.

Now Corollary 7.7 reads: If G is big and $(n-1, 12) = 1$ then $n \nmid o(G)$.

We now eliminate Type I groups by observing that their orders lose out in the competition or they are impossible because they are perfect.

To eliminate Type II groups we proceed as in Proposition 7.8. G is a $(2, 3, \lambda)$ of order $\mu\lambda$ where $n \nmid o(G)$. $2p - 2 = (\mu/6)(\lambda - 6)$. Consequently, there is an integer τ which divides 24 and $\lambda = \tau n + 6$, and $\mu = 24/\tau$. Finally there is an integer k dividing λ so that $\lambda \mid k(\lambda, \mu)$ and $k \leq \mu$.

If $\tau = 1$ we see that $(\lambda, \mu) = 1$ since n is prime, and so $\lambda = k$. It follows that $n < 18$ and so λ has the possible values 17, 19 and 23. Since all these groups would be perfect, they are eliminated.

If $\tau = 2$ we see that $(\lambda, \mu) = 4$, and so $\lambda \leq 48$, or $n < 21$. The possible values for n are 19, 17, 13 and 11, and the possible values for λ are 44, 40, 32 and 28. The first three possibilities lead to groups that are too small.

Thus we are led to consider the possibility of a $(2, 3, 28)$ of order 336 on a W_{23}. If we denote G by $\langle a, b \rangle$ with $c = ab$ and $d = ba$, then $G = \langle c, d \rangle$ and the normal subgroup $\langle c \rangle \cap \langle d \rangle$ must have order 4. The quotient of G by this normal subgroup would then have to be a $(2, 3, 7)$ of order 84, a contradiction.

The larger values of τ are easily eliminated and we see that our

big group must be a (2, 4, λ). For these groups we have:

$$2p - 2 = 4n = \mu(\lambda - 4)/4 \quad \text{or} \quad 16n = \mu(\lambda - 4) .$$

Since $n \nmid o(G)$ we see that there is an integer τ so that $\lambda - 4 = \tau n$, and so $\lambda = \tau n + 4$, $\mu = 16/\tau$ and $o(G) = 8\left(2n + (8/\tau)\right)$. The proof will be completed by showing that $\tau = 1$ or 2 is impossible.

We let $G = \langle a, b \rangle$, $c = ab$ and $d = ba$, as before. We must first show that $G \neq \langle c, d \rangle$. Assuming the contrary, we use the transfer onto the central subgroup $\langle c \rangle \cap \langle d \rangle$ to find an integer k, dividing λ, $k \leq \mu$, so that $\lambda \mid \mu k$. μ is always a power of 2. If $\tau = 1$, λ is odd. Now G is simultaneouly a (2, 4, λ) and a (λ, λ, 2). Thus the abelianized G must be cyclic with order dividing 2 and λ. Thus we reach the contradiction that G is perfect. If $\tau = 2$, we see that $(\lambda, \mu) = 2$, $\lambda \leq 16$ and so n is too small. We conclude that the index of $\langle c.d \rangle$ in G is 2.

Now refer to the diagram accompanying Proposition 7.9.

First assume $\tau = 2$. $\lambda = 2n + 4$, and so λ is not divisible by 4. Consequently, $k = 2$ or 3. But if $k = 2$, $G/\langle c \rangle \cap \langle d \rangle$ is a (2, 4, 2) of order 16, a contradiction. If $k = 3$ then $3 \mid (2n + 4)$ which implies that $3 \mid (n - 1)$, in which case the Type III groups are not in the competition.

Now assume $\tau = 1$, $\mu = 16$, $\lambda = n + 4$, and $k \leq 8$. Since n is prime, k must be odd. Let $G' = G/\langle c \rangle \cap \langle d \rangle$. $k = 3$ implies that G' is a (2, 4, 3) of order 48, a contradiction. $k = 5$ implies that G' is a (2, 4, 5) of order 80, again a contradiction since it implies that a W_3 admits an automorphism of order 5. If $k = 7$, we have a (2, 4, 7)

of order 112. Let c' and d' be the elements in G' corresponding to c and d in G. $o(\langle c', d' \rangle)$ is 56, and since S_7 is not normal in $\langle c', d' \rangle$, S_2 is. $o(S_2) = 8$ and being characteristic in $\langle c', d' \rangle$, it is normal in G'. S_2 has as non-identity elements

$$\left\{ c'^{-k} b'^2 c'^k \;\middle|\; k = 0, 1, \ldots, 6 \right\},$$

and so S_2 is seen to be an elementary abelian group of order 8. G'/S_2 acts faithfully on S_2 as a dihedral group of order 14. This is the final contradiction since the automorphisms of S_2 form the simple group of order 168, a group which does not have as a subgroup a dihedral group of order 14.

This completes the proof of Theorem 7.17.

With the work of Paul Hewitt we know $N(p)$ for all $p = n + 1$, or $2n + 1$ where n is a prime integer.

Orders of non-cyclic simple groups $< (168)^2 = 28224$

$$\begin{aligned}
60 &= 2^2.3.5 \\
168 &= 2^3.3.7 \\
360 &= 2^3.3^2.5 \\
504 &= 2^3.3^2.7 \\
660 &= 2^2.3.5.11 \\
1092 &= 2^2.3.7.13 \\
2448 &= 2^4.3^2.17 \\
2520 &= 2^3.3^2.5.7 \\
3420 &= 2^2.3^2.5.19 \\
4080 &= 2^4.3.5.17 \\
5616 &= 2^4.3^3.13 \\
6048 &= 2^5.3^3.7
\end{aligned}$$

$6072 = 2^3.3.11.23$

$7800 = 2^3.3.5^2.13$

$7920 = 2^4.3^2.5.11$

$9828 = 2^2.3^3.7.13$

$12180 = 2^2.3.5.7.29$

$14880 = 2^5.3.5.31$

$20160 = 2^6.3^2.5.7$

$25308 = 2^2.3^2.19.37$

$25920 = 2^6.3^4.5$

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